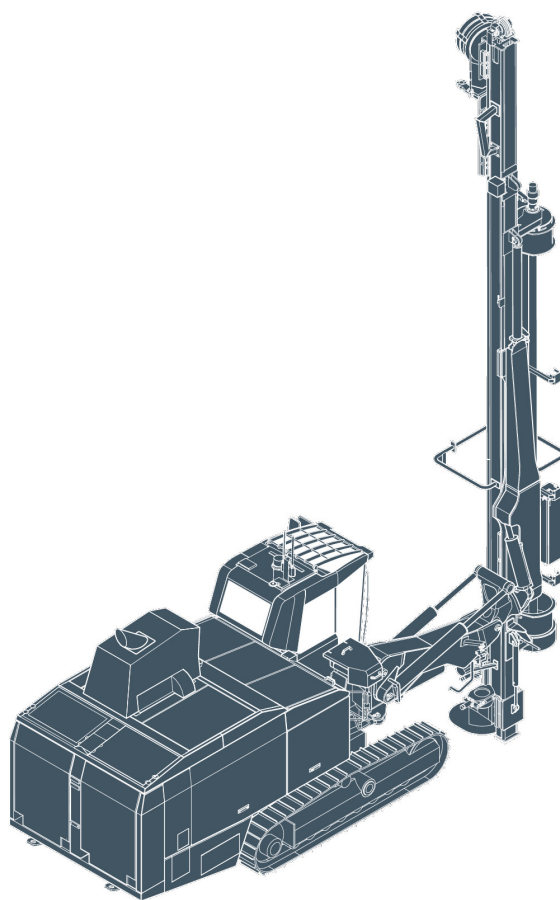


Control System

SmartROC D50/D55/D60/D65 (RCS 6.29)



Original Instructions

2025-04-30 | No: TIS0006203.003 en-US

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1 Introduction

1.1 Safety First

Always read the information in the safety manual before starting to use the machine or before starting any maintenance work.

1.2 The Purpose of This Publication

The control system manual is intended to give:

- Description of the control system operational and service menus.
- Description of the control system functions and parts.
- Instructions in maintenance and calibration of the monitoring function of the control system.
- Instructions in applicable settings.

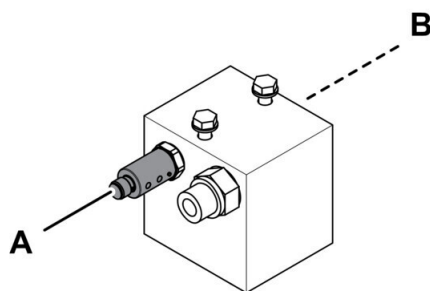
1.3 Target Group

The information in this publication is intended for everyone involved in the operation, and maintenance of the machine and related equipment. All personnel are expected to have basic competence regarding mining methods, construction methods, and the equipment used for that kind of operation.

1.4 Illustrations

The illustrations shown in this document are for illustration purpose only. The specific product or machine configuration can vary.

1.4.1 Graphic Elements in Illustrations



A	A reference with a solid line is pointing at a part that is visible in the illustration.	B	A reference with a broken line is pointing at a part that is hidden and not visible in the illustration.
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1.4.2 Images on the Display

The display Graphical User Interface (GUI) images in this document are part of the Rig Control System (RCS) software. All images are for reference purposes only. Menu views, GUI, and informational content can change with the revision of the software or firmware.

1.5 Feedback and Contact Information

Epiroc works actively to develop and continuously improve its products and associated documentation. Provide your feedback on products and documentation by contacting your local supplier at <https://www.epiroc.com>, under 'CONTACT'.



NOTE: Epiroc values customer feedback. If you give feedback via email, make sure that you use a valid email address.

Postal Address:

Epiroc Rock Drills AB
SE-701 91 Orebro
Sweden

2 System Description

2.1 Rig Control System

The Rig Control System (RCS) is a system that monitors and controls different machine functions and conditions. The RCS is intended to:

- Check necessary and essential parameters
- Present necessary information to the operator
- Execute orders given from the operator controls.

The system consists of various electronic modules, all of which are independent computers that communicate with each other using Controller Area Network (CAN) technology. For communication between the system and the operator, there is a display that is controlled from the operator panel.

The number of electronic modules varies depending on the options the machine is equipped with.

Any use other than what is listed as intended use is classified as misuse. Misuse of the control system can lead to malfunctions and is therefore not permitted.

NOTICE

Damage to Control System

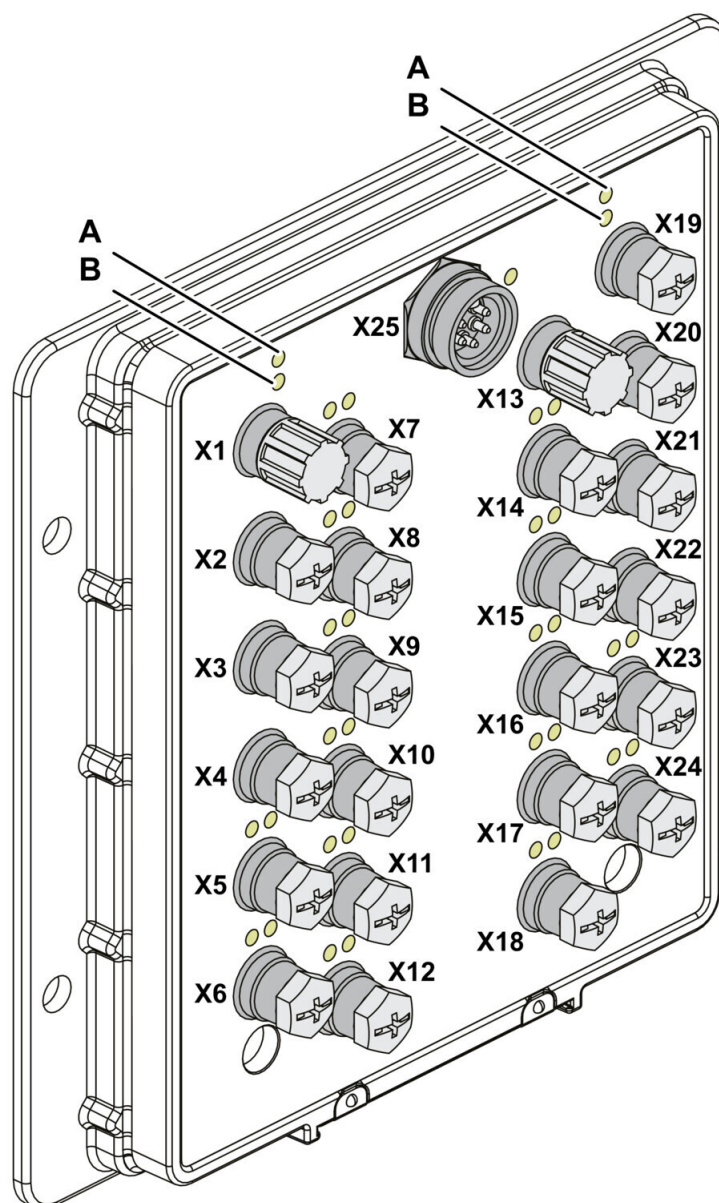
Misuse of the control system can lead to the control system becoming inoperative or damage the machine and associated equipment.

- ▶ Read and understand the instructions before using the control system.
- ▶ Follow the instructions and use the control system as intended.
- ▶ If there are any ambiguities regarding the instructions, contact Epiroc maintenance personnel.

2.2 System Hardware

2.2.1 I/O Module

The I/O module receives information from various sensors and provides each valve with the correct signal to obtain the necessary motion or function.



I/O Module

A	5 V
B	24 V
X1, X19	CAN in (X1), CAN out (X19)
X2, X3, X4, X20, X21, X22	Analog inputs
X5, X23	Digital Input (DI)/ Digital Output (DO) Output/Input with a transistor Negative-Negative-Negative (NPN) encoder
X7, X8, X9, X10, X14, X15, X16, X24	DI/DO Output signal with Pulse Width Modulation (PWM).
X11, X12, X17, X18	DI/DO Output with a transistor Positive-Negative-Positive (PNP)

X13	Address plug
X25	Power supply module

The I/O module will only turn on the 24 V power supply when CAN communication is established with the display. The 24-V LED indicator is red when communication is established.

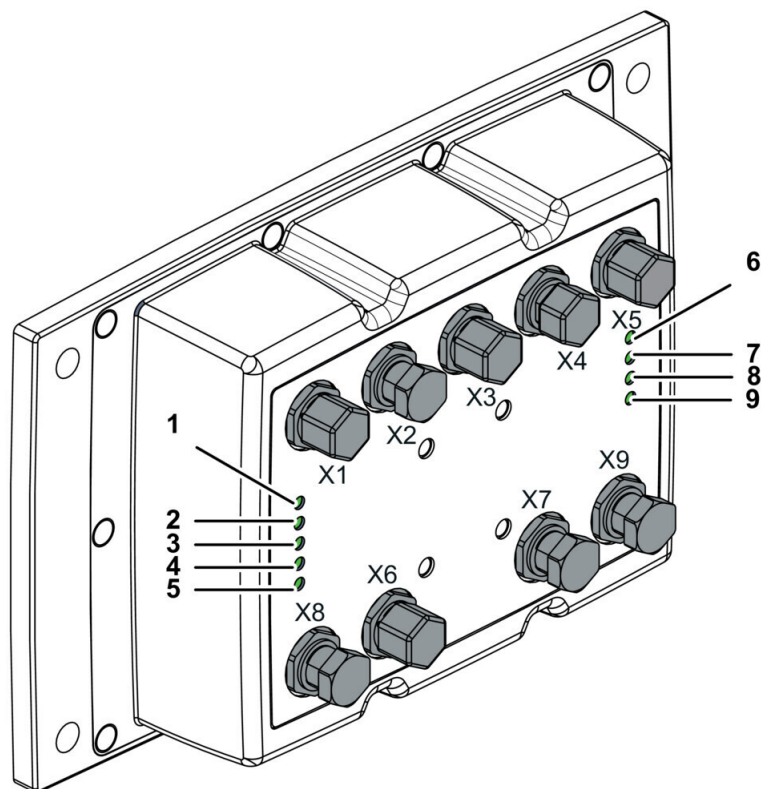
X25 LED Functions

Color	Description
Green	- Indicates 5 V and 24 V supply voltage. - When the system starts, the LED flashes twice per second. - When CAN communication is in progress, the LED flashes once per second. - When there is no CAN communication or the I/O module is starting up, the LED flashes five times per second.
Red	Indicates short circuit to ground of sensor supply voltage. The supply voltage must be reset after one second when the short circuit is broken.

Digital Inputs and Outputs

Contact	Off	Green	Orange	Red
PNP output	Load not activated	- Open circuit - Output not activated	Output activated and actuated	Output activated, but short circuit to ground
PNP input	- Input voltage low - No signal on input	- Voltage on input - Input active	Does not arise if the port is initialized as input	Does not arise if the port is initialized as input
NPN output	Not activated	- Output activated externally - Short circuit to ground	The output is activated and actuated	Output activated, but short circuit to ground
NPN input	- No signal Input - Voltage too high	Input activated (closed to ground)	Does not arise if the port is initialized as input	Does not arise if the port is initialized as input
PWM input	- Not activated - Load present	- Open circuit - Output not activated	Output activated and actuated	Output activated (low current) or short circuit to ground
PWM output	- Input voltage low - No signal on input	- Voltage on input - Input active	Does not arise if the port is initialized as input	Does not arise if the port is initialized as input

2.2.2 Application Module



X1	Power supply voltage	X2	CAN network 2
X3, X4	CAN network 1	X5	Address plug
X6	COM2	X7	USB / COM1
X8	Video	X9	Ethernet

Pin configuration

Contact	Type	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5
X1	Four-pin M12, male contact	+24 V supply voltage	+24 V supply voltage	GND	GND	
X2, X3, X4	Five-pin M12, male contact (X2, X4), or female contact (X3)	NC (Not Connected)	CAN +24 V	CAN GND	CAN Hi	CAN
X5	Five-pin M12, male contact	Bit 2	Bit 1	GND	Bit 0	Bit 3

Each address bit is connected to GND (ground). The address plug can be connected to four different addresses.

LED Functions

LED	Color	Description
Module status D16	Green	- When the system starts, the LED flashes two times per second. - When CAN communication is in progress, the LED flashes once per second. - When CAN communication is absent or the I/O module is about to start, the LED flashes five times per second.
CAN status network 1 (D14) and network 2 (D15)	Red	The LED flashes when communication is established.
	Black	No communication.
Supply voltage CAN network 1 (D19) and CAN network 2 (D20)	Green	Steady green light means that supply voltage is maintained.
	Black	No supply voltage.

2.2.3 RCS Standard Sensors

The control system consists of a number of sensors. The sensors are connected to the I/O unit and the LCD display through CAN BUS communication. The system is operated and presented from the operator panel and the LCD display.

2.2.4 Radio Remote Receiver

The Radio Remote Receiver is mounted on the machine. The Radio Remote Receiver converts radio signals to CAN open messages and communicate between the Radio Remote Control and the operator panel.

2.3 Operator Menus

The operator has access to several control system menus. In some menus, different settings and selections can be made, in others only information is provided to the operator.

Operator menus and their submenus are different depending on:

- The options that the machine is equipped with.
- The authorization that the operator is logged in with.

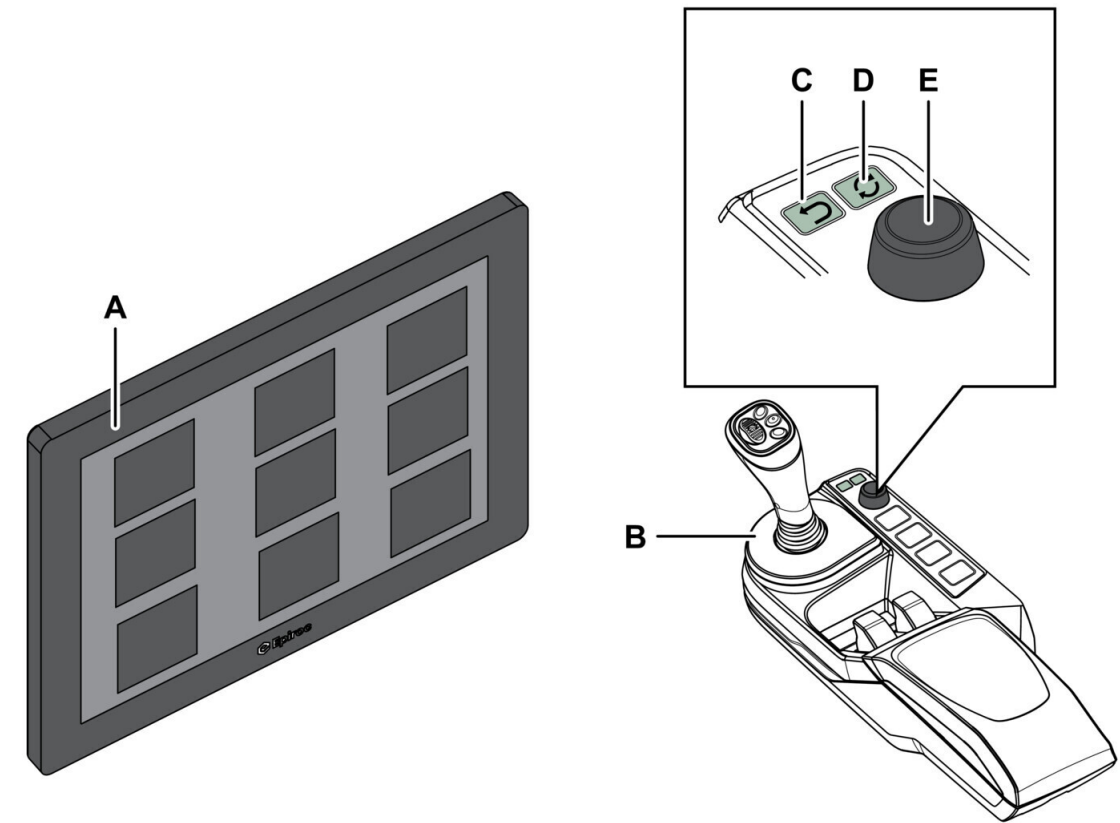
2.4 Access Levels

This manual describes the two access levels that are available to the end user.

- Operator (OP) — Provides basic access to the system
- Service (SE) — Provides advanced access to the system.

2.5 Menu Navigation

To read or change values in the different menus for drilling, the operator must activate the correct menu. To activate a menu, either press directly on the menu on the touch screen or use the right-hand operating control to move between menus.

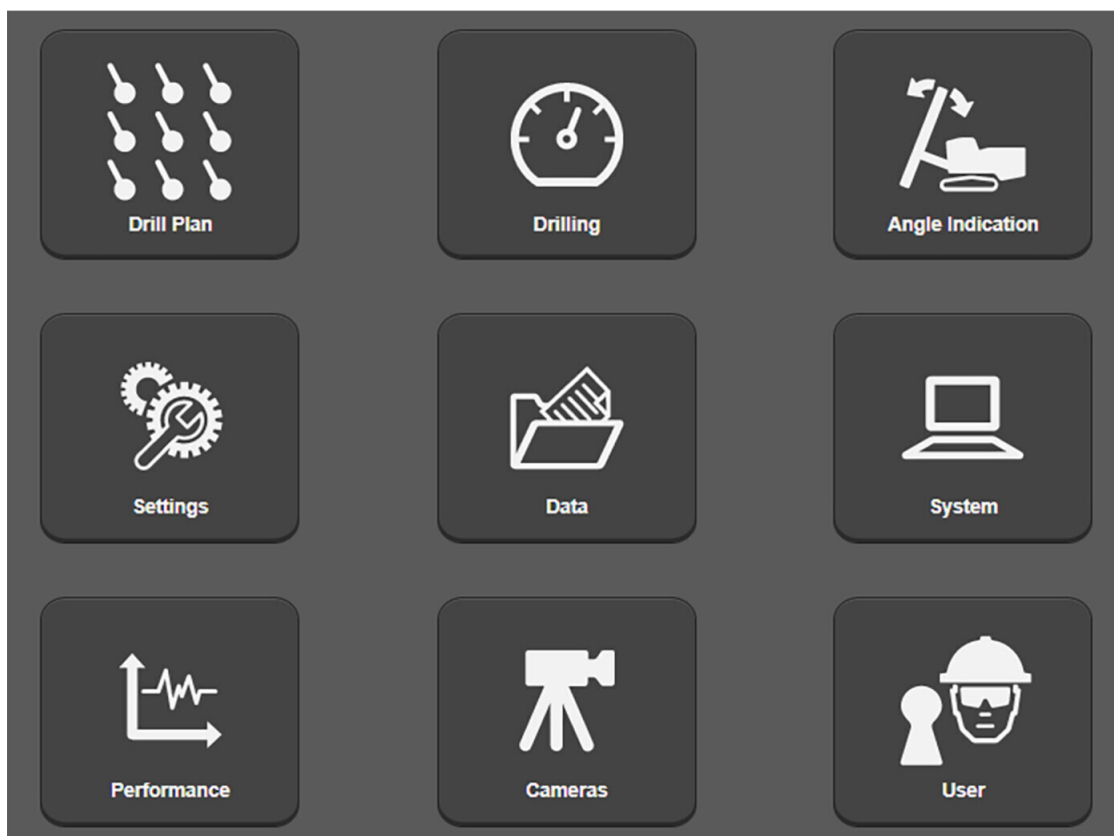


A	Display. The display is a touch screen and used to navigate in the menus.	B	Right-hand operating control
C	Esc key Back one step in the menu tree. If any changes have been made in a menu, press the Enter key (E) first to save the settings before moving back one step.	D	Toggle button Switches between two pages, chosen by the last times the button was used. Press to save the current menu page and directly be moved to last previous saved menu page.
E	Navigation knob and Enter key Using the navigation knob is not recommended, use the touch screen to navigate in the menus.		

2.5.1 Home Menu

The **Home** menu opens up when the control system is started.

The menu buttons offer access to various menus related to rig settings, setup, performance, operations, and recorded on-board data.



Example of a **Home Menu**

	Drill Plan This menu shows information about the drill plan. In this menu, drill plans can be created, added, and saved.		Drilling This menu shows information about pressures and flows during drilling.
	Angle Indication This menu displays the machine angle orientation and feed angle.		Settings This menu shows the machine settings.
	Data This menu is used to import and export data to and from the machine. Production plans, production logs, and system logs.		System This menu shows information about modules, event logs, and levers. The Administration menu shows the configuration of the machine.
	Performance This menu shows the statistics about the machine.		Cameras This menu is used for camera settings.
	User This menu is used to log in to higher access levels and to select menu language.		

2.5.2 User Menu

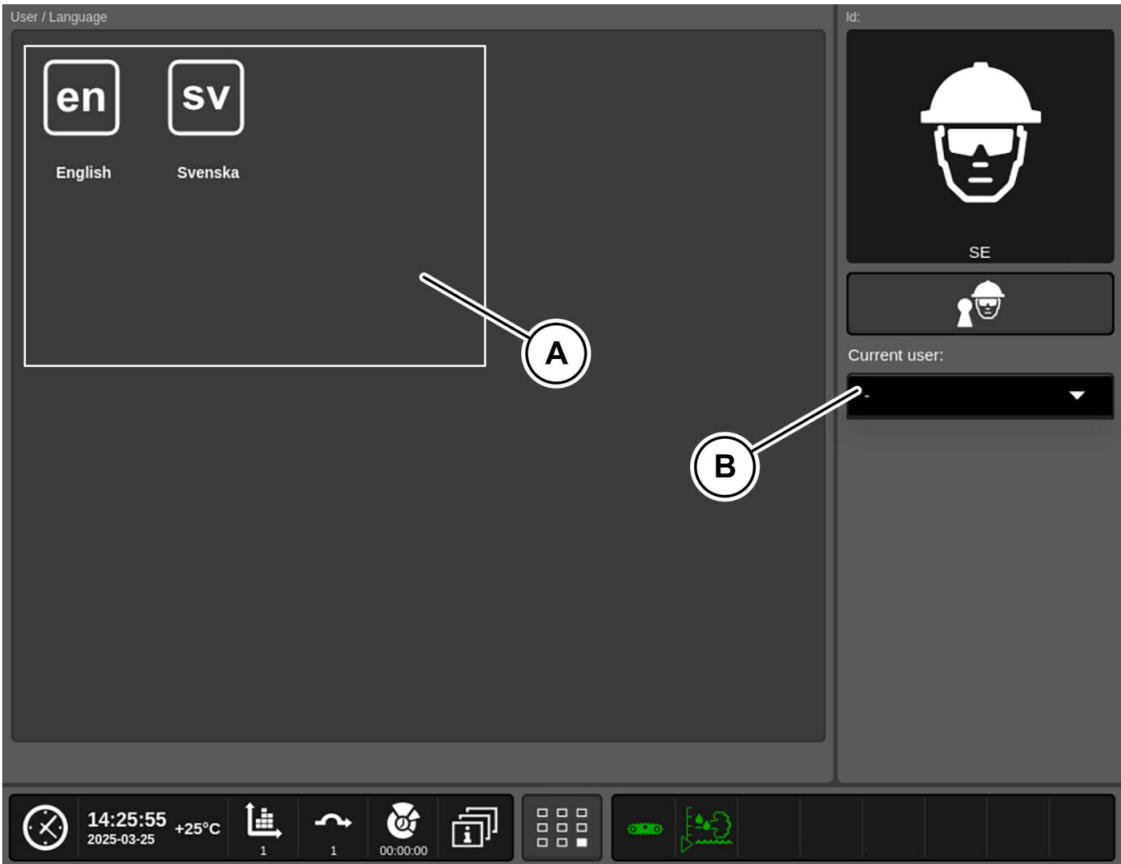
NOTICE

Malfunction of the Control System

Incorrect use of the service menus can cause the system to malfunction.

- ▶ Only maintenance personnel that have completed the training for the Epiroc drilling system are allowed to use the service menus.

The **User** menu is used to access Service level (SE) and to select menu language. The menu languages available are shown in the work panel.



Example off a **User Menu**

A	Tap on the relevant language symbol in the language menu to set the required language.	B	Select current user from the drop-down list.
	Press the button to open the Login pop-up menu, where the access level can be changed. Operator (OP) access level is the default operator level when the RCS starts.		

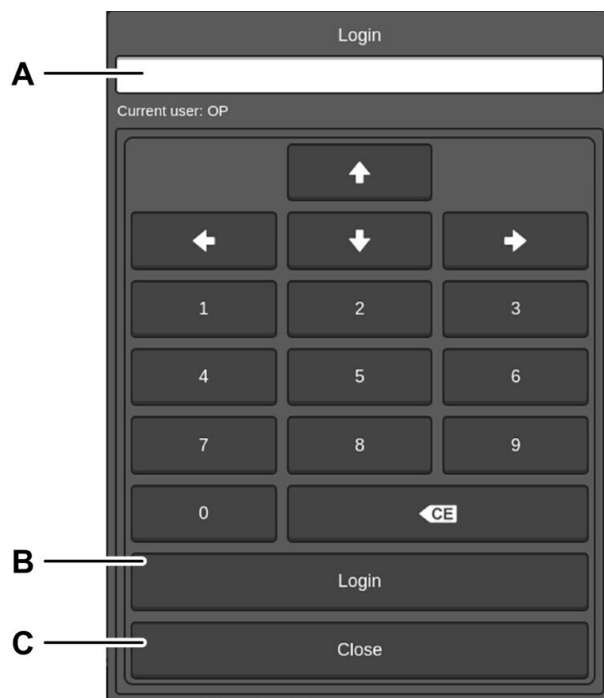
Login to a Higher Access Level

NOTICE

Malfunction of the Control System

Incorrect use of the service menus can cause the system to malfunction.

- Only maintenance personnel that have completed the training for the Epiroc drilling system are allowed to use the service menus.



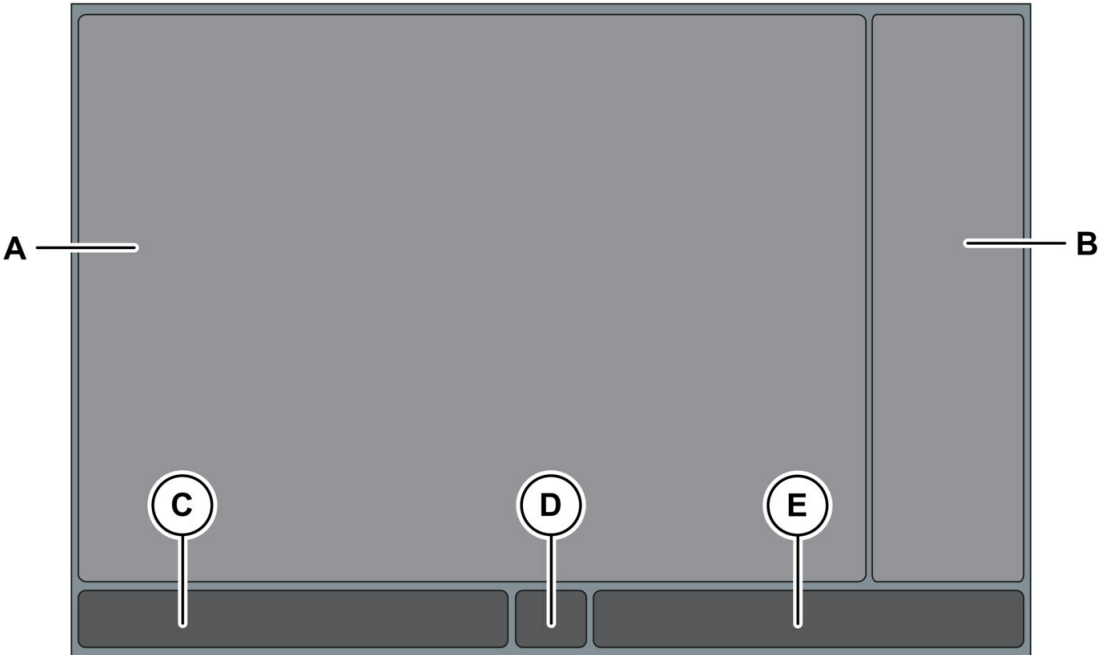
1. Open the **User** menu and press the button, .
2. Enter the access code (A) and then press **Login** (B).



NOTE: If the access code does not work and you need a new access code, contact Epiroc.

3. Press **Close** (C) to close the pop-up menu.

2.5.3 Menu Layout



A	Work panel • The work panel shows data and available settings.	B	Tool panel In the tool panel, settings and underlying menus are accessed for the work panel (A).
C	Information bar • The information bar provides, delay codes, last 10 event logs, quick access to interlocks, and information in messages from the monitoring of the system.	D	Home • The number of squares on the button reflects the number of menus in the current Rig Control System (RCS). • Select the Home button to navigate back to the Home menu.
E	Status bar • The status bar shows information, faults, and warnings from the monitoring of the system. • The information, faults, and warnings are shown as symbols. To see more information, tap on the status bar.		

NOTE: Some menus can differ from the standard menu layout.

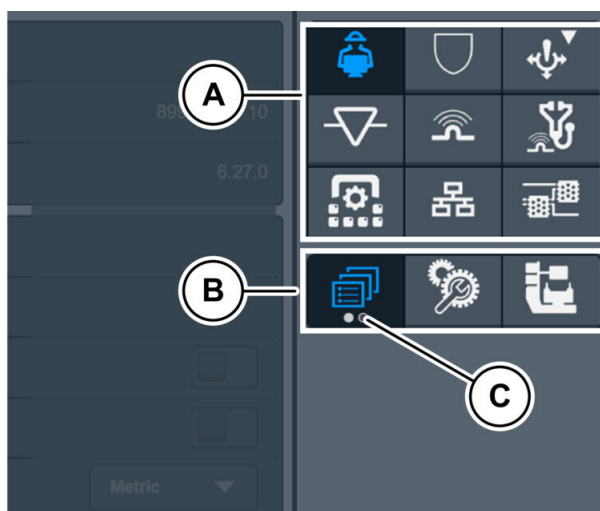
Rig Control System (RCS) Button Status



Example of a Menu

- If the symbol on the button is blue (A), the button is selected and active. The name of this button is shown in the top-left corner of the screen (C).
- If the symbol on the button is white (B), then the button is inactive and not selected.
- A dot (D) under any selected button shows that more menus are available.

2.5.4 Access Underlying Menus



Example of a Menu

- Click the menu button in field (A) to select an underlying menu.
- Click the parameter button in field (B) to select the type of parameters to be shown.
- Click the dot (C) to see more parameters.

2.6 System Monitoring

The control system continually monitors several different conditions on the machine. The system also monitors whether different functions are On or Off.

The control system gives necessary information regarding the system and alarms when any fault occurs in the system.

2.6.1 Alarms in Control System

The control system gives necessary information concerning the system in three different levels.

- Fault
- Warning
- Information

They are presented to the operator by:

- Text messages in a separate window on the display screen.
- Symbols in the status bar.

The operator must acknowledge the text messages to close the window.

2.6.2 Description of Status Bar

Information on the various conditions monitored by the control system, is presented with symbols in the status bar. The symbols come on, and in different colors which indicates different information levels:

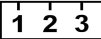
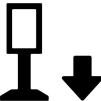
- Fault, is indicated in red. The system detects a condition that requires immediate attention.
- Warning, is indicated in yellow. The system detects a condition that should be attended to as soon as possible.
- Information, is indicated in green when the function is active, or in grey when not activated.

Information event symbols are indicated in white.

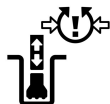
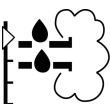
2.6.3 Symbols in the Status Bar

Symbol	Function	Information Level
	General warning - fault symbol. Always with printout in event log.	• Yellow - Warning • Red - Fault
	Emergency stop. The symbol lights up red when any one of the emergency stops has been activated or not reset. The diesel engine is switched off automatically.	• Red - Fault
	Low hydraulic oil level. The diesel engine turns off automatically if a fault occurs.	• Yellow - Warning • Red - Fault
	Low coolant level. The diesel engine turns off automatically if a fault occurs.	• Yellow - Warning • Red - Fault
	High compressor temperature. The diesel engine turns off automatically if a fault occurs.	• Yellow - Warning • Red - Fault

Symbol	Function	Information Level
	Clogged air filter, diesel engine.	Yellow - Warning
	Clogged air filter, compressor.	Yellow - Warning
	Clogged hydraulic oil filter. The symbol stays on until the Rig Control System (RCS) is restarted.	Yellow - Warning
	Diesel engine temperature high. The diesel engine turns off automatically if a fault occurs.	- Yellow - Warning - Red - Fault
	Warning from one of the sensors that are connected to the Electronic Control Module (ECM). See engine status menu for more info.	Yellow - Warning
	The diesel engine has gone down to reduced power.	Yellow - Warning
	Unknown fault from Electronic Control Module (ECM), see Suspect Parameter Number (SPN) and Failure Mode Identifiers (FMI) codes in engine status menu.	Yellow - Warning
	High or low fuel pressure. The diesel engine turns off automatically if a fault occurs.	- Yellow - Warning - Red - Fault
	Low oil pressure in diesel engine. The diesel engine turns off automatically if a fault occurs.	- Yellow - Warning - Red - Fault
	High charge air temperature.	Yellow - Warning
	High fuel temperature. The diesel engine turns off automatically if a fault occurs.	- Yellow - Warning - Red - Fault
	High engine speed.	Yellow - Warning
	High or low hydraulic oil temperature. The diesel engine turns off automatically if a fault occurs.	- Yellow - Warning - Red - Fault

Symbol	Function	Information Level
	Length sensor not calibrated. This symbol is displayed when the control system starts and stops when the rock drill cradle has been calibrated.	Yellow - Warning
	Low Hammer Electric Controlled Lubrication (HECL) pressure. Drilling is switched off automatically.	Yellow - Warning
	Support leg activated.  NOTE: <i>Tramming is not allowed if the support leg is down (yellow symbol).</i>	- Yellow - Warning. Support leg activated. - Green - system OK. support leg on ground. (Only on Operator Station (OPS) not Rig)
	Support leg out.  NOTE: <i>Tramming is not allowed if the support leg is down.</i>	Green - information
	Support leg in.  NOTE: <i>Tramming is not allowed if the support leg is down.</i>	Green - information
	Low fuel level. The symbol is activated when the fuel level drops below the value set in the Settings menu.	Yellow - Warning
	Time for engine service.	Yellow - Warning
	Time for rock drill service.	Yellow - Warning
	Lever or levers not in neutral.	Yellow - Warning
	Incorrect positioning data obtained from angle sensor.	Yellow - Warning

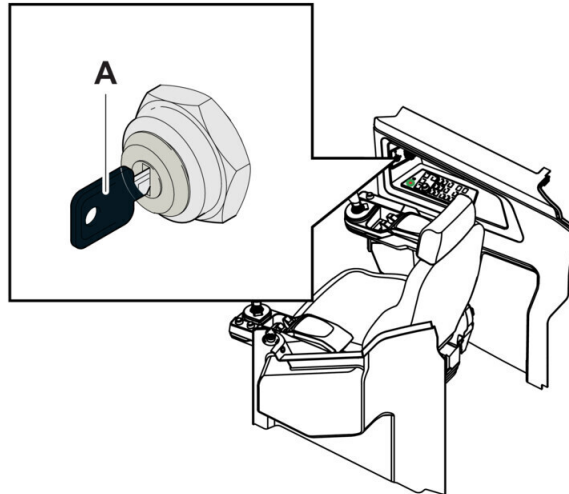
Symbol	Function	Information Level
	Lever fault: Incorrect values from lever or levers. See event log for more information.	Yellow - Warning
	File transfer initiated. Goes out automatically when finished saving. Do not remove the USB flash drive while the symbol is displayed. Green information Measurement While Drilling (MWD) log is saved on the USB flash drive.	Yellow - Warning
	Time to regrind drill bit.	Yellow - Warning
	Quality log has been corrupted because the operator has added one more hole than the maximum allowed. Create a new correct drill plan to clear the warning symbol.	Yellow - Warning
	Locked collaring.	Green - Active
	Auto-drilling locked. This symbol is green while the function is activated.	Green - Active
	Option Green - the automatic rod handling system is active. Gray - Inactive or option not activated.	- Gray - Inactive - Green - Active
	Automatic positioning system - Gray - Inactive or option not activated. - Green - automatic positioning system is active.	- Gray - Inactive - Green - Active
	Laser receiver - Gray - no hit, or option not activated. - Green - laser receiver has received a hit on the laser plane.	- Gray - Inactive - Green - Active
	Cleaning screw sampler.	Green - Active
	Loading screw sampler.	Green - Active
	Reverse Circulation (RC) kit.	Green - Active

Symbol	Function	Information Level
	Breakout table jaws Green - breakout table jaws activated.	Green - Active
	The anti-jamming function is activated due to high rotation pressure.	Yellow - Warning
	This symbol is shown if the cradle stops earlier than expected at M7 or M8 because the drill steel supports are closed.	Yellow - Warning
	Damper pressure out of limit	Yellow - warning
	Diesel Exhaust Fluid (DEF) tank empty, engine shutdown (warning level 3). DEF level critical (warning level 2). DEF level low (warning level 1).	- Red - Fault - Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V
 (Red)	DEF tank empty, engine shutdown (warning level 3). DEF level critical (warning level 2). DEF level low (warning level 1).	- Red - Fault - Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V
	Exhaust filter warning.	Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V
	Indicates Exhaust filter error.	Red - Fault  NOTE: Only applies to Tier 4 / Stage V
	Indicates that the DEF level is < 80%.	Green - Information  NOTE: Only applies to Tier 4 / Stage V
	Indicates that the DEF level is < 80%.	Green - Information

Symbol	Function	Information Level
		 NOTE: Only applies to Tier 4 / Stage V
	Indicates that the DEF level is > 60%.	Green - Information  NOTE: Only applies to Tier 4 / Stage V
	Indicates that the DEF level is > 40%.	Green - Information  NOTE: Only applies to Tier 4 / Stage V
	Indicates that the DEF level is > 20%.	Green - Information  NOTE: Only applies to Tier 4 / Stage V
	High Exhaust Temperature (HEST). Illuminates when regeneration has started.	Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V
	Automatic regeneration blocked. Used when there is a risk of personal injury or damage to property.	Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V
	High soot content in the Diesel Particulate Filter (DPF). Shows when the soot level is close to the maximum level.	Yellow - Warning  NOTE: Only applies to Tier 4 / Stage V

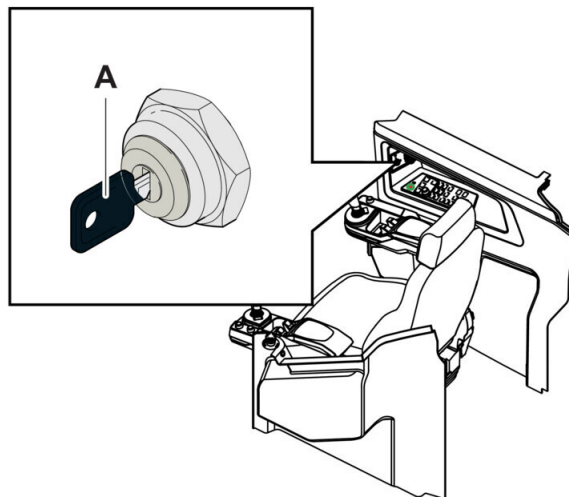
3 Start and Stop the Control System

3.1 Start the Control System



- Turn the key (A) from position 0 to position 1 to start the control system on the machine.

3.2 Stop the Control System

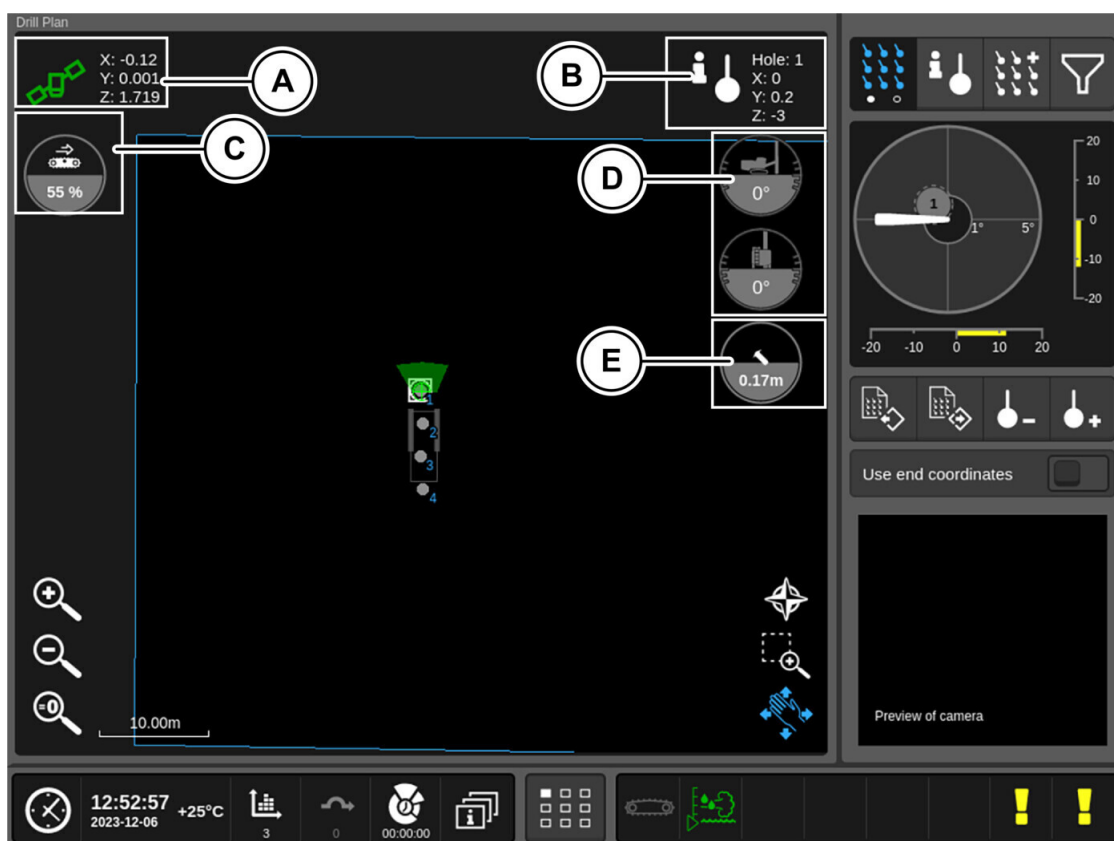


- Turn the key (A) from position 1 to position 0 to stop the control system. The display goes off after a few seconds.

4 Description of Menus

4.1 Drill Plan Menu

The drill plan menu shows the current drill plan. On the tool panel to the left, there are parameters for adjusting the functions of the drill plan.



Example of a Drill Plan Menu

A	The satellite symbol shows the status of the satellite contact. • Yellow: System not OK (for example, calculated baseline error, base station error or COM error). • Green: System OK. • Red: HNS is deactivated or error has occurred. X, Y, and Z show the position of the drill bit.	D	A gray symbol shows that the angle is within the pitch or roll limit value for the machine. A yellow symbol shows that the pitch or roll angle is less than the maximum value but greater than the minimum limit value.
B	Information showing the coordinates of the selected hole.	E	Indicates distance and direction from the bit to the selected hole.

C	Shows the percentage of speed on the belts and gear. One arrow is shown for low gear and two arrows are shown for high gear with respective percentage shown underneath.		
	Tool panel for drill plan handling.		Tool panel for hole Information.
	Tool panel for templates and settings.		Tool panel for filters.
	Zoom in image on screen.		Zoom out image on screen.
	Reset zoom		Tap the symbol and then mark the area to zoom in.
	Tap the symbol, the image on the screen can now be moved around.		Tapping this symbol rotates the drill plan toward the north or towards the direction of the drill plan.

4.1.1 Tool Panel for Drill Plan Handling

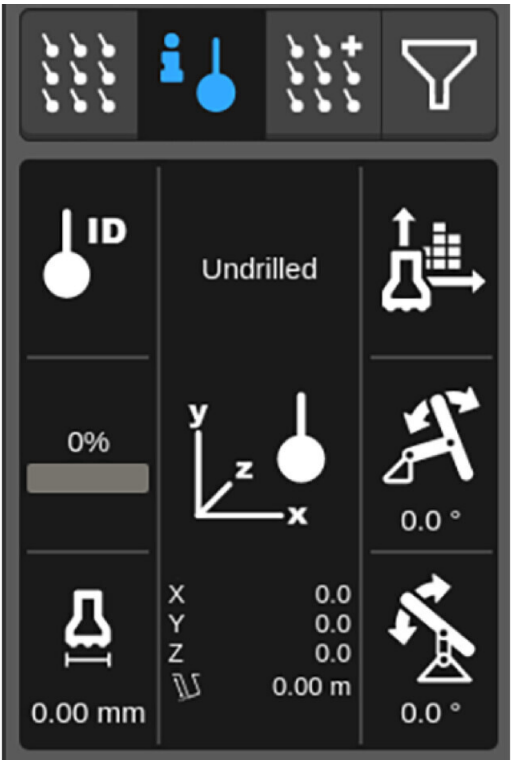


NOTE: Information about selected drilling is only shown if the drill holes and the drill plan are properly created and saved.



A	Shows the accuracy of the drill hole position relative to the drill bit and the angle of the feed beam according the required feed angle.	B	Use end coordinates Activate to aim for the hole bottom.
	Select to load a drill plan		Select to save a drill plan
	Select to delete a marked hole		Select to add a hole. In the pop-up menu that opens set the parameters for the hole. • Blast direction • Hole angle • Level • Drillbit diameter

4.1.2 Tool Panel for Drill Plan Hole Information



Example of a Tool Panel with **Drill Plan** hole information

	This field shows the ID number of the selected hole.		Shows the drilled % of the selected hole.
	Shows the drill bit diameter.		This field shows the selected hole number and its position against the origin of the drill plan: • X: Vertical position • Y: Horizontal position • Z: Height position • : Hole length Select a hole in the drill plan to view the hole in the tool panel.
	MWD logging is on when the symbol is green.		Feed dump angle
	Feed swing angle		

4.1.3 Drill Plan Overview Menu

When a drill plan is loaded this menu shows the information for all included holes in the drill plan, for instance, status and planned start and end coordinates.

Drill Plan

DP_10X10

Hole Id	Status	Planned start	Planned end	Actual start
a0b2b679c7	Undrilled	[-18.00 14.00 0.00]	[-18.00 14.00 -20.00]	-
a002cf69a0	Undrilled	[-16.00 18.00 0.00]	[-16.00 18.00 -20.00]	-
a66cc3393a	Undrilled	[-6.00 16.00 0.00]	[-6.00 16.00 -20.00]	-
a303c81b9a	Undrilled	[-4.00 2.00 0.00]	[-4.00 2.00 -20.00]	-
a5498ecb63	Undrilled	[0.00 6.00 0.00]	[0.00 6.00 -20.00]	-
aed898125a	Undrilled	[-16.00 6.00 0.00]	[-16.00 6.00 -20.00]	-
afa08abbe9	Undrilled	[-10.00 12.00 0.00]	[-10.00 12.00 -20.00]	-
b7fccdeea9	Undrilled	[-18.00 10.00 0.00]	[-18.00 10.00 -20.00]	-
b9dd77a620	Undrilled	[-4.00 18.00 0.00]	[-4.00 18.00 -20.00]	-
b0872da4d9	Undrilled	[-12.00 0.00 0.00]	[-12.00 0.00 -20.00]	-
b1251a0ad7	Undrilled	[-10.00 4.00 0.00]	[-10.00 4.00 -20.00]	-

09:27:28
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1 2 00:00:00

4.1.4 Tool Panel to Create Drill Plan Templates

Id:

Create template

Template Name

Filename

Burden m

Spacing m

Number of rows

Holes per row

Save

This tool panel is used to create a **Drill Plan** template.

Template Name: Enter the name of the template in this box.

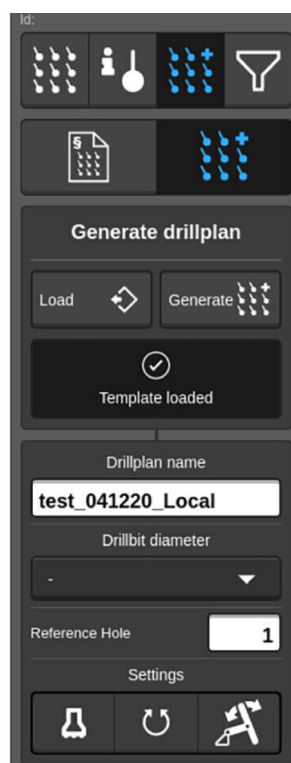
Burden: Set the burden distance between rows in the drill plan.

Spacing: The distance between the holes.

Number of rows: The number of rows in the template.

Holes per row: The number of holes per row in the template.

4.1.5 Tool panel to Load and Generate Drill Plan



This tool panel is used to load and generate Drill plan settings.

Load: Tap to open the pop-up menu to select a drill plan template.

Generate: Tap to generate drill plan.

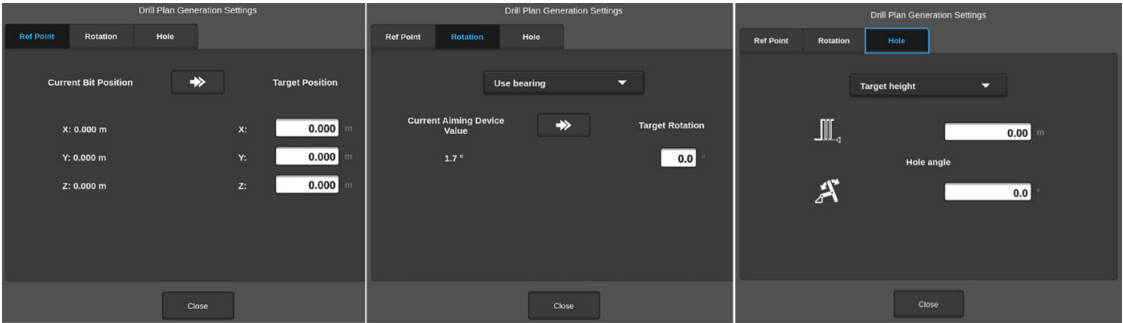
Drillplan name: Add the name of the drill plan. If a name is not added, the system creates an automatic name with the template name plus the actual date and time of the drill plan creation.

Drillbit diameter: Diameter of drill plan holes.

Reference Hole: Set the hole that is used as a reference point for the drill plan.

Settings: Tap on the buttons to open the pop-up menu **Drill Plan Generation Settings** and set the data for the drill plan.

Drill Plan Generation Settings



Example of a Drill Plan Generation Settings Pop-Up Menu.

Ref Point: Use to set the reference point for where to place the drill plan.

Rotation: Use the compass bearing to set the blast direction.

Hole: Use to set the **Target height**, **Hole Depth**, and angle of the holes.

4.1.6 Tool Panel for Drill Plan Filters

The filter menu is used to set which functions are shown in the work panel. A line over the symbol indicates it is not visible in the work panel.



Example of a Tool Panel with Filter Settings

	Status of the satellite		Information status
	Tramming		Rig front tilt inclination
	Rig lateral inclination		Drill holes
	Undrilled holes		Subdrilled holes
	Hole numbers		

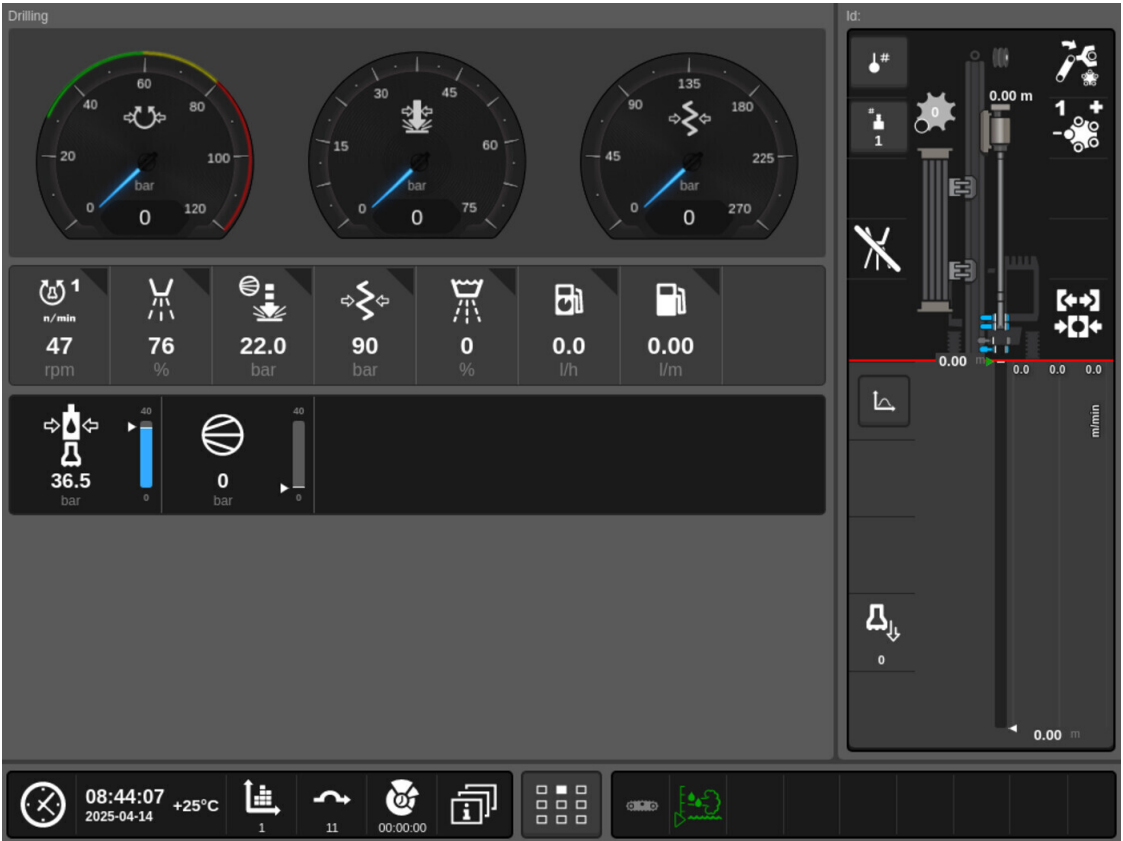
4.2 Drilling Menu

The Drilling menu consists of:

- Pressure gauges
- Information and status of drilling functions
- Drilling menu Tool panel



NOTE: Different symbols are shown on the screen view, depending on the configuration of the machine.

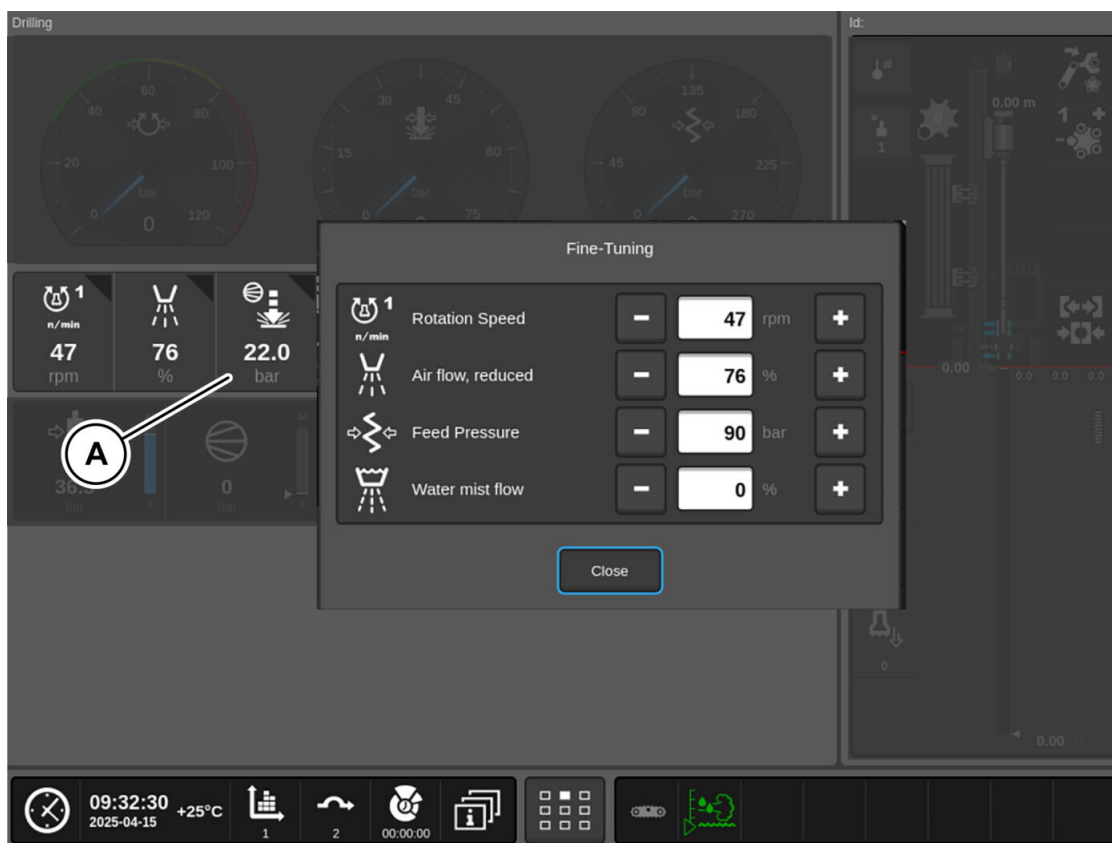


Example of a Drilling Menu

	Rotation unit speed Shows the set target for rotational speed of the drill bit.		Air flush Shows the set target for percentage of air flow.
	Air pressure, collaring drilling Shows the set target for air pressure for collaring.		Air pressure, full drilling Shows the set target for air pressure for full drilling.
	Feed pressure Shows the set target for feed pressure while forward and reverse feed is being used.		Water mist Shows the set target for water flow in the water mist system (%).
	Dust Collector Shows the set target in % of suction effect.		Fuel consumption Shows the fuel consumption (liters per hour).

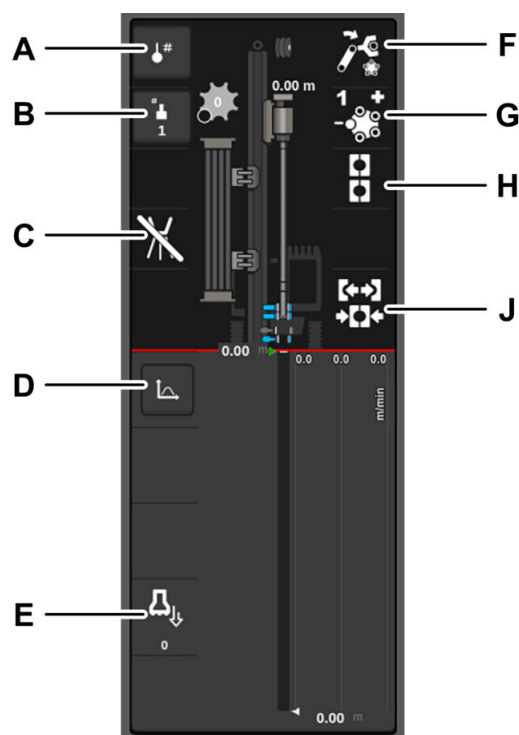
	Fuel consumption Shows the fuel consumption (liters per minute).		HECL lubrication pressure Shows the lubrication pressure in the drill string.
	Air Pressure Shows the current pressure in the pressure tank (Bar).		

4.2.1 Adjust Drilling Settings



1. Open the **Drilling** menu.
2. Select on one of the adjustable drilling parameter buttons (A) to open the pop-up menu for **Fine-Tuning**.
3. Adjust the values using the + and - buttons or enter a value in the numerical box.

4.2.2 Tool Panel for Drilling Menu



Example of a Tool Panel for a Drilling Menu.

A	Select to open pop-up window Quick Drilling Settings . Here parameters can be adjusted for Subdrilling and Target Hole Length .	B	Shows the number of rods in the drill center (in the hole). Select to open pop-up window Rod count settings .
C	Shows that air flushing valve is open or closed. A line over the symbol shows that the flushing valve is closed.	D	Select to open pop-up window Quick Drilling Settings .
E	Shows the rate of penetration while drilling. When the hole is completely drilled, it shows the average penetration rate of the drilled hole.	F	Rod Handling System (RHS) arms position. In or out of the magazine.
G	Show the magazine slot number that is aligned with the magazine door.	H	•  ,  - Shows if breakout table jaws are open or closed. •  - RC mode
J	Drill rod support position. Symbol changes if drill support open or closed.		

Quick Drilling Settings for Hole

Quick Drilling Settings

#

Target hole ID:

Subdrilling:

0.0

m

Target Hole Length:

0.0

m

Drill hole comments

Close

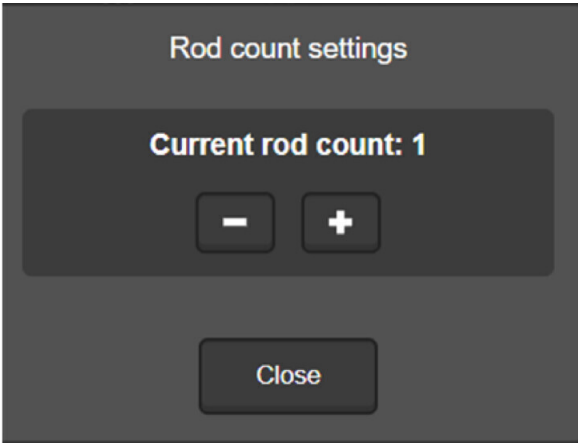
Ok

Quick Drilling Settings for Hole

Target hole ID	Shows the hole number.
Subdrilling	The distance set increases the hole depth.
Target Hole Length	Length of the hole. - HNS used - the value is calculated. - HNS not used - the operator enters the value.
Drill hole comments	In this window, a comment for the hole can be entered.

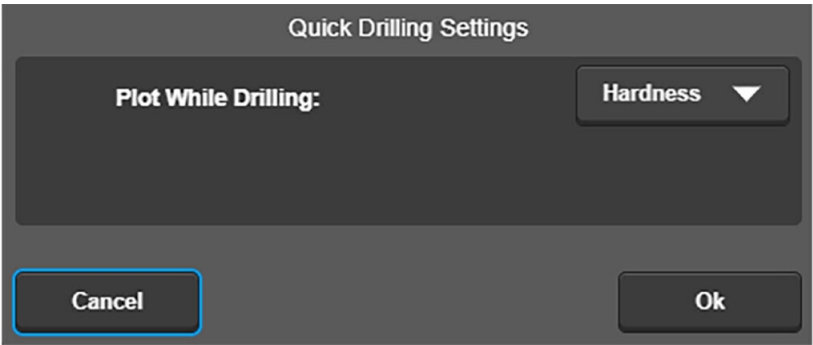
Rod Count Settings

The Rod count settings pop-up menu is used to override the rod counter.



	Press - to remove a rod from the counter.		Press + to add a rod to the counter
---	---	---	-------------------------------------

Quick Drilling Settings



Quick Drilling Settings *pop-up menu*

Plot While Drilling	Select which Measure While Drilling (MWD) data to display in the Drilling menu, during the actual time of drilling.
----------------------------	---

4.2.3 Drilling Menu Gauges



Adjust by tapping on the gauges or use the control panel.

The gauges in the Drilling Menu show:

- Rotation
- Percussion
- Feed pressure

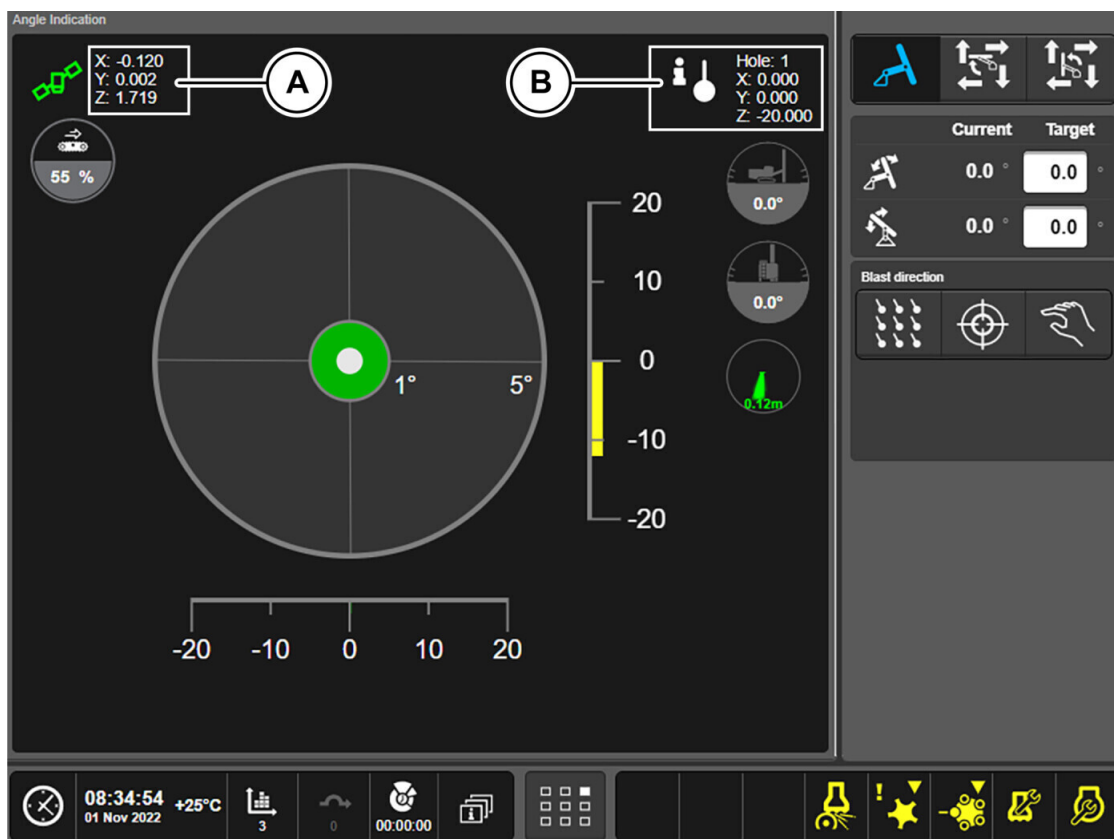
The needle of the gauge changes, in accordance with incoming actual value from the sensors.

Values must not reach the red zones during operation.



NOTE: If values reach the red zones during operation, protections are activated and stop the operation.

4.3 Angle Indication Menu

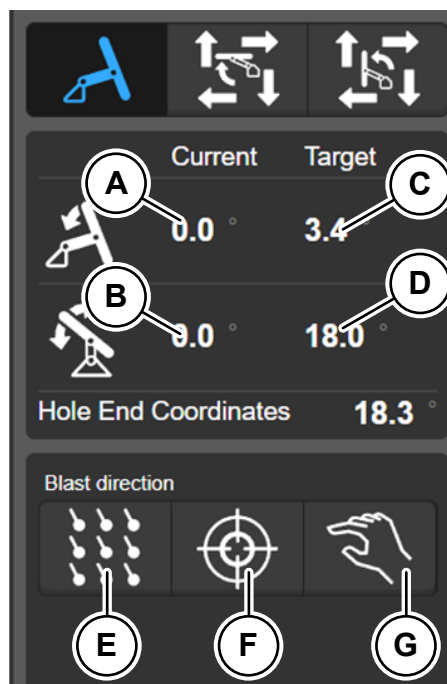


Example of an Angle Indication Menu

A	Shows the position of the drill bit.	B	Information showing the coordinates of the selected hole.
	The satellite symbol shows the status of the satellite contact. - Green: System OK. - Yellow: System not OK (for example, calculated baseline error, base station error or COM error). - Red: HNS is deactivated.	- A gray symbol shows that the angle is within the pitch or roll limit value for the machine. - A yellow symbol shows that the pitch or roll angle is less than the maximum value but greater than the minimum limit value.	
	Shows the percentage of speed on the belts and gear. One arrow is shown for low gear and two arrows are shown for high gear with respective percentage shown underneath.		Indicates the direction and the distance from the drill bit current position and the selected hole.

	Angle Indication tool Panel		Smart fold condition tool panel
	Smart unfold condition tool panel		

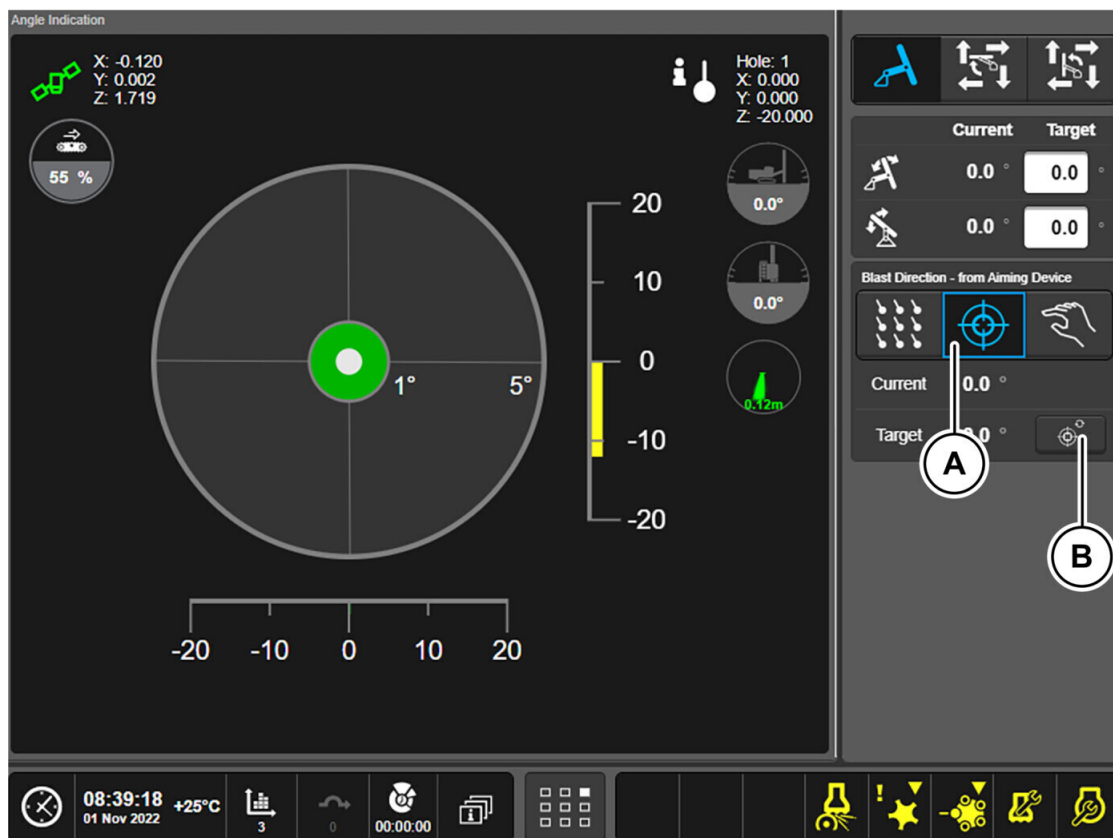
4.3.1 Angle Indication Tool Panel



Example of an **Angle indication tool panel**

A	Shows the actual forward and backward inclination of the feed. When the feed is positioned perfectly, the input value and the actual value are the same.	B	Shows the actual left and right inclination of the feed. When the feed is positioned perfectly, the input value and the actual value are the same.
C	Desired inclination forward or backward. Set the input value for the inclination angle in relation to the reference point (aiming device).	D	Desired inclination left or right. Set the input value for the inclination angle in relation to the reference point (aiming device).
E	Blast direction from drill plan	F	Blast direction from aiming device (GPS compass or HNS System).
G	Manual blast direction		

Set Heading with GPS Compass



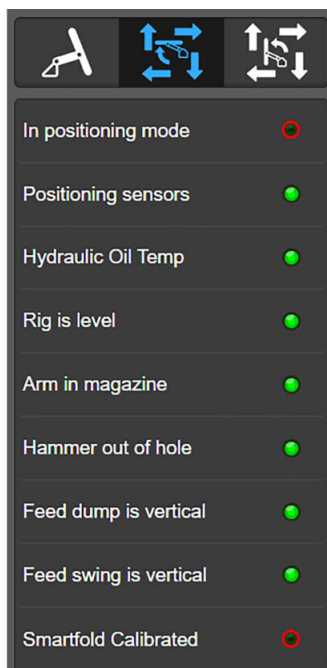
Precondition ✓ The GPS compass is activated.

1. Open the **Angle Indication** menu.
2. Turn the aiming device to the direction needed.
3. Tap on button (A) to view the set compass direction.
4. Tap on button (B) to lock the compass direction value.



NOTE: The aiming direction remains the same until the operator selects a new direction.

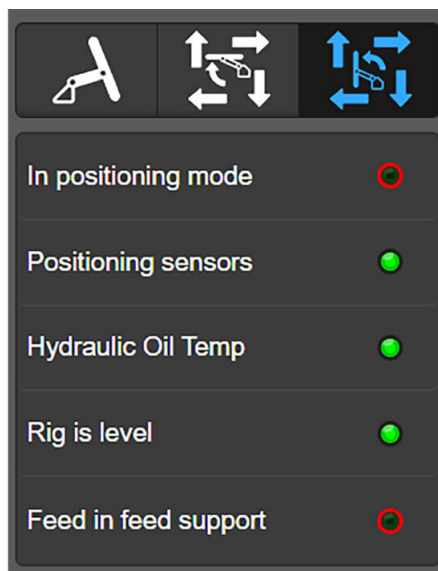
4.3.2 Smart Fold Condition Tool Panel



Example of a **Smart fold** condition tool panel

The **Smart fold** condition tool panel shows which conditions are met to start the automatic feed fold function (moving the feed to tramming position).

4.3.3 Smart Unfold Condition Tool Panel

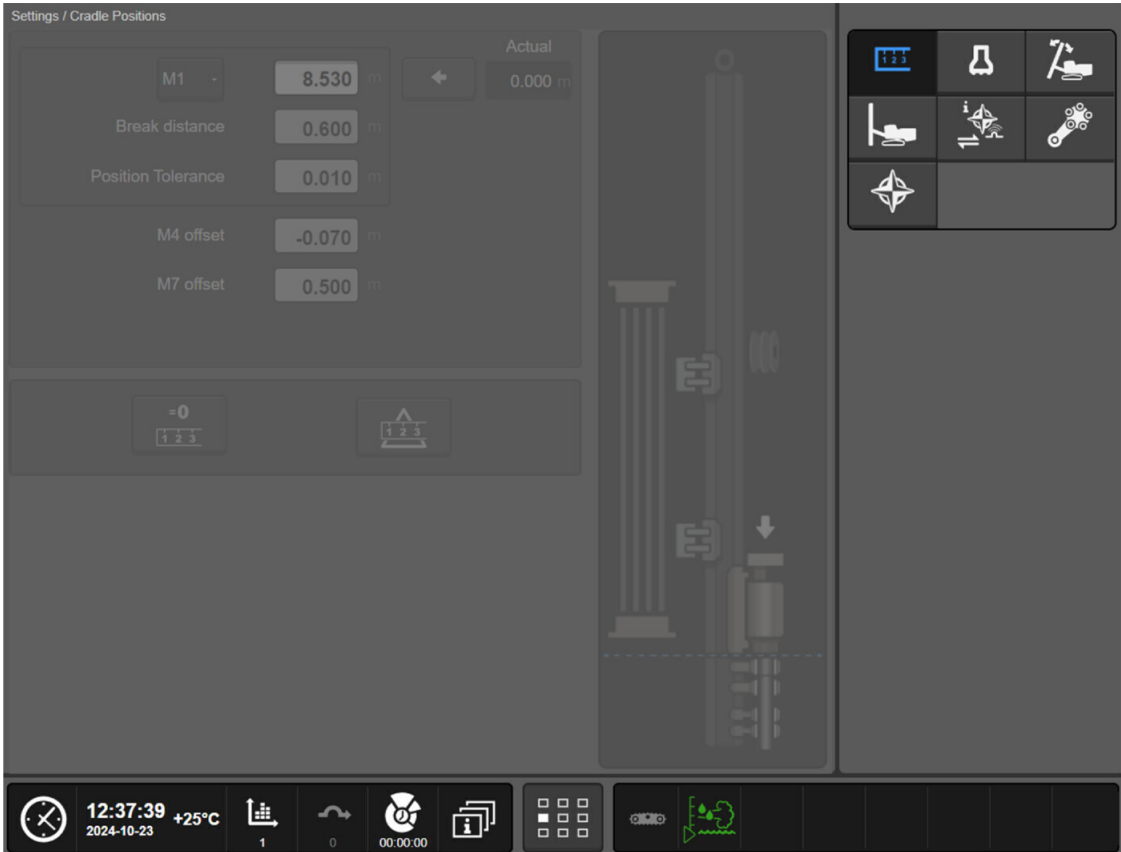


Example of an **Smart unfold** condition tool panel

The **Smart unfold** condition tool panel shows which conditions are met to start the automatic feed unfold function (prepared for drilling).

4.4 Settings Menu

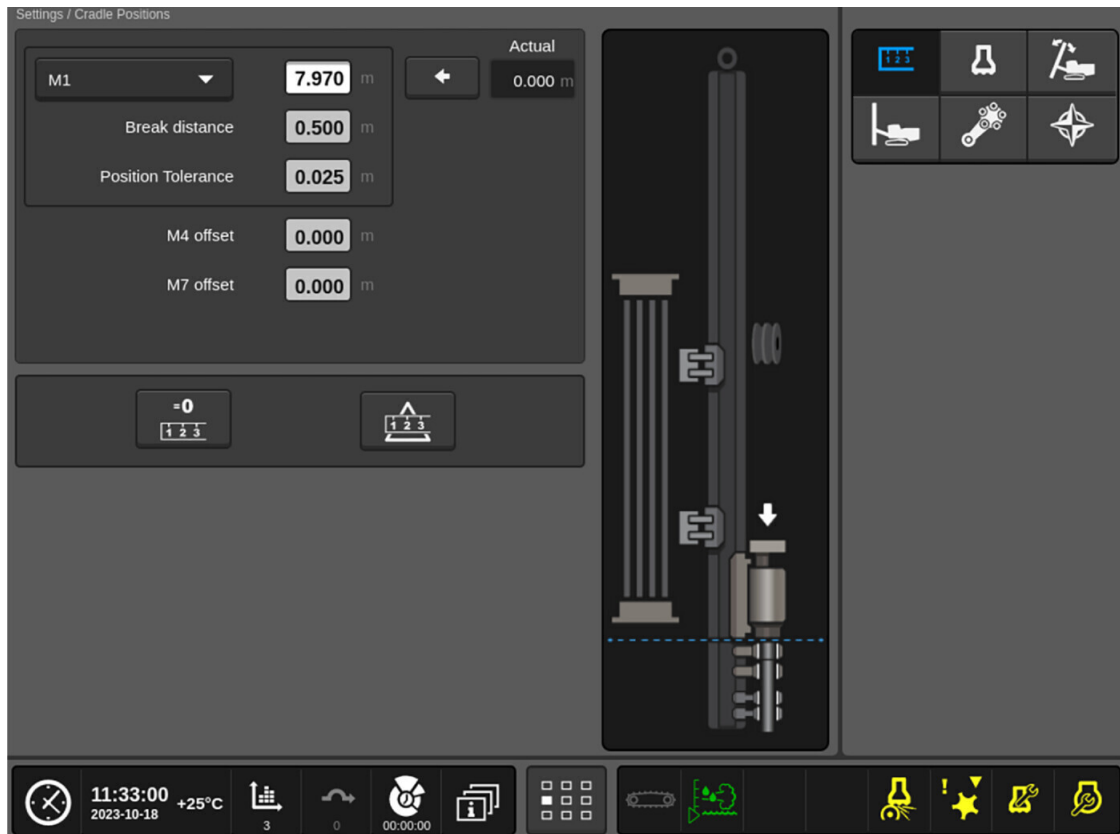
In the **Settings** menu tool panel, there are several submenus. Via the submenus, the different settings for each function are reached. The accessible submenus depend on rig type, set of options on the rig and user access level of the person that is operating the system (OP, SE, AC).



Example of a Settings Menu

	Cradle Positions menu.		Drill Settings menu.
	Positioning settings menu.		Rig settings menu.
	Road model menu		Rod Handling System (RHS) settings menu.
	Navigation menu.		

4.4.1 Cradle Positions Menu



Example of a **Cradle Positions Menu**

M1: Position to stop drilling, just before hitting the mechanical stop.

M2: Position for breaking the adapter-rod joint in the breakout table.

M3: Position for rapid feed stop forward.

M4: Position for rapid feed stop reverse, stop at the correct position to break the rod-rod joint. Used when extraction mode is activated and during One Hole Automatics (OHA).

M5: Position when the rod can be transferred to the magazine. Used when extraction mode is activated and during One Hole Automatics (OHA).

M6: Position for rapid feed stop backward.

M7: Position for rapid feed stop backward on rod number 1 when upper drill support is closed. Done to avoid the hammer hitting the upper drill support.

M8: Position where the cradle stops after the last drill rod is extracted to avoid collision of the bit with the lower drill support.

Break distance: Shows brake distance for rapid feed stop.

Position tolerance: If the cradle position is closer than the tolerance distance to M2, M4, M5, or M6 the cradle cannot be moved backward.

M4 offset: Position where the cradle stops after the last drill rod is extracted.

M7 offset: Position where the upper drill support opens during extraction of rod number 1.

Current cradle position: Shows the current cradle position in relation to reset point. Tap on the button  to make the current value take on the **Actual** value setting.

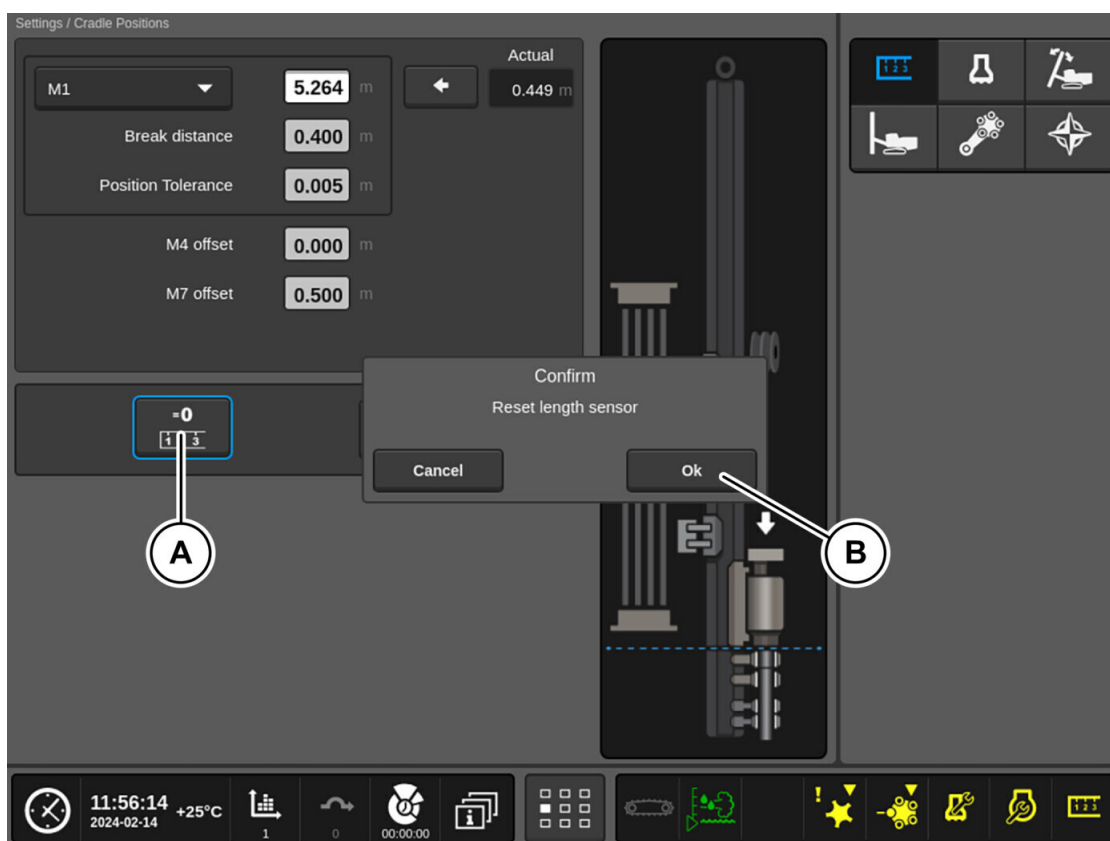
=0
: This button is used to reset the length sensor. Reset the length sensor if the current position is not 0.0 in relation to the rock drill at the rear mechanical stop end position.

: This button can be used if the length sensor has stopped working. None of the pre-programmed stops work but the rig can be used until the length sensor is working. Use rapid feed with great care as none of the stops are working.

Calibrate the Length Sensor

The length sensor needs calibrating in the following cases:

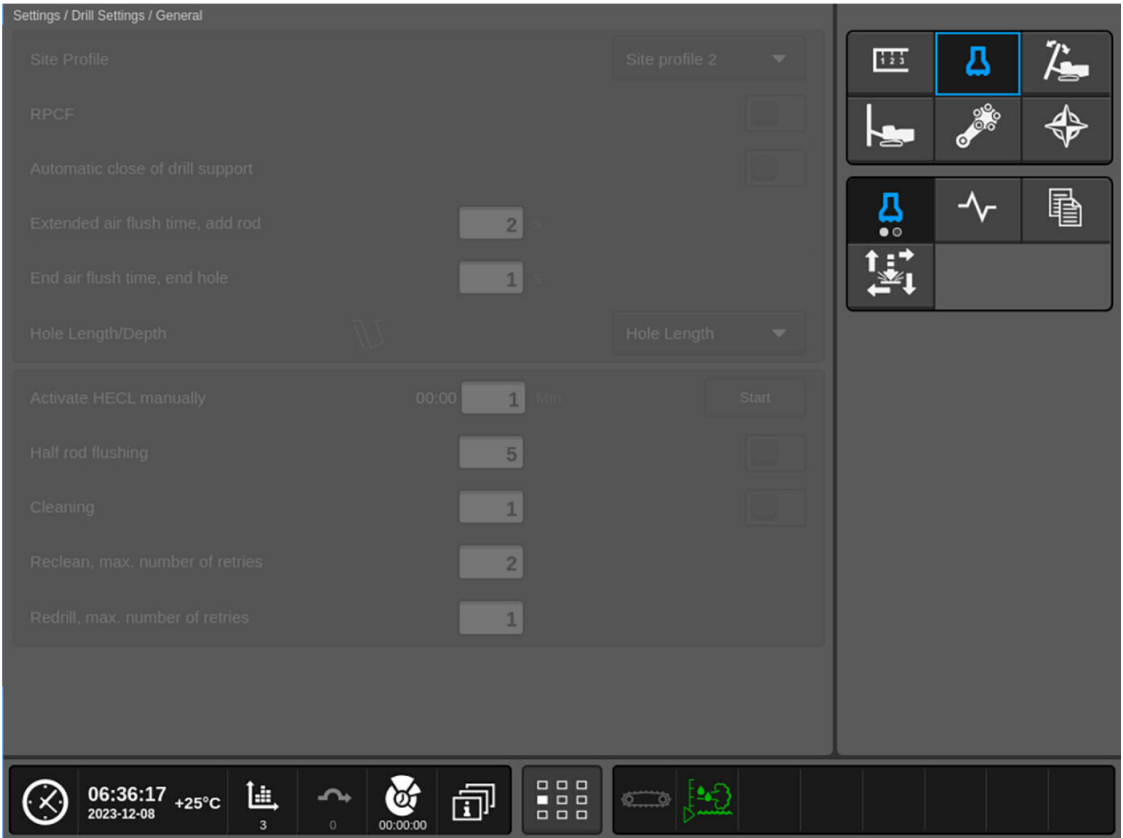
- If M-stop positions of the cradle are not correct.
- When the warning symbol  is shown.
- After any maintenance of the chain feed.



1. Change user to service (SE) in the **User** menu.
2. Open **Settings > Cradle Positions** menu.
3. Reverse the rock drill to the mechanical stop with drill feed or unthreading.
4. Tap on button (A) and then tap on **OK** (B) to reset the length sensor.

4.4.2 Drill Settings Menu

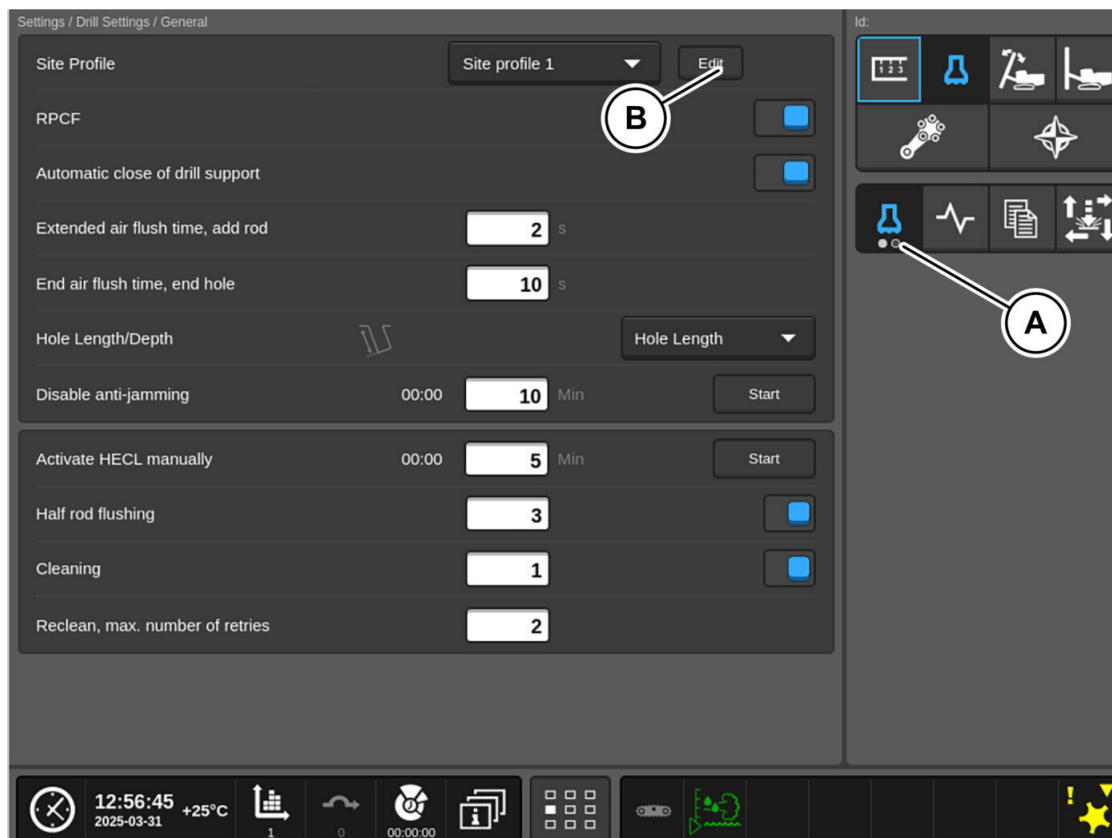
The **Drilling** settings menu tool panel has several submenus. Depending on which submenu is selected different parameters and functions can be adjusted and activated.



Example of a Drill Settings menu

	Select this button to select functions and to set parameter values for general drilling settings.		Select this button to view actuations of the rig components for the selected menu.
	Select this button to access the drilling Parameters menus (only valid for SE level or higher).		Select this button to set functions and to set parameter values for automatic collar-ing.

General Drill Settings Menu



Example of a General Drill Settings Menu



NOTE: The menu image shows some of the described parameters. Select button (A) to see the other parameters described here.

Site Profile: List of preset site profiles.

Use the **Edit** button (B) to edit the name of the site profile. Tap on the button and a pop up window with keyboard is shown, enter the new name for the site profile and select **Ok**.

RPCF (Rotation Pressure Controlled Feed): Activate this function to lower the feed pressure when the rotation pressure increases over an adjustable parameter. Normal setting is active

Automatic close of drill support: The drill rod support closes automatically when drilling starts.

Extended air flush time, add rod: Factor in seconds that the operator can change. The setting affects the flushing time before adding new rods. If the setting is 3 seconds, the flushing time in the end of the first rod is 3 seconds. The flushing time in the end of the second rod is 6 seconds. The flushing time in the end of the third rod is 9 seconds. The rod number multiplies the factor.

End air flush time, end hole: When the required hole depth is reached, the flushing air remains on for the time set in this parameter.

Hole Length/Depth: Use this function to select a method of measurement **Hole Depth** or **Hole Length**.

Disable anti-jamming: Disable the anti-jamming function for the set amount of time.

Activate HECL manually: Use this function to manually activate the Hammer Electric Controlled Lubrication (HECL) pump. The pump remains active during an adjustable period of 1-90 minutes. For the pump to work, the engine must be on.

Half Rod Flushing: Activate this function to enable half rod flushing. The process starts at drill rod number 1 and is activated for every rod until the drill rod number that is set in the menu (only automatic drilling).

Cleaning: Activate this function to enable cleaning. The process starts at drill rod number 1 and is activated for every rod until the drill rod number that is set in the menu (only automatic drilling).

Reclean, max. number of retries: The max number of clean cycles done if the rotation pressure is above the adjustable **Reclean/redrill, rotation pressure limit**.

Redrill, max. number of retries The maximum number of redrills that are done if the rotation pressure is above the the adjustable **Reclean/redrill, rotation pressure limit**.

Reclean/redrill, rotation pressure limit: While cleaning, if the rotation pressure goes above the limit set, a reclean/redrill is done.

Start screw sampler: Activate the screw sampler option.

Reverse feed distance after coupling: Use this function to set the distance that the rotation unit backs up before starting to drill.

Rotation delay time, auto collaring: Feed and percussion start earlier than rotation in auto collaring. Time depends on the setting in the box.

Automatic collaring: Select to use the function **Automatic Collaring**. **Automatic Collaring** is a function for drilling the first part of a hole. The basic function is to drill and back up the first rod repeatedly to build up the hole collar.

Automatic rc Vibrator: The reverse circulation (RC) vibrator starts when the lower knife valve opens.

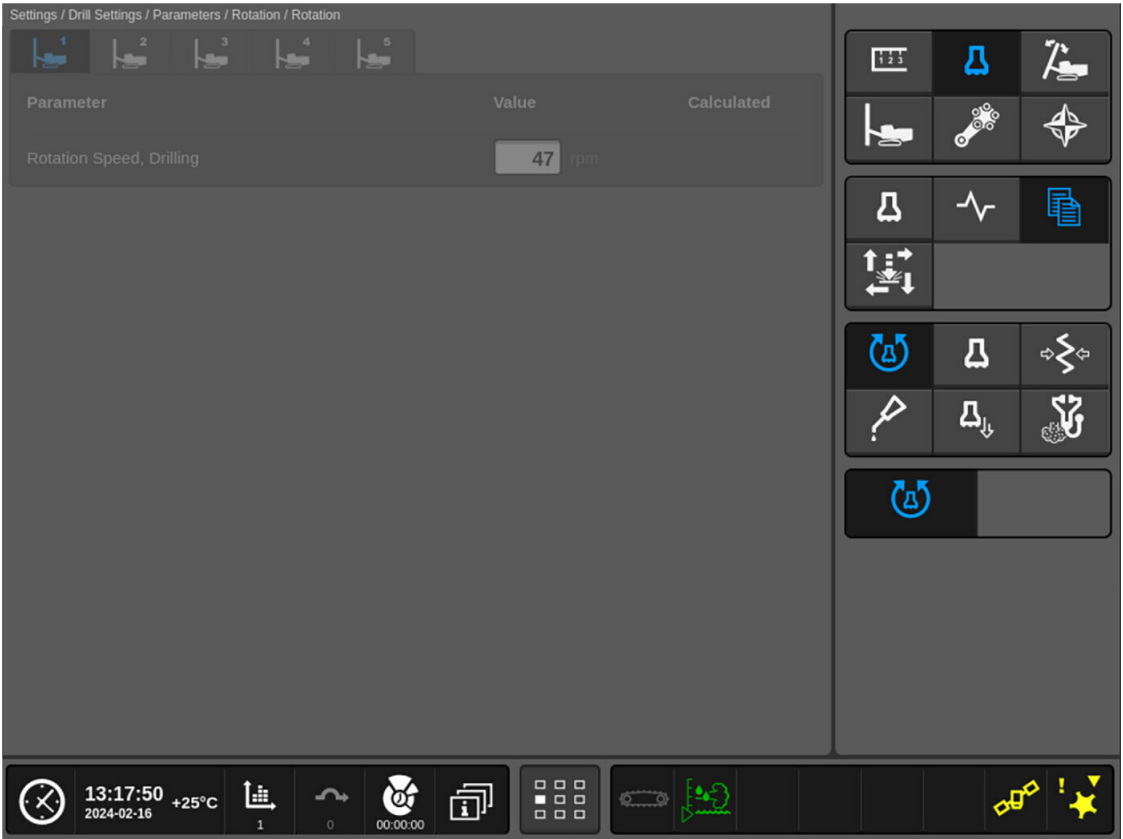
Emergency mode RC: Select to activate RC functions for all drill and position settings.

Disable anti-jamming: Disable the anti-jamming function for the set amount of time.

Full air flow during collaring: Select this function to activate full air flow from the start of drilling.

Rod weight compensation: Compensate feed pressure when drilling with more than one rod. Feed drilling pressure decreases for each rod that is added in the hole.

Parameters Menu

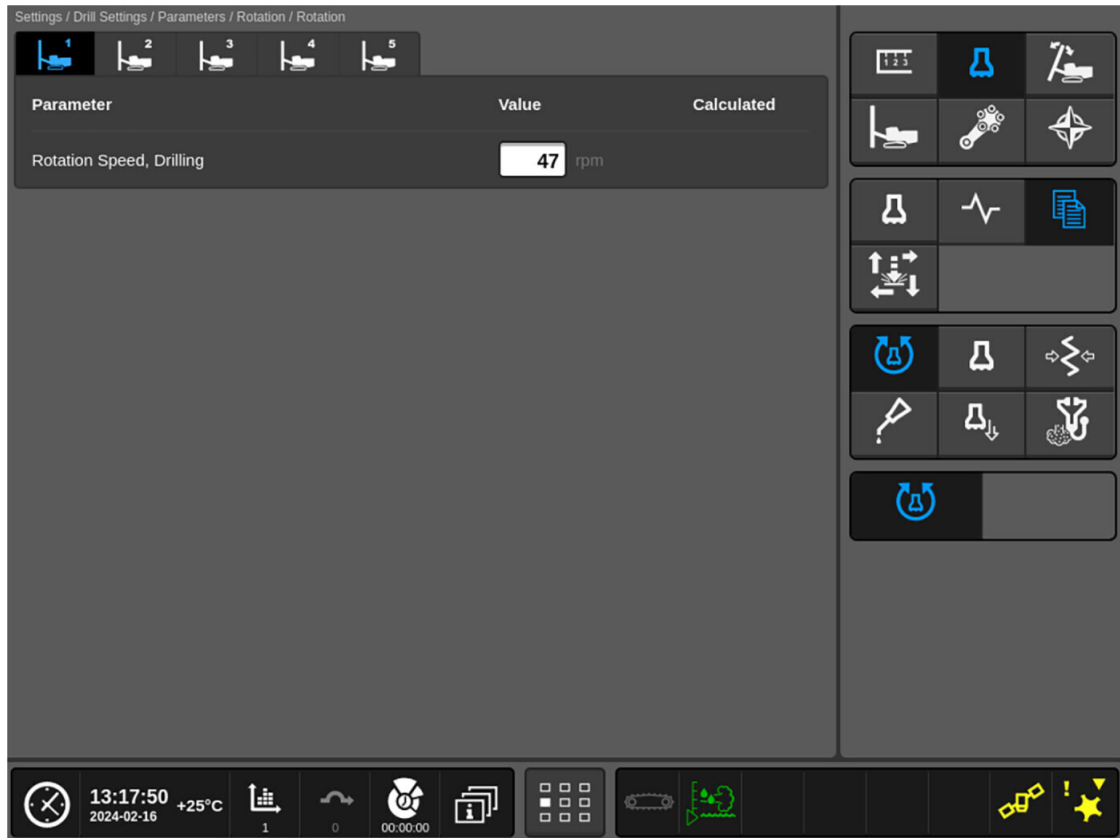


Example of a **Parameters** menu for drill settings.

	Select to open the Rotation Parameters menu		Select to open the Site Profile menu.
	Select to open the Feed Pressure menu.		Select to open the Lubrication menu.
	Select to open the Penetration Rate menu.		Select to open the Screw sampler menu.
	Select to open the DCT menu.		

Rotation Parameters Menu

This menu can only be accessed at SE level or higher.



Example of a Rotation parameter menu

Rotation Speed, Drilling: Drill bit RPM

Rotation Pressure, Drilling: Pressure when **RPCF** starts to reduce the feed pressure.

Pressure Increase, Jamming: Pressure increase when jamming is detected and when the feed direction switches to reverse during drilling.

Pressure Decrease, Free: Pressure decrease from when jamming is detected and when the feed direction switches to forward during drilling

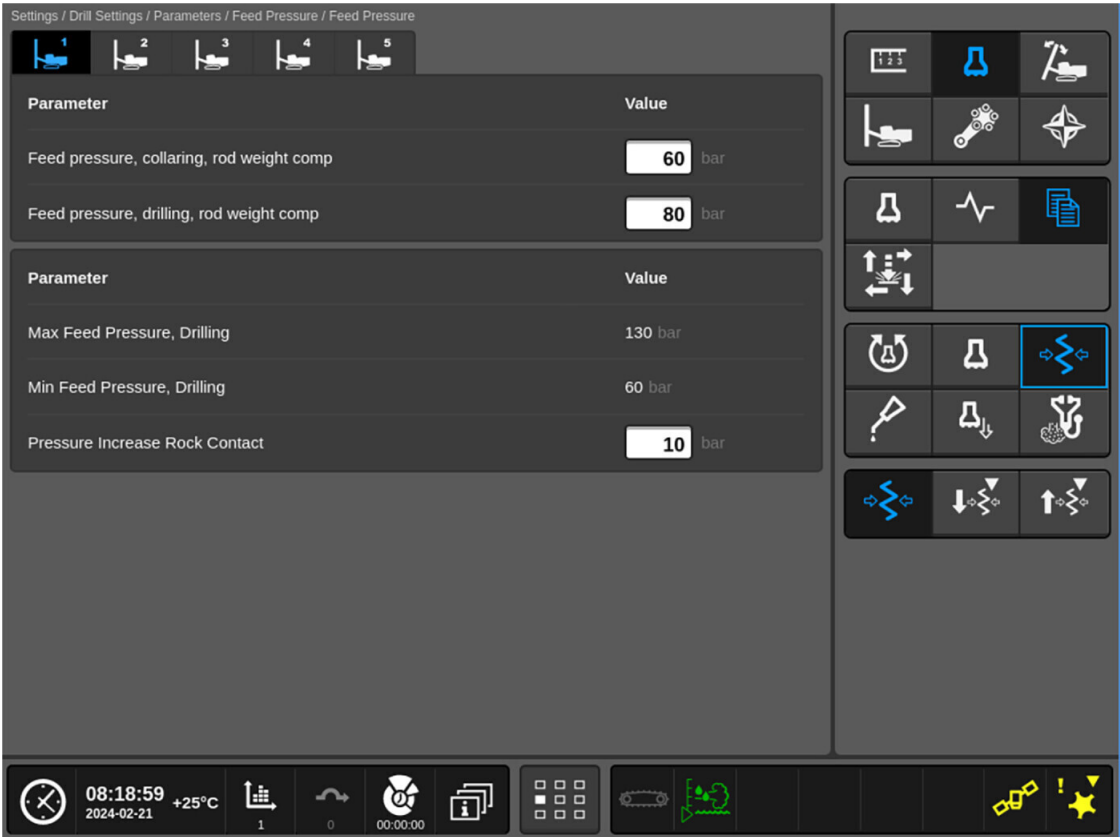
Site Profile Parameters Menu



Example of a Site Profile Parameter Menu

In the **Site Profile** parameter menu, settings for one site profile can be copied to another site profile.

Feed Pressure Parameters Menu



Example of a Feed pressure Parameters Menu.

	Feed pressure parameters menu.		Calibration forward feed pressure menu.
	Calibration backward feed pressure menu.		

- Feed pressure, Collaring:** The pressure is best adjusted during collaring. Lock collaring with the slide button at the bottom right of the drilling lever. The feed pressure for collaring must be between 30-60 bar depending on the type of feed and the drill bit diameter.
- Feed Pressure, Drilling:** Switch to drilling with the slide button at the top right of the drilling lever. The feed pressure for drilling must be between 60-120 bar depending on the type of feed and the drill bit diameter.
- Max Feed Pressure, Drilling:** The value that is shown here is the maximum value that is permitted for feed pressure drilling. The feed pressure cannot be set higher than this value.
- Min Feed Pressure, Drilling:** The value that is shown here is the minimum value that is permitted for feed pressure drilling. This value is the lowest feed pressure that Rotation Pressure Control Feed (RPCF) is allowed to regulate down to.
- Pressure Increase Rock Contact:** This pressure indicates when the drill bit has good contact with the ground or rock when collaring in automatic drilling.

Steps for Calibration of Forward Feed Pressure

Settings / Drill Settings / Parameters / Feed Pressure / Calibration forward

Parameter	Value	Parameter	Value
50 mA	29 bar	0 mA	1.3 bar
100 mA	37 bar		
150 mA	42 bar		
200 mA	56 bar		
250 mA	68 bar		
300 mA	79 bar		
350 mA	94 bar		
400 mA	108 bar		
450 mA	125 bar		
500 mA	139 bar		
550 mA	152 bar		
600 mA	166 bar		

Calibration

Precondition

- ✓ The diesel engine is on.
- ✓ The rig is in drilling mode.
- ✓ The hydraulic oil is at working temperature (more than 20 °C).

1. Open the **Setting > Drill Settings > Parameters > Feed Pressure > Calibration forward** menu.
2. Position the cradle against the front mechanical stop on the feed.
3. Tap **Calibration** and wait until the pop-up message **Calibration completed** is shown.
4. Tap **Close** to close the pop-up message.

Steps for Calibration of Backward Feed Pressure

Settings / Drill Settings / Parameters / Feed Pressure / Calibration backward

Parameter	Value	Parameter	Value
50 mA	34 bar	0 mA	1.3 bar
100 mA	47 bar		
150 mA	60 bar		
200 mA	77 bar		
250 mA	100 bar		
300 mA	122 bar		
350 mA	142 bar		
400 mA	167 bar		
450 mA	190 bar		
500 mA	213 bar		
550 mA	232 bar		
600 mA	249 bar		

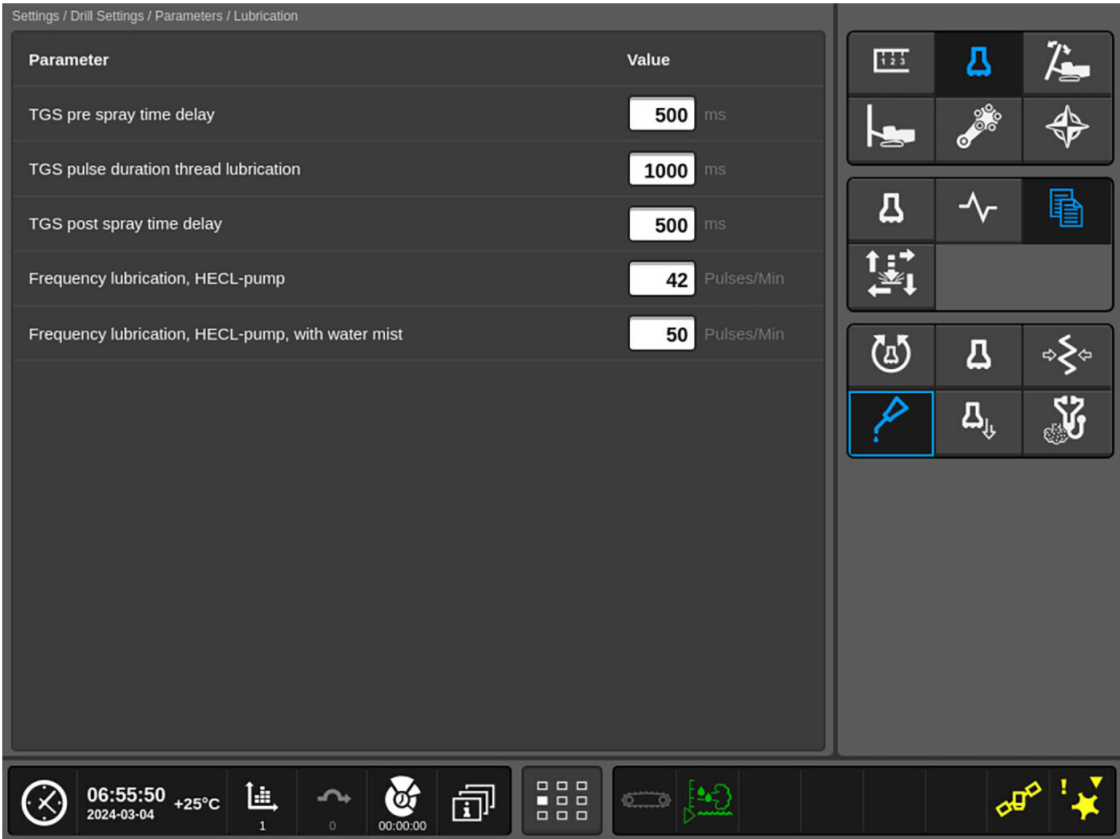
Calibration

Precondition

- ✓ The diesel engine is on.
- ✓ The rig is in drilling mode.
- ✓ The hydraulic oil is at working temperature (more than 20 °C).

1. Open the **Setting > Drill Settings > Parameters > Feed Pressure > Calibration backward** menu.
2. Position the cradle against the rear mechanical stop on the feed.
3. Tap **Calibration** and wait until the pop-up message **Calibration completed** is shown.
4. Tap **Close** to close the pop-up message.

Lubrication Parameter Menu



Example of a Lubrication parameter menu

TGS pre spray time delay: The spray time of air before the grease spray of threads.

TGS pulse duration thread lubrication: The spray time (ms) of grease during thread lubrication.

TGS post spray time delay: The spray time of air after the grease spray of threads.

Frequency lubrication, HECL-pump: The frequency of oil release in pulses/ minute.

Frequency lubrication, HECL-pump, with water mist: The frequency of oil release in pulses/ minute, when using water mist during drilling.

Penetration Rate Menu

Settings / Drill Settings / Parameters / Penetration Rate

1

2

3

4

5

Function	Value
Retract drill bit at low penetration	☐
Minimum allowed penetration rate, drilling	0.001 m/min
Minimum allowed penetration rate, collaring	0.001 m/min

Function	Value
Delay, low penetration rate detection	3.0 s
Back distance at low penetration, drilling	0.050 m
Back distance at low penetration, collaring	0.050 m

Function	Value
Strategy for handling cavities	No action
Maximum allowed penetration rate	2.00 m/min

1 2 3

06:00:20
2024-02-19 +25°C

1

1

00:00:00

Example of a Penetration Rate Menu

Retract drill bit at low penetration: If the penetration rate is too low during the time set, the cradle is retracted.

Minimum allowed penetration rate, drilling: Minimum allowed speed at which the bit advances while drilling.

Minimum allowed penetration rate, collaring: Minimum allowed speed at which the bit advances during collaring.

Delay, low penetration rate detection: After the set delay time the rig stops, reverses the feed and then continues drilling.

Back distance at low penetration, drilling: Distance to back up the cradle at low penetration during drilling.

Back distance at low penetration, collaring: Distance to back up the cradle at low penetration during collaring.

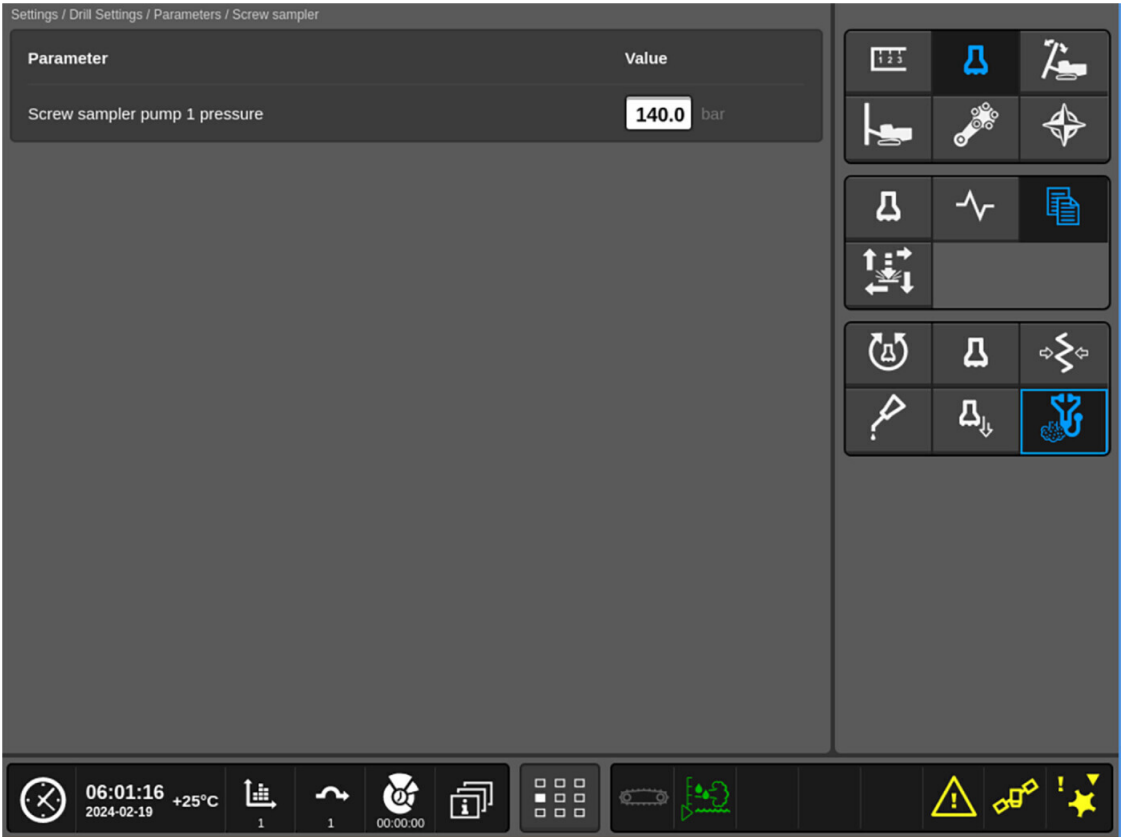
Strategy for handling cavities: Select strategy for handling cavities, select **No action** or **Stop feeding forward**.

Maximum allowed penetration rate: Maximum allowed speed at which the bit advances during the **Cavity detection delay** time.

Cavity detection delay: Set the time from when the speed set in **Maximum allowed penetration rate** is detected until action is taken by the RCS.

Max anti jam retries - low penetration rate: Maximum amount of anti-jamming (Low penetration rate) retries allowed before Auto Drill Plan Execution (ADPE) is stopped.

Screw Sampler Parameters

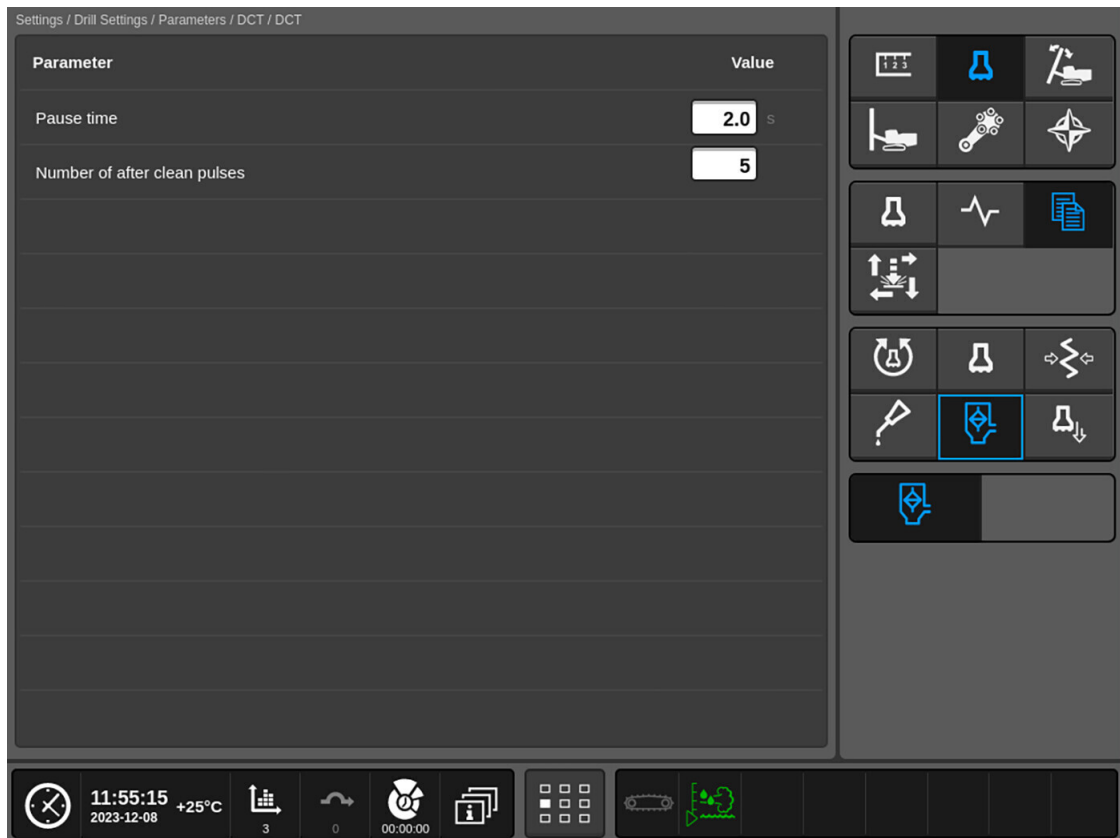


Example of a Screw sampler Menu

Screw sampler pump 1 pressure: The pressure demanded for pump 1 to activate the screw sampler function.

NOTE: The **Screw sampler cleaning time** and the **Screw sampler loading time** are set in the **Settings > Rig** menu.

DCT Parameters Menu



Example of a DCT Parameter Menu

Pause Time: Adjustable parameter for the interval between the clean pulses.

Number of after clean pulses: Number of clean pulses after drilling stop. The number of clean pulses can be set depending on the condition of the rock, that is, moisture content, weight, and so on.

Automatic Collaring Menu

Settings / Drill Settings / Automatic collaring / Automatic collaring

Parameter	Value
Collar length	3.00 m
Collar final cleaning sequence	☒
Collar complete auto detect	☒
Time limit rock detection after overburden	5.0 s
Collar min length	0.50 m
Auto airflush cut off	☒
Manual air flush cut off depth	1.00 m
Auto collaring steps	☒

Example of an Automatic collaring Menu

Collar length: Maximum length of automatic collaring counted from the rock contact point. Maximum length adjustable is the drilled length on rod 1. Automatic collaring does not continue on rod 2

Collar final cleaning sequence: When the hole building is complete, a cleaning sequence is done before drilling is continued.

Collar complete autodetect: Activate to monitor the penetration rate. If the penetration rate goes below the **Penetration rate solid rock** for longer than the time set in **Time limit rock detection** after overburden, it switches to full drilling.

Time limit rock detection after overburden: How long time the penetration rate must be below the set limit to identify the rock as solid. The penetration rate settings are found in each step in the auto collaring settings

Collar min length: The drill will auto collar to this depth regardless of penetration rate.

Auto airflush cut off: Use this parameter to avoid muck blowing back into the drill hole and destroying the hole collar when removing the drill bit from the drill hole. When this parameter is selected and removing the last rod or cleaning, the air flush stops automatically at the collar length that was detected in **Collar complete auto detect**.

Manual air flush cut off depth: If **Auto air flush cut off** is not selected, the air flush stops at this set distance when removing the last rod or cleaning hole is activated.

Air evacuation - Air burst prohibit: Activate to release the air pressure in the drill string before unthreading when drilling in auto.

Air evacuation time limit: Set value for time limit of air evacuation.

Air evacuation pressure limit: Set pressure limit for air evacuation.

Auto collaring steps: Select to show the **Collar step** menu.

Automatic Collar Step Menu

Parameter	Value
Forward distance	0.50 m
Number of cycles per collar step	2 1 m
Increase of length for each cycle	0.50 m
Move backward to M4+offset	☐
Backward distance (if not M4+offset is selected)	0.50 m
Air forward on at rock contact	☐
Air at forward feed	30 %
Air at backward feed	0 %
Rotation on at backward feed	☒
Pause after backward feed	0.0 s
Penetration rate, solid rock	1.0 m/min
Rotation delay time, auto collaring	0.0 s

Example of a an Automatic Collaring > Collar Step Menu

Use the **Edit** button (A) to edit the name of the overburden. Tap on the button and a pop up window with keyboard is shown, enter the new name for the overburden and select **Ok**.

Step 1, Step 2, and Step 3: The automatic collaring is done in three steps. Each of the three collar steps has its own set of parameters. When the automatic collaring sequence is started, the rig begins by using the values in collar step 1, progressing on to collar steps 2 and 3.

Forward distance: Each **Collar step** starts by moving the drill bit forward this length.

Forward distance is measured from the rock contact point (collar step 1) or from the point where the previous step ends (collar step 2 or 3).

Number of cycles per collar step: Use this function to set the number of times that forward and backward movement is done for each collar step. The number to the right is the end length (from rock contact point) for the selected collar step.

If collar **Step 3** is finished before the **Collar length** is fully drilled, the last part of the collar is drilled with collar settings excluding forward and backward movement.

Increase of length for each cycle: A distance that the **Forward distance** is increased with at each repetition. This parameter can be set to 0.

Move backward to M4 + offset: If **Move backward to M4 + offset** is selected, the cradle reverses to position M4 + M4 offset between each forward cycle. Position M4 is above ground level. Position **M4 offset** is set in the menu **Settings > Cradle Positions**.

Backward distance (if not M4 + offset is selected): This parameter is used if **Move backwards to M4 + offset** is not selected. When the forward movement is ready, backward movement is started for this distance.

Air forward on at rock contact: If this selection is active the air is off when moving forward until the measured feed pressure exceeds the calibrated idle feed pressure with the value **Air on rock contact detect level**. Then air flow is started according to **Air at forward feed**.

Air flow is always started when percussion is on (controlled by damper pressure) independent of if rock contact on feed pressure is detected or not.

The air remains on for **Air on time after rock contact** when the rock contact is lost. The air is always turned on when the bit is less than 0.3 m from the drilled bottom of the hole.

Air at forward feed: This parameter is the air flow at forward feed. **Air at forward feed** can be set to minimum 10%. This parameter is to prevent that percussion is on without air.

Air at backward feed: This parameter is the air flow at backward feed. If the parameter is set to 0%, air is off.

Rotation on at backward feed: Rotation is on at backward feed when this selection is active.

Pause after backward feed: It is possible to set a pause time after backward feed where the feed is stopped and air is on according to **Air at backward feed** with no water mist.

Penetration rate, solid rock: Shows the drilling rate for solid rock. If the penetration rate during automatic collaring is below this value, collaring steps finish and the hole continues being drilled at full drilling.

Rotation delay time, auto collaring: Feed and percussion start earlier than rotation in auto collaring. Time depends on the setting in the box.

Water mist on at forward feed: Selection if water mist is on at forward feed.

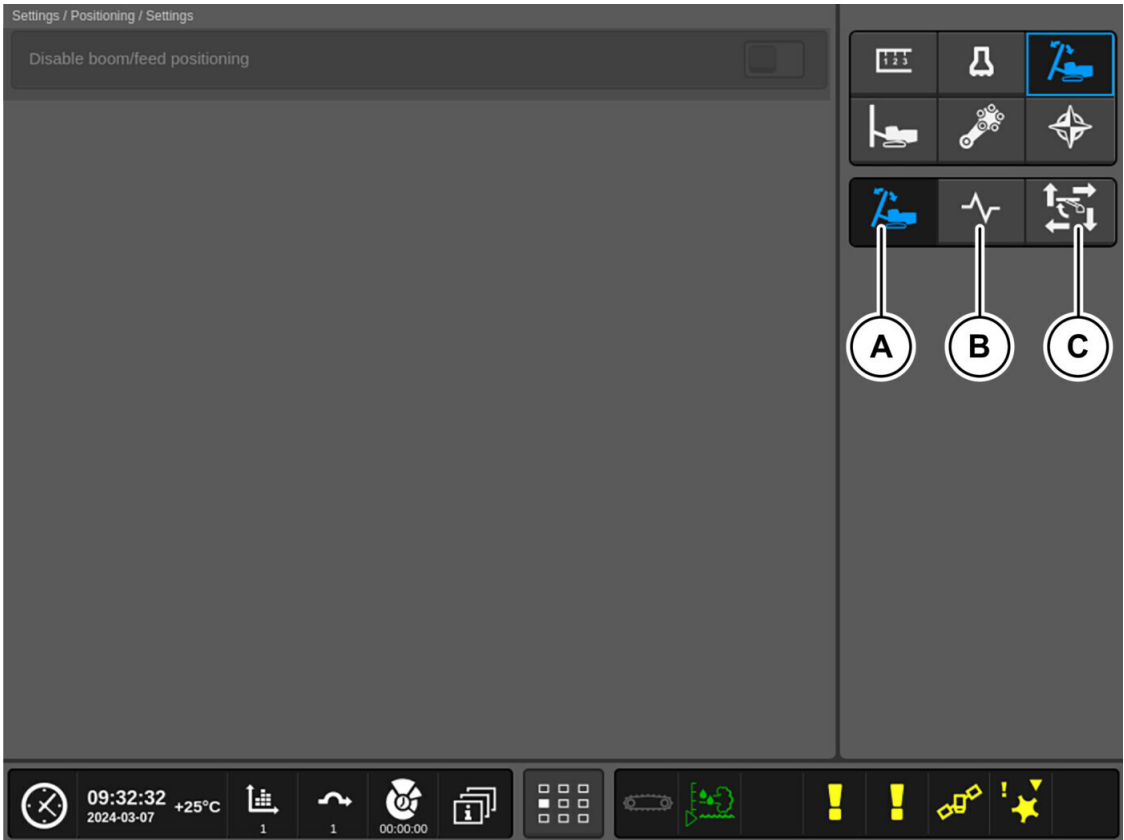
Water mist on at backward feed: Selection if water mist is on at backward feed. This selection is only valid if **Air at backward feed** is selected.

Water mist pump speed: Use to adjust the water flow that is injected in the flushing air. The speed of the water mist pump can be adjusted between 0-100%. This parameter is only valid if water mist is selected.

Rock contact, feed pressure: Specifies the pressure level that the control system interprets as rock contact.

Air on time after rock contact: This function helps avoid sudden short shocks which can lead to blockages in the hole. The air is always switched on when the drill steel is less than 0.3 m from the drill hole base to protect against shocks.

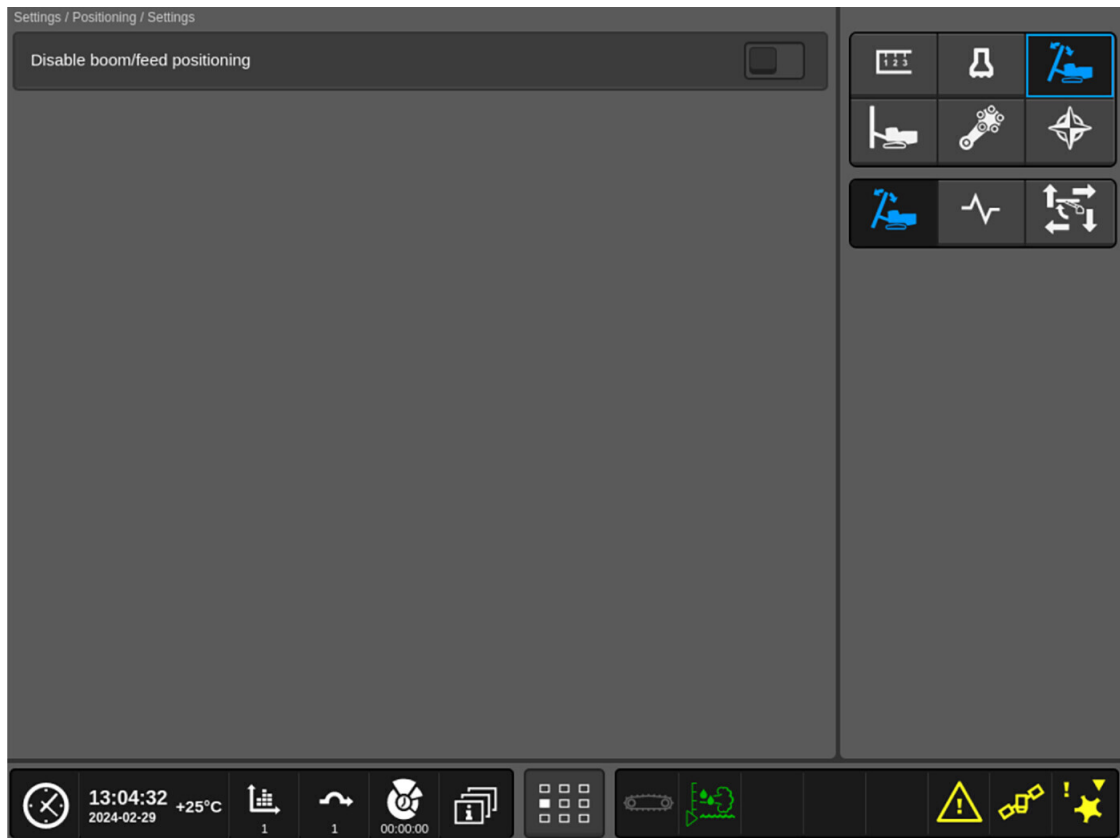
4.4.3 Positioning Menu



Example of a Positioning Menu

A	Select to open the Settings menu for Positioning.	B	Select to open the Actuation menu for Positioning. This menu shows values for: • Boom Swing • Boom lift • Feed extend • Feed dump • Feed swing
C	Select to open Smart fold menu for positioning.		

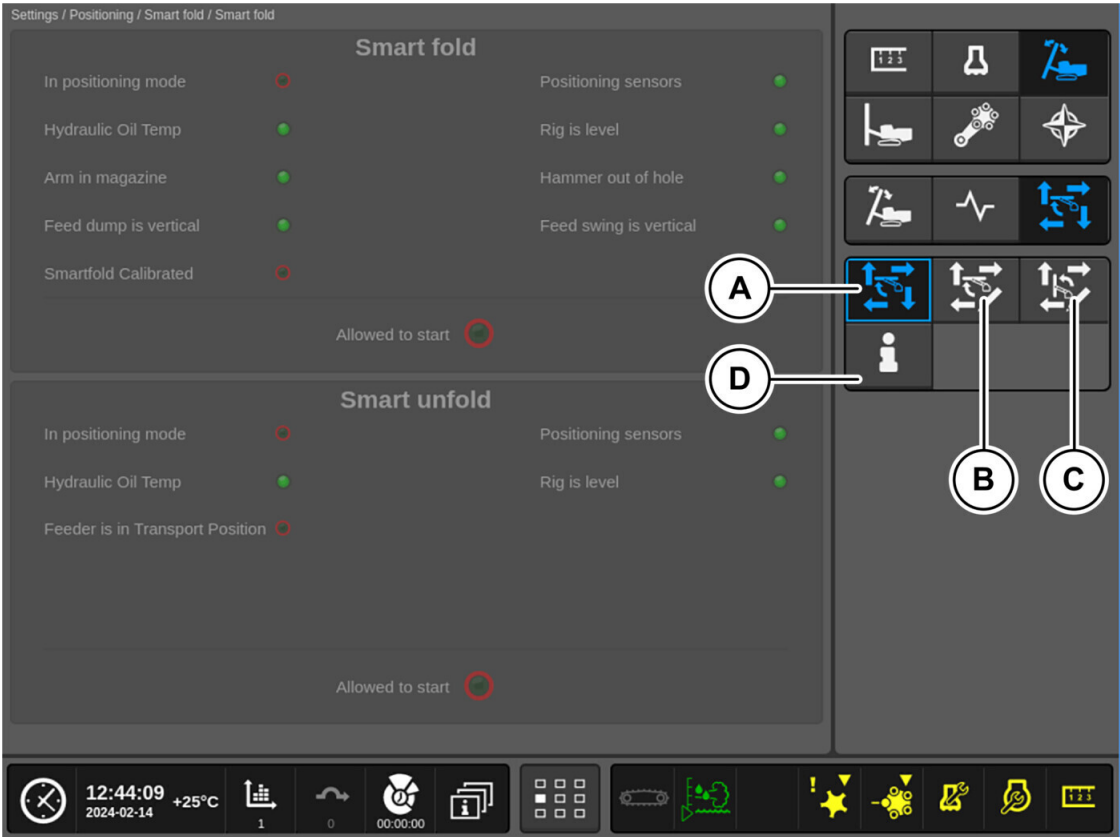
Settings Positioning Menu



Example of a **Settings** Positioning Menu

Disable boom/feed positioning: Possibility to disable the feed and boom positioning. This setting can be useful during machine transport.

Smart Fold Positioning Menu

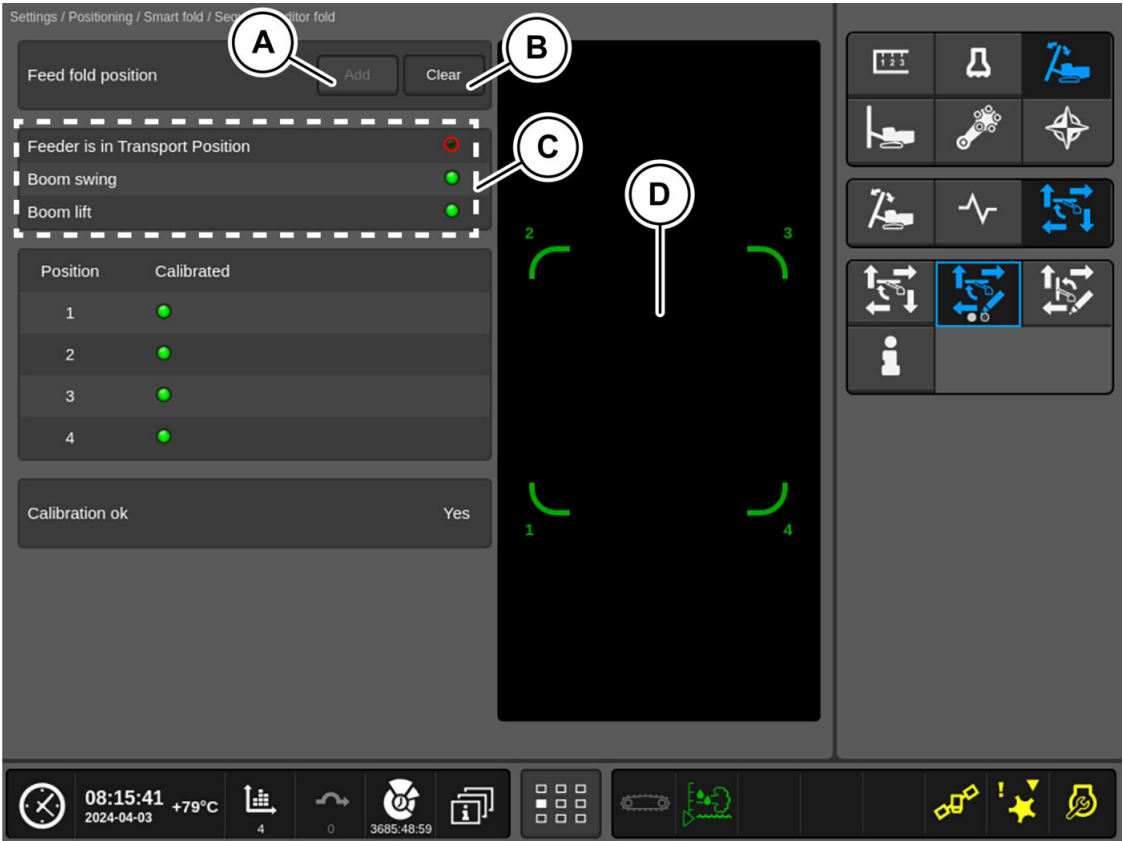


Example of a Smart fold Menu

A	The Smart fold menu shows the conditions for allowing the function to start (moving the feed to tramming position or unfolding).	B	Select to open the Sequence editor fold menu and calibrate the folding sequence (access level SE or higher).
C	Select to open the Sequence editor unfold menu (access level SE or higher).	D	Select to open the Smart fold Information menu

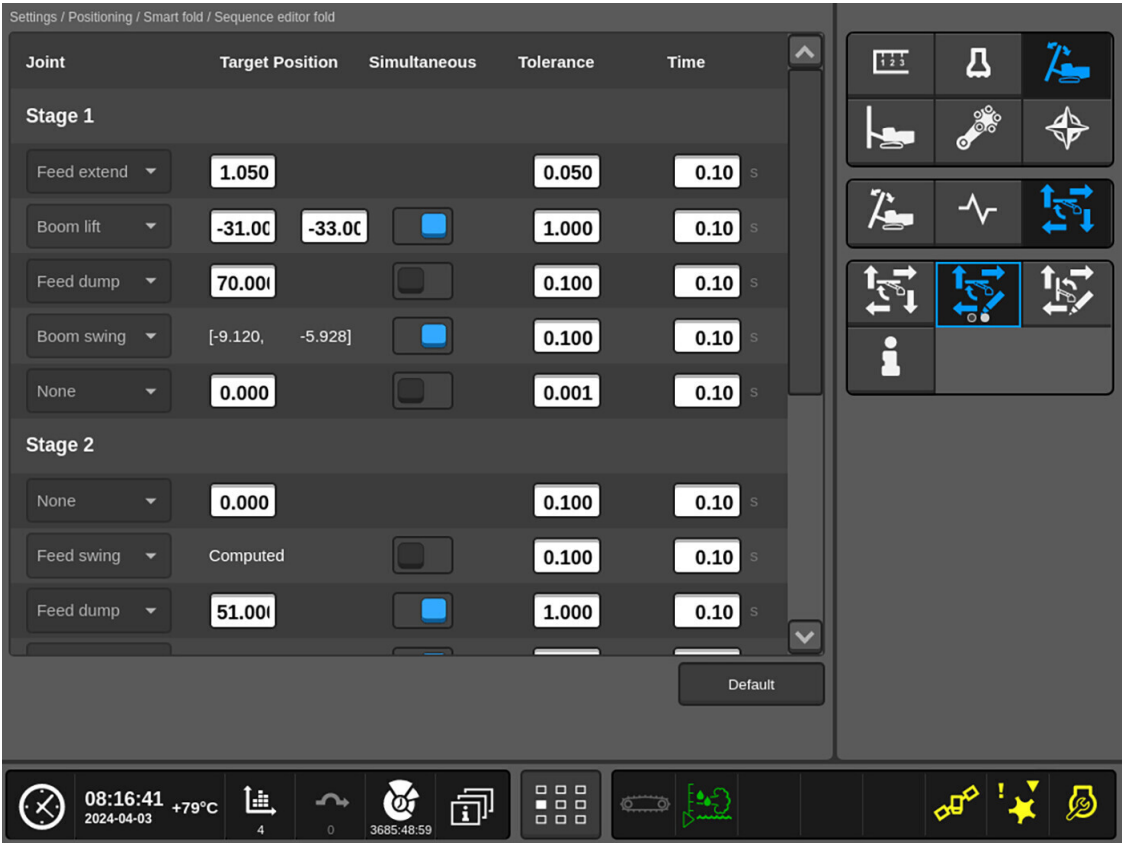
Sequence Editor Fold Menu

In the **Sequence editor fold** menu, the Smart fold is calibrated. Extreme values can be mapped out for boom swing and boom lift with the feed in the feed support. At the same time making sure that a collision between the feed and the rig is avoided.



Example of a **Sequence editor fold** Menu (menu number 1)

A	The Add button is used to add the calibration space.
B	The Clear button is used to clear the setting and start from the beginning.
C	The lamp icons are lit up when the condition is fulfilled. Feeder is in Transport Position means that the inductive sensor in the support is activated since the feed is there. When going between the calibration points, the Boom swing and Boom lift symbols are lit up. The lit symbols indicate that they are moved enough from the last calibration point to avoid all points being taken in the same position.
D	Arc table 1. Boom lift as low as possible (or needed) with boom swing to the left. 2. Boom lift as high as possible with boom swing to the left. 3. Boom lift as high as possible with boom swing to the right. 4. Boom swing as low as possible with boom swing to the right.



Example of a **Sequence editor fold Menu** (menu number 2)

The feed and boom movement are set in the sequence editor. The order is preset and cannot be changed.

Sequence Editor Unfold Menu

Settings / Positioning / Smart fold / Sequence editor unfold

Parameter	V-value	Target Position
Boom swing	-180	0
Boom lift	51.029	0
Feed swing ang	-0.327	0
Feed dump ang	102.546	0
Feed extension	0.678	0

Example of a Sequence editor unfold Menu (menu number 1)

Settings / Positioning / Smart fold / Sequence editor unfold

Joint	Target Position	Simultaneous	Tolerance	Time
Feed dump ▾	30.00	☐	6.800	0.10 s
Feed extend ▾	1.100	☐	0.010	0.10 s
Feed dump ▾	45.00	☒	6.500	0.10 s
Boom lift ▾	-19.00	☒	2.500	0.10 s
Boom swing ▾	0.000	☐	0.200	0.50 s
Boom lift ▾	-28.00	☒	1.000	0.10 s
Feed swing ▾	0.000	☒	0.500	0.30 s
Feed dump ▾	102.00	☒	2.000	0.30 s
None ▾	0.000	☐	0.001	0.10 s

Default

Example of a Sequence editor unfold Menu (menu number 2)

Information Menu for Smart Fold

All the calibration angle combinations for positions 1–4 in the **Sequence editor fold** menu are shown in the **Information** menu.

Settings / Positioning / Smart fold / Information

Boom swing	Boom lift	Feed swing	Feed dump
-5.928000	-1.020588	9.465255	2.150590
-6.042000	-3.855556	9.485989	4.116601
-9.120000	-3.855556	11.523420	4.403743
-9.120000	-0.680392	11.544710	2.150590

Calibration ok: Yes

Calibration error: 0.132 °

Last computed feed swing: 0.00 °

Last computed feed dump: 0.00 °

FS = 5.548 - 0.656*BS + 0.000 *BL FD = 1.564 + 0.000 *BS - 0.698*BL

08:20:34
2024-04-03

+79°C

4

0

3685:48:59

*Example of an **Information** Menu for Smart Fold*

Calibration ok	The parameter Calibration ok must show Yes to be able to run, it is the same value as in the calibration view.
Calibration error	Calibration error must be as small as possible. Having a larger difference in angles between the calibration points has a positive effect.
Last computed feed swing Last computed feed dump	Last computed feed swing and Last computed feed dump shows the parts from the last automatic folding step where the selected combination of boom lift and boom swing generates the values shown here to reach the support within the calibrated angle space.

Calibrate Smart Fold

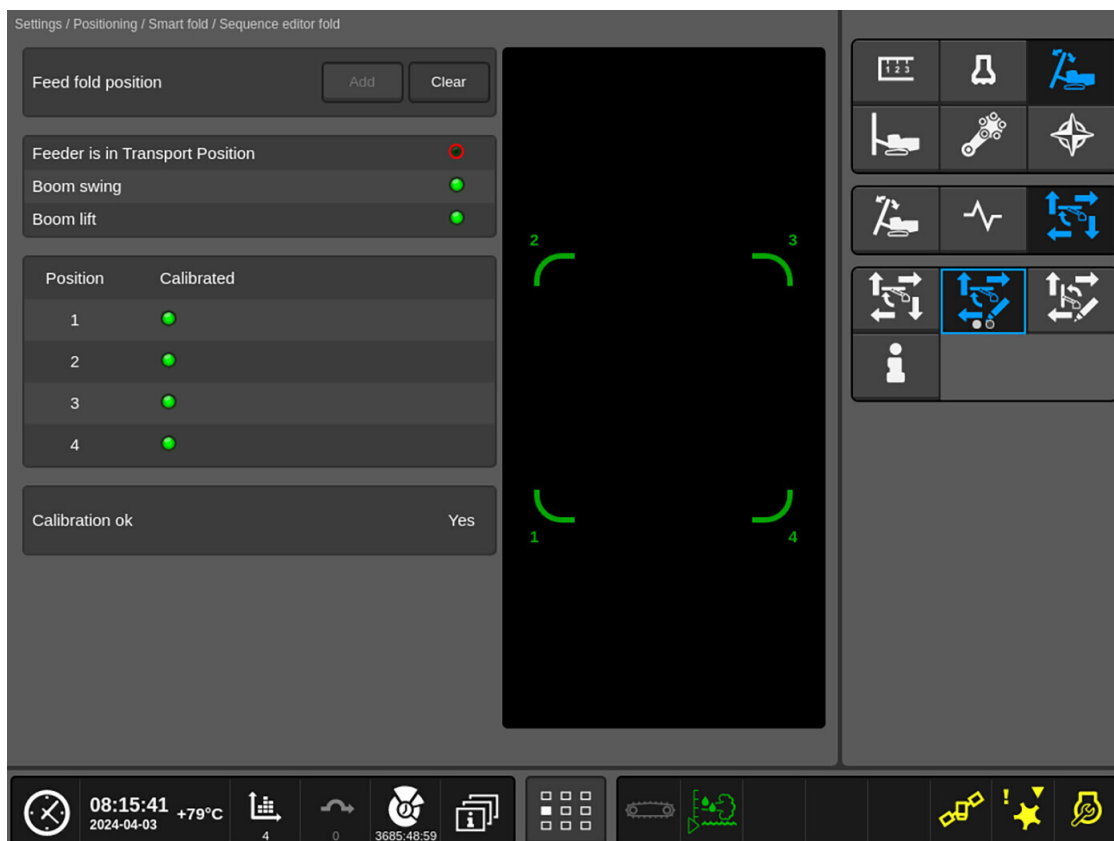
NOTICE

Damage to Rig

Calibrating the **Smart fold** end position to a position above the transport position increases the height of the rig so it exceeds the transport height. If the rig is loaded on a lorry, it can damage the rig.

- Make sure that the rig has the correct transport height before transporting on a lorry.

This menu can only be accessed at SE level or higher.



1. Position 1

- Boom lift low and boom swing to the left, place the feed in the feed support.
- Make sure that there is some space between the feed and the cab.
- Make sure that the feed extension sensor is not too close to the roof of the canopy.
- Press **Add** when you have the position that is needed and all lamps are green.

Step result

➔ The red circle for position 1 turns green.

2. Position 2

- Move boom lift upwards and keep boom swing to the left. The boom lift lamp turns green when ready.
- Place the feed back in the support with help of feed swing and dump.
- Press **Add** and check that the lamp for position 2 turns green.

3. Position 3

- Move boom swing to the right and keep boom lift high. The boom swing lamp turns green when ready.
- Place the feed back in the support with help of feed swing and dump.
- Press **Add** and check that the lamp for position 3 turns green.

4. Position 4
- a. Move boom lift back down and keep boom swing to the right. The boom lift lamp turns green when ready.

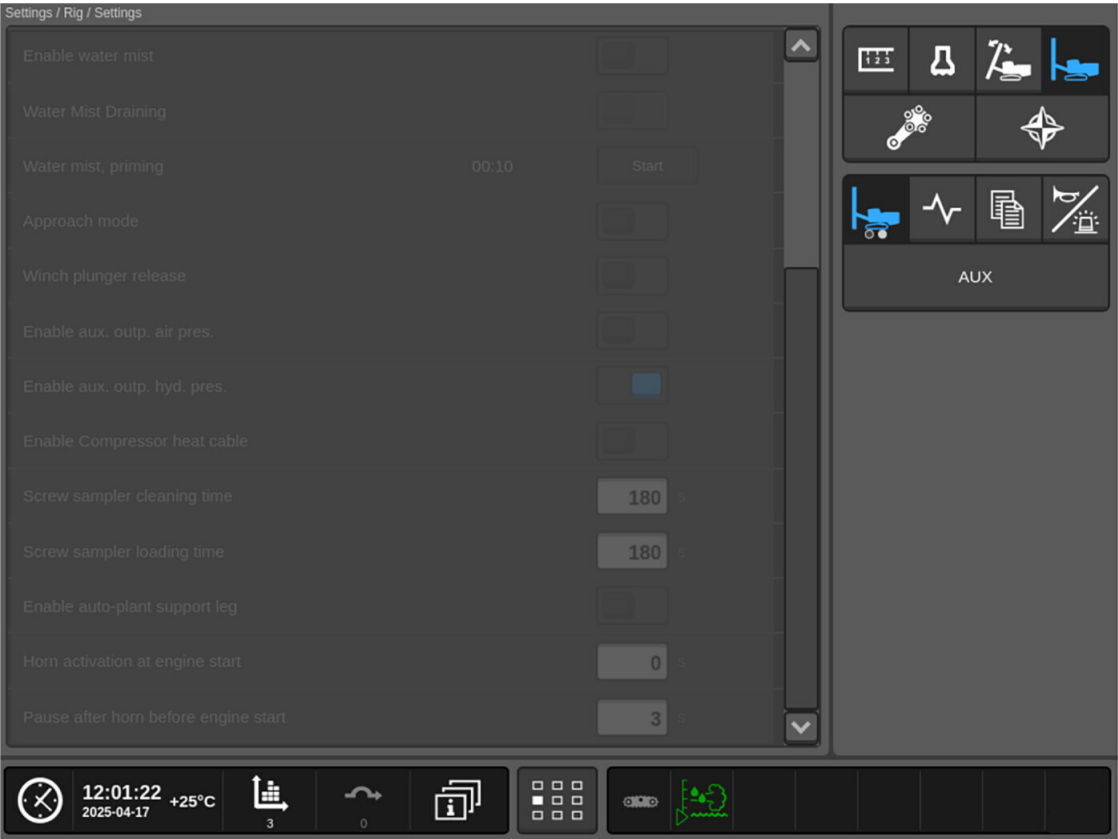
b. Place the feed back in the support with help of feed swing and dump.

c. Press **Add** and check that the lamp for position 4 turns green.

Step result

➔ Calibration ok turns to OK

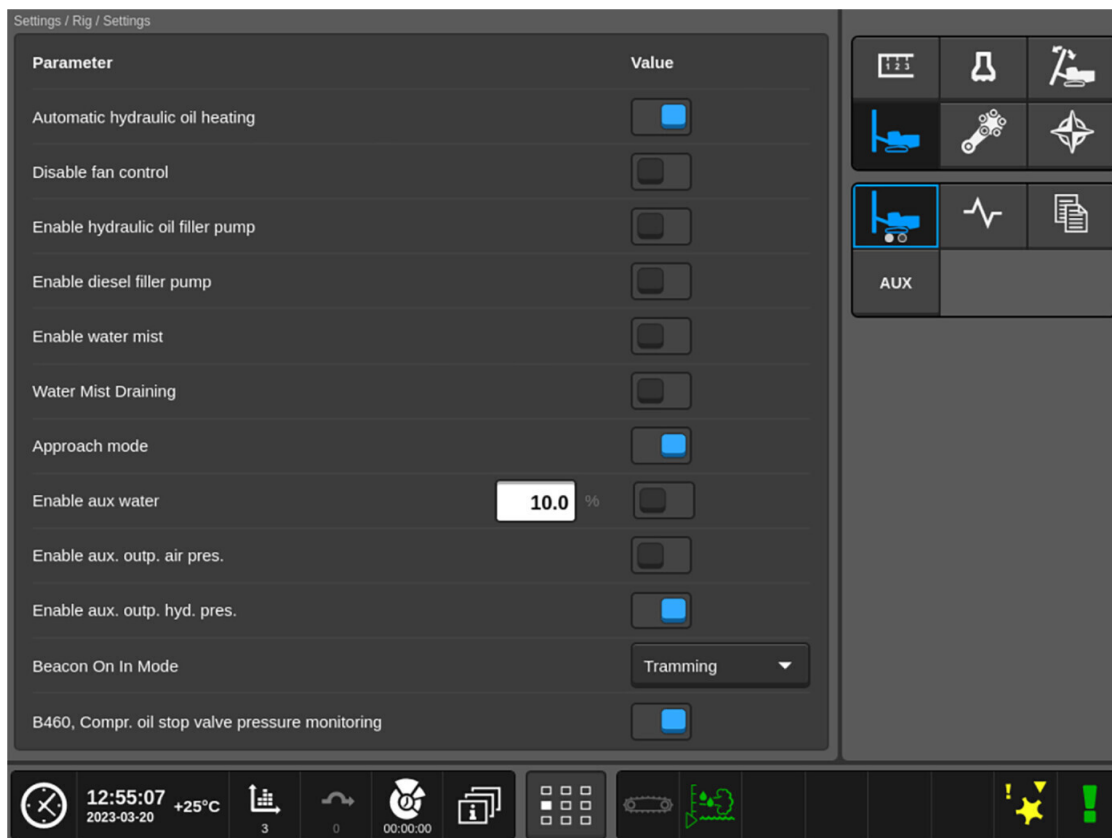
4.4.4 Rig Settings Menu



Example of a Rig Settings Menu

	Select to open the Settings Rig menu.		Select to open the Rig Actuations menu.
	Select to open the Rig Parameters menu (SE level or higher).		Select to open the Safe to board menu (SE level or higher).
AUX	Select to open the Auxiliary Output menu (SE level or higher). This menu is used to modify the output of the 3 auxiliary buttons on the control panel.		

Settings Rig Menu



Example of a Rig Settings Menu

Automatic hydraulic oil heating: Select this function to activate automatic hydraulic oil heating.

Disable fan control: The fan cooler runs at full speed when the fan control is disabled.

Enable hydraulic oil filler pump: Enable to activate the start button of the hydraulic oil filler pump.

Enable water mist: Activate to enable the use of water mist.

- Enabled - it is possible to use water mist, and the warning level indicator shows when the level is low.
- Not enabled - water mist cannot be used, and the warning level indicator is not shown when the level is low.

Water mist, draining: Activate this function to enable water to be drained from the water mist pump.

Water mist, priming: A function to add water to the water mist circuit and to the water mist pump.

Approach mode: Used to activate and deactivate approach mode for Safe to Board. All movements of the rig are disabled when active. The status bar shows if **Approach mode** is active or not active.

Enable aux water: Possibility to use external water for the pressure washer. **Enable water mist** must be deselected.

Enable aux. outp. air pres.: Activate this function to increase the engine rpm and air pressure.

Enable aux. outp. hyd. pres.: Activate this function to increase the engine rpm and pump 1 pressure.

Beacon On In Mode: Select in what mode that the beacon is active.

- **Ignition On** - Beacon Light On, as soon the key is in position 1.
- **Tramming Mode** - Beacon Light On, as soon the position mode knob is in tramming mode.
- **Tramming** - Beacon Light On as soon the rig starts to tram.

B460, Compr. oil stop valve pressure monitoring: Shuts down the engine if the pressure in the oil stop valve drops below 0 bar.

Enable auto-plant support leg: If you select this parameter, extending the support leg can be done automatically.

Horn activation at engine start: When pressing the engine start button, the signal horn is active for 5 seconds before the engine starts. The time for how long the signal horn is active before the engine starts can be adjusted with parameter **Horn activation at engine start**. Set it to 0 to disable the horn before engine start.

Pause after horn before engine start: It is possible to set a pause time after the activation of the signal horn and before the engine starts if **Horn activation at engine start** is used. The engine will start set number of seconds after the activation of the signal horn. The engine start button must be held down for at least the horn activation time and the pause time combined. If the start button is released before this combined timeout, the engine will not be started.

Enable diesel filler pump: Enable to activate the start button of the diesel filler pump.

Screw sampler cleaning time: The time that the screw sampler cleans before taking a sample.

Screw sampler loading time: The time that the screw sampler loads sample to the container.

Rig Actuations Settings

Settings / Rig / Actuators / Pumps

Function	Raw Value	Module	Contact	Marking
Pump 1 pressure control	0	D510	X7	Y100
Pump 2 displacement control	600	D511	X8A	Y190
Pump 2 pressure control	0	D510	X16	Y191
Pump 3 pressure control	0	D510	X15A	Y192
Pump 3 displacement control	600	D515	X14A	Y134

AUX

08:45:29
2023-10-25 +25°C

3

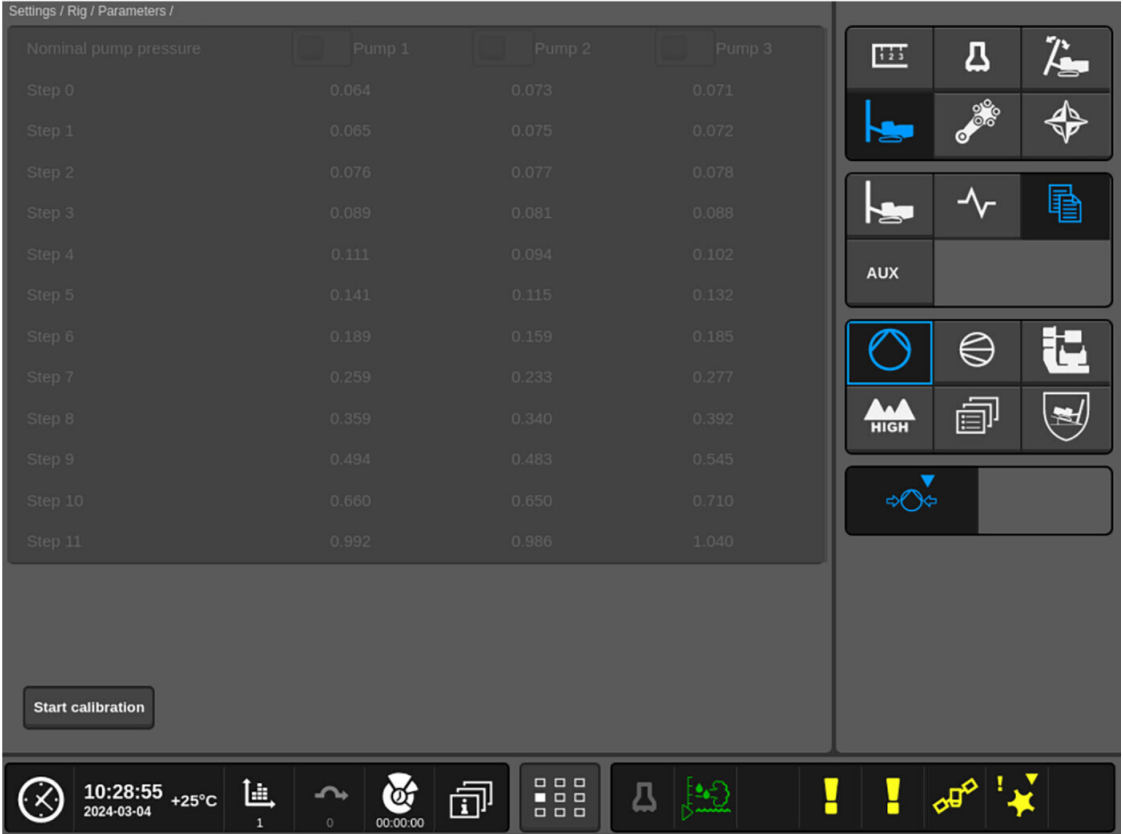
0

00:00:00

Example of a Rig Actuation Setting Menu

	Select this button to open the menu showing actuations for pump settings.		Select this button to open the menu showing actuations for wagon frames.
	Select this button to open the menu showing actuations for cooling fans.		Select this button to open the menu showing actuations for BenchREMOTE.

Rig Parameters Menu



Example of a Rig Parameters Menu (Access Level SE or Higher).

	Select this button to open the menu for pump pressure calibration.		Select this button to open the menu for Air control parameters.
	Select this button to open the menu for BenchREMOTE parameters.		Select this button to open the menu showing High altitude parameters.
	Select this button to open the menu for Other parameters.		Select this button to open the menu for Rig inclination parameters.

Pump Pressure Calibration Menu

Settings / Rig / Parameters / Pumps / Pres calibration

Nominal pump pressure	☐ Pump 1	☐ Pump 2	☐ Pump 3
Step 0	0.073	0.058	0.068
Step 1	0.078	0.060	0.073
Step 2	0.085	0.064	0.078
Step 3	0.104	0.070	0.090
Step 4	0.137	0.082	0.114
Step 5	0.179	0.109	0.160
Step 6	0.239	0.163	0.253
Step 7	0.327	0.264	0.380
Step 8	0.464	0.393	0.517
Step 9	0.609	0.555	0.675
Step 10	0.774	0.728	0.842
Step 11	1.105	1.075	1.177

Start calibration

AUX

HIGH

10:49:25
2023-03-24 +25°C

3 0 00:00:00

!

Example of a **Pumps > Pres calibration** Menu

This menu can only be accessed at SE level or higher.

Calibration can be done for all three pumps from the **Pres calibration** menu.

- Calibration is important for the machine performance, the pumps must be calibrated every 250 diesel engine hours or more frequently if needed.
- Calibrate the pumps after replacement to new pumps.
- All three pumps can be calibrated at the same time. One pump can also be calibrated separately.

Calibrate the Pump Pressure

Settings / Rig / Parameters / Pumps / Pres calibration

Nominal pump pressure	☐ Pump 1	☐ Pump 2	☐ Pump 3
Step 0	0.073	0.058	0.068
Step 1	0.078	0.060	0.073
Step 2	0.085	0.064	0.078
Step 3	0.104	0.070	0.090
Step 4	0.137	0.082	0.114
Step 5	0.179	0.109	0.160
Step 6	0.239	0.163	0.253
Step 7	0.327	0.264	0.380
Step 8	0.464	0.393	0.517
Step 9	0.609	0.555	0.675
Step 10	0.774	0.728	0.842
Step 11	1.105	1.075	1.177

A

Start calibration

10:49:25
2023-03-24 +25°C

3 0 00:00:00

! ⚠

Example of a Pres calibration Menu

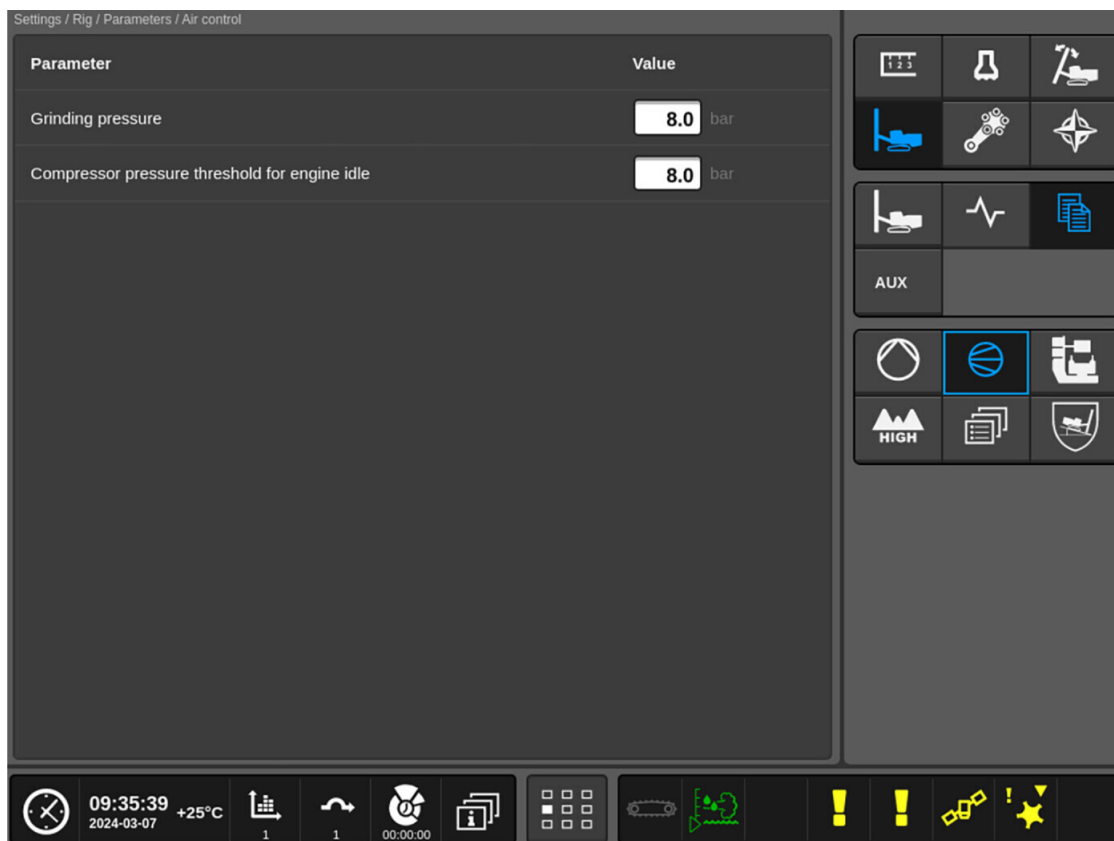
- Precondition**
- ✓ Drilling mode is selected.
 - ✓ The hydraulic oil has reached the working temperature (recommended above 30 °C).
 - ✓ No functions are activated. All pumps are in idle mode.

1. Open the **Settings > Rig > Parameters > Pumps > Pres calibration** menu and select the pumps that must be calibrated.
2. Tap on **Start calibration** (A).

- End result**
- ➔ The calibration results are shown after completed calibration.
The calibration values are automatically logged to the system.

Air Control Parameter Menu

This menu can only be accessed at SE level or higher.

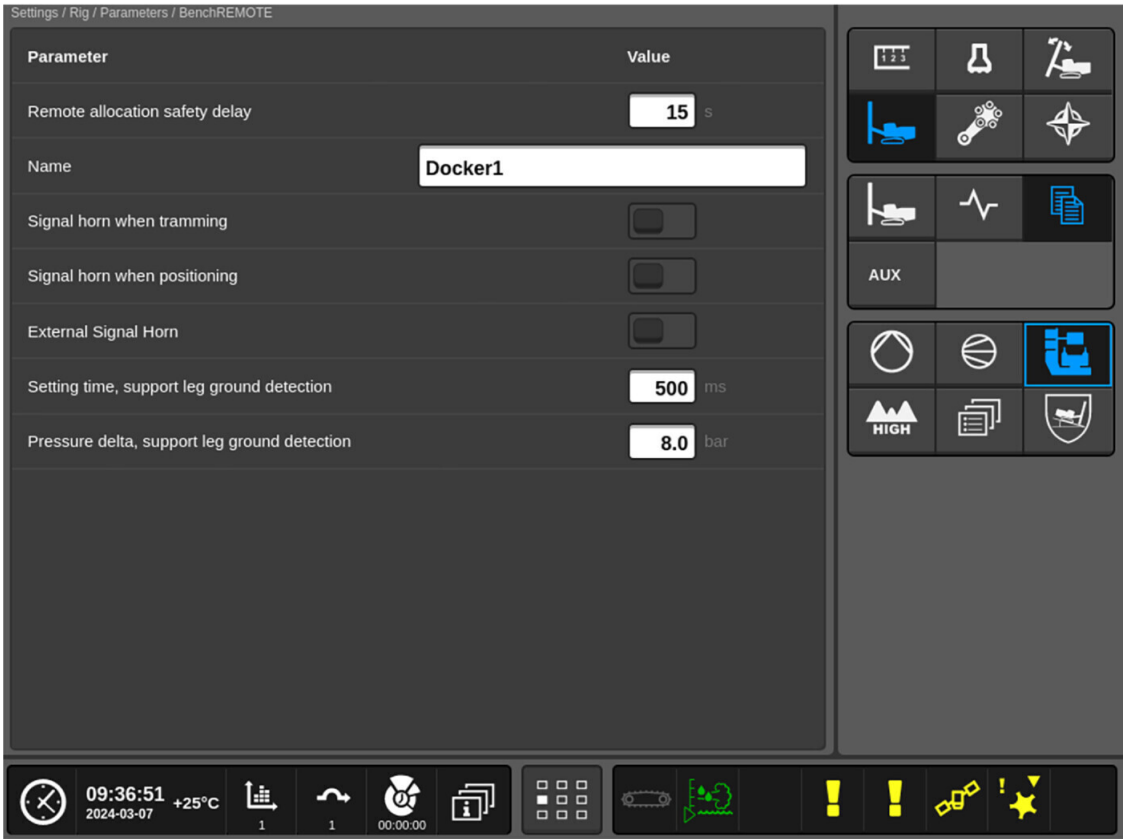


Example of an Air control parameter menu

Grinding pressure: Use this parameter to set the pressure when activating auxiliary air.

Compressor pressure threshold for engine idle: Use this parameter to set compressor tank pressure level for allowing low idle speed of the engine. If the pressure is higher than around 8 bar, the engine is not strong enough to keep running when at low idle.

BenchREMOTE Parameter Settings



Example of a BenchREMOTE Parameter settings menu

Remote allocation safety delay: Set the delay time from when the machine is set in remote mode to when the machine can be remotely operated.

Name: The machine name that is displayed in the OPS.

Signal horn when tramming: Tick the box to sound the signal horn during tramming.

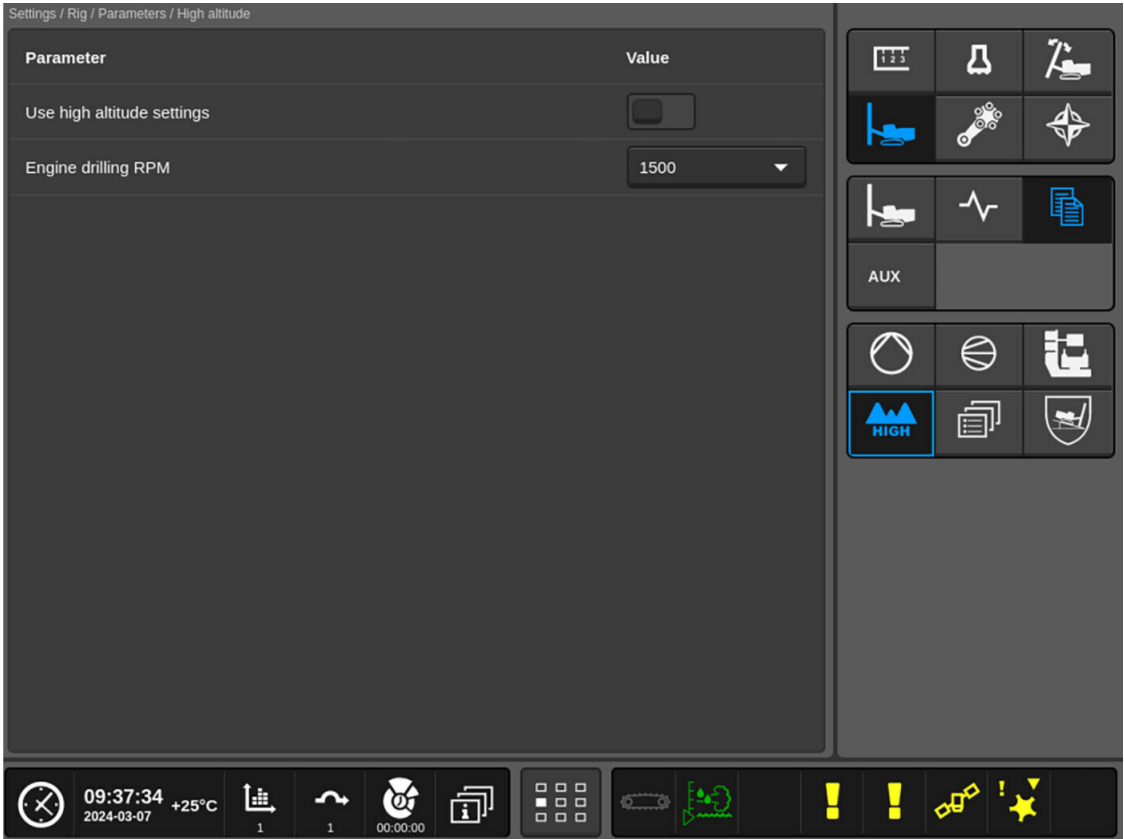
Signal horn when positioning: Tick the box to sound the signal horn during positioning.

External Signal horn: Activate this function if an external larger signal horn is used.

Setting time, support leg ground detection: Time that must elapse for the value set in **Pressure delta, support leg ground detection** to register with the rig.

Pressure delta, support leg ground detection: Shows how much the pressure on the support leg must be increased for the rig to know its reached ground level.

High Altitude Parameter Settings

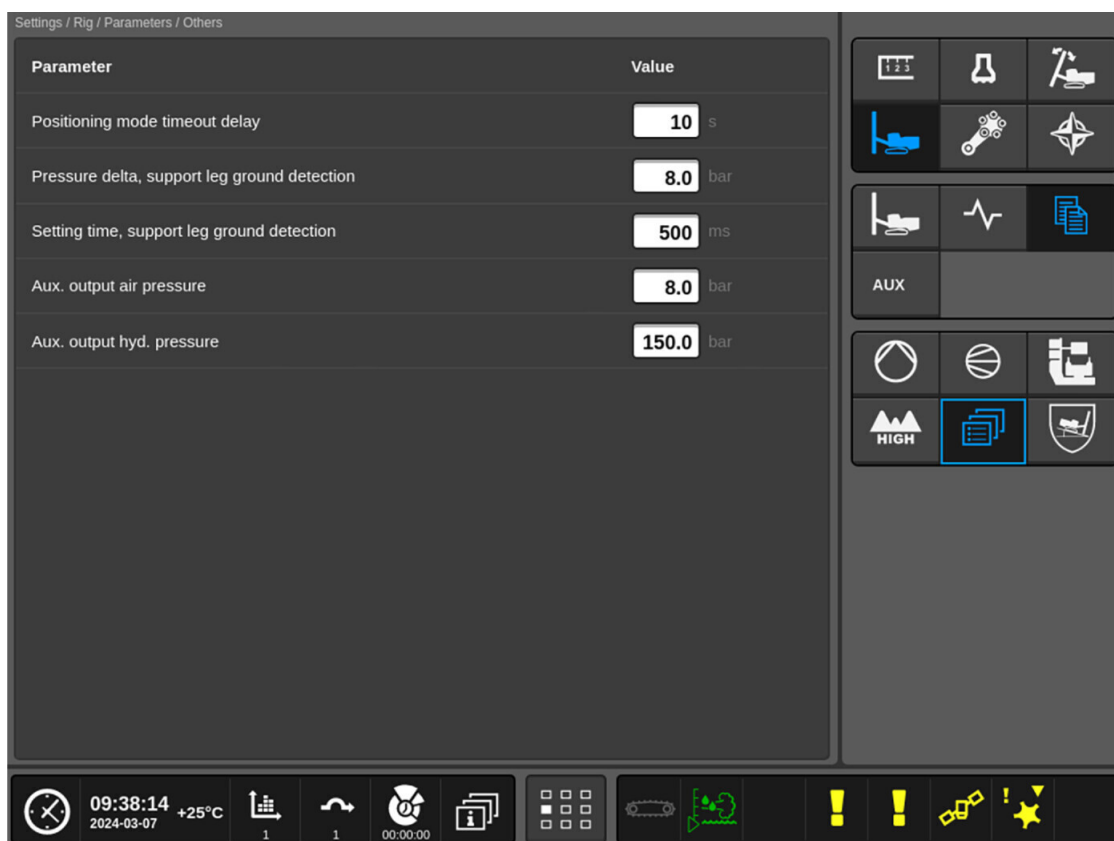


Example of a High altitude Parameter Settings Menu.

Use high altitude settings: This feature is used for machines working at high altitude where oxygen level is lower. The engine increases the idle rpm in order not to stall in some working scenarios.

Engine drilling RPM: Select suitable engine RPM to prevent the engine from stalling.

Others Parameters Settings



Example of an Others Parameters Menu

Positioning mode timeout delay: The time for which the pump is loaded, as well as the time for which the engine is in raised rpm after lever movement.

Pressure delta, support leg ground detection: Shows how much the pressure on the support leg must be increased for the rig to know its reached ground level.

Setting time, support leg ground detection: Time that must elapse for the value set in **Pressure delta, support leg ground detection** to register with the rig.

Aux. output air pressure: This parameter is used to set the compressor tank pressure when activating auxiliary air for running optional equipment like air guns and grinders.

Aux. output hydraulic pressure: Use to set the desired hydraulic pressure to pump 1. The value is set to 150 bar by the system, it is adjustable 20–200 bar.

Rig Inclination Parameters Menu

This menu can only be accessed at SE level or higher.

WARNING

Tipping Risk

The tipping over of a machine can cause serious injury or death.

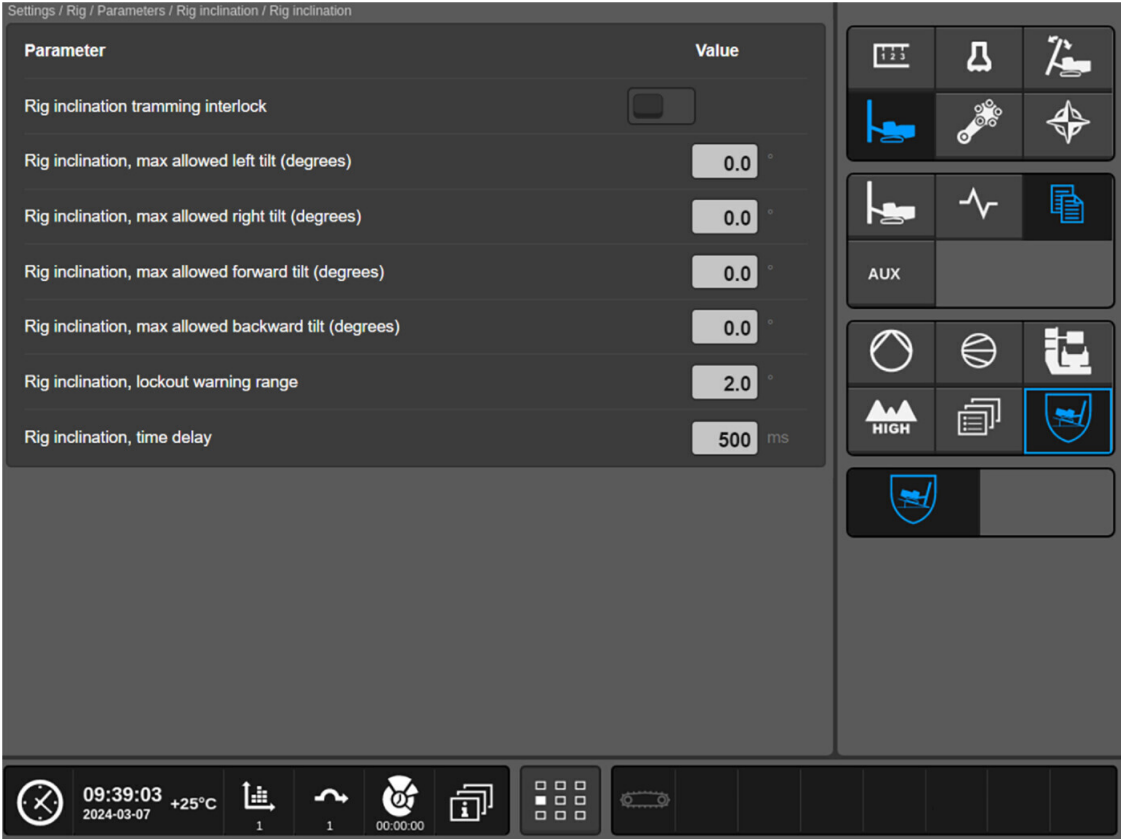
- ▶ Never exceed the maximum allowed inclination angles for the machine.
- ▶ Never combine inclination angles.

!

NOTE: The rig inclination feature is set to off at delivery from the factory.

!

NOTE: Default values on tilt angles are 0 degrees, any changes that are made are at the users own risk. The maximum inclinations of the machine vary depending on boom and feed position. Read the operation manual for allowed inclinations.



Example of a Rig inclination Parameter Menu (Access Level SE or Higher)

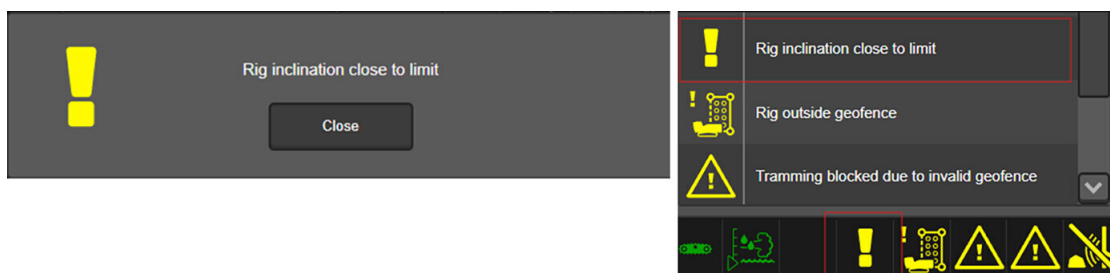
Rig inclination tramming interlock	Use to activate the rig inclination function.
Rig inclination, max allowed left tilt (degrees)	Use to set maximum allowed rig inclination to the left. If the maximum value is reached, the tramming function shuts off.
Rig inclination, max allowed right tilt (degrees)	Use to set maximum allowed rig inclination to the right. If the maximum value is reached, the tramming function shuts off.
Rig inclination, max allowed forward tilt (degrees)	Use to set maximum allowed rig inclination forward. If the maximum value is reached, the tramming function shuts off.
Rig inclination, max allowed backward tilt (degrees)	Use to set maximum allowed rig inclination backward. If the maximum value is reached, the tramming function shuts off.

Rig inclination, lockout warning range	Use to set the value for when the pop-up message Rig inclination close to limit is shown. The value must always be positive and default is 2 degrees.
Rig inclination, time delay	Use to set the time for how long the rig inclination can be outside allowed limits before warning.

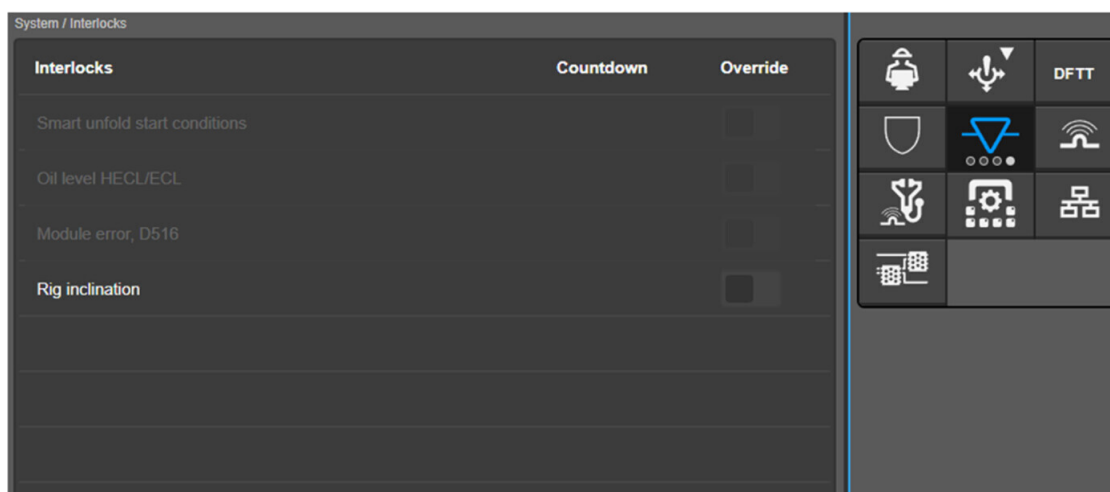
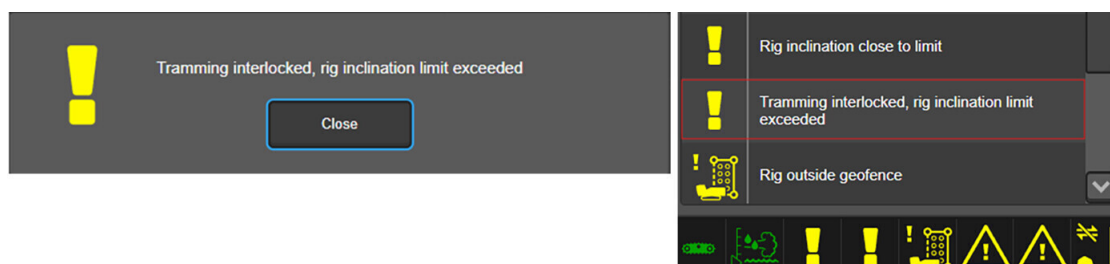
Example:

Rig inclination, max allowed forward tilt (degrees) is set to 10 degrees and the **Rig inclination, lockout warning range** is set to 3 degrees.

- If tilting reaches 7 degrees, a warning pop-up message and a warning in the status bar are shown. Tramming is allowed.



- If the maximum tilt value is exceeded, a warning pop-up message and a warning in the status bar are shown. Tramming is not allowed. Tilt backward to allow tramming or override the rig inclination in the **System>Interlocks** menu (requires access level SE or higher).



System>Interlocks Menu (Access Level SE or Higher)



NOTE: Read the operation manual for allowed maximum inclinations. Do not exceed these values.

Safe to Board Menu

The operating status for the Safe to Board (STB) lights is configured in the **Setting > Rig > Safe to Board** menu.



Example of a Safe to board Menu

State	Description
Manual/Local	The machine is in manual mode and cannot be allocated by a remote station
Remote switched	The machine is in remote mode and can be allocated by a remote station
Remote allocated	The machine is in remote mode and allocated by an operator
Autonomous	The machine is operation autonomously without operator intervention
Approach mode	The machine is in approach mode. When forced into approach mode, the machine becomes safe to approach as movement can only occur if the the mode is exited.
Approach mode exit	When a remote operator exits approach mode, there is a time delay with light and sound signals. The signals warn nearby people that the rig is going to be remotely operated and they must leave the area or switch the machine to local mode.

Beacon Color	Beacon status	Description
There are four colors red, amber green and blue.	Dim	The light flashes every 1000 milliseconds.

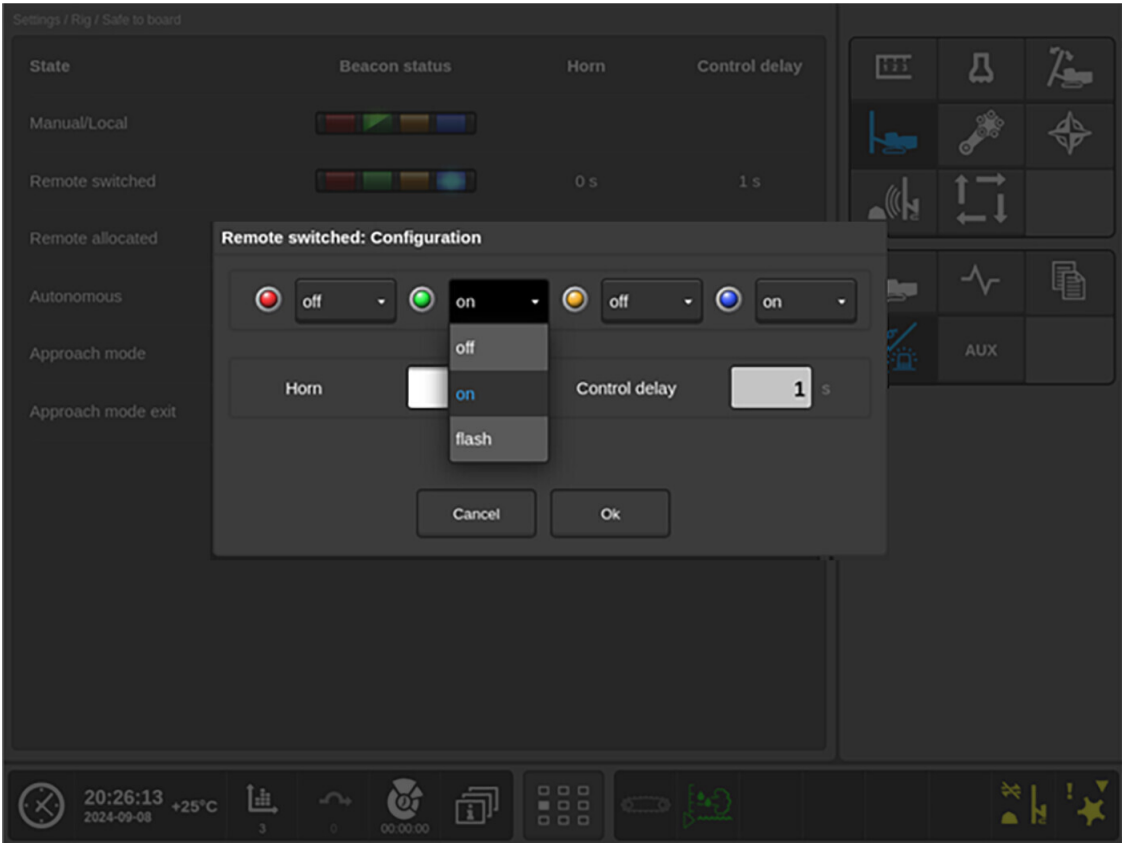
Beacon Color	Beacon status	Description
	Lit	The light constantly illuminates.
	Half dim / Half lit	The light is not illuminated.

Horn	Description
The horn is used to alert the operator of different conditions.	No horn - The horn is not enabled
	Complete cycle - The horn activates during the leveling operation.
	Warning time - The horn activates before the leveling operation for the set amount of time.

Control delay

Control Delay is the amount of time (in seconds) that inhibits machine functions as the machine enters a different STB state.

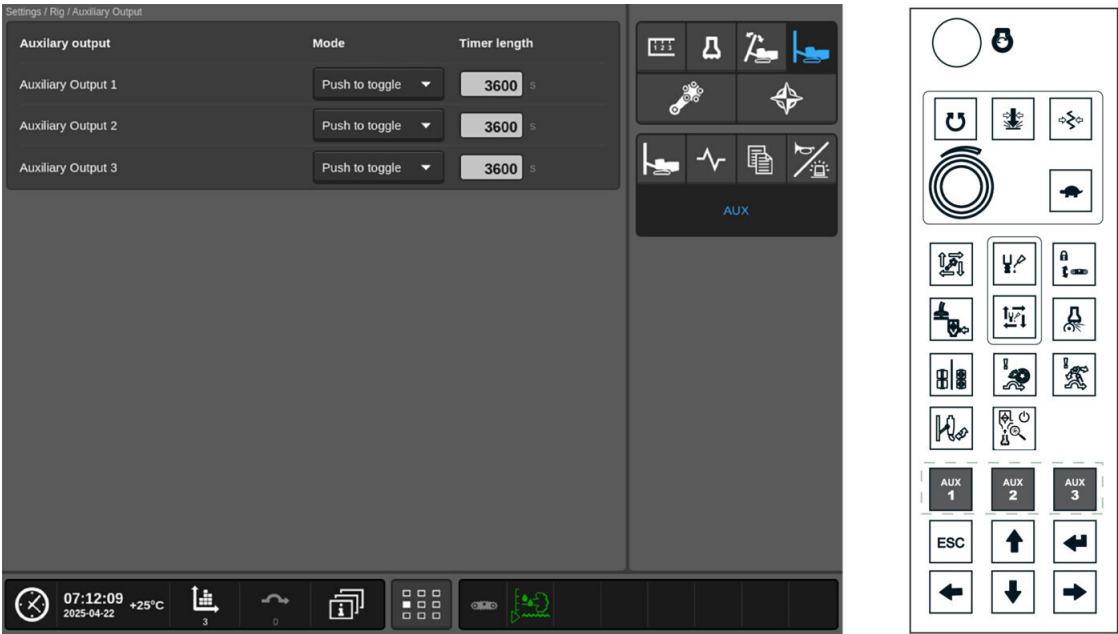
Safe to Board Configuration



Example Safe to board Configuration

Beacon status	The drop-down list contains three alternative behaviors for indicator lamps. • on • off • flash
Horn settings	The signal horn can be set to sound for a specified time when a certain transition is entered.
Control delay	Control Delay is the amount of time that inhibits machine functions as the machine enters a different STB state. When control delay is active, an event symbol is shown in the RCS GUI system tray.

Auxiliary Output Menu



The **Auxiliary Output** menu is used to set the functionality of the auxiliary buttons on the control panel.

Dropdown Box Settings	Auxiliary Button Function
Push to toggle	Press the button once to toggle between ON and OFF.
Push and hold	Press and hold the button to turn ON. Turn OFF when the button is released.
Timer	Press once to turn ON. A countdown timer starts with a specific time set. When the countdown time has elapsed, it turns OFF the equipment. The timer can be set between 1–9999 seconds.

4.4.5 Road Model Menu

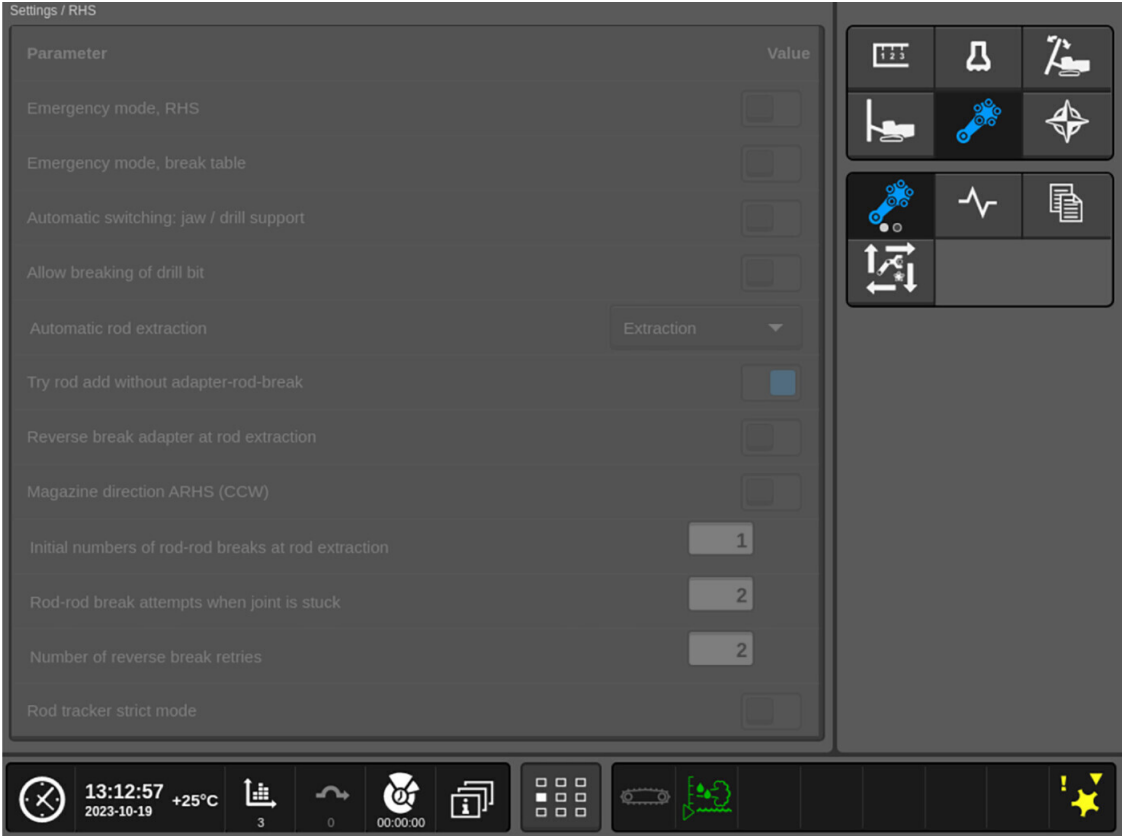
Settings / Road model

Current state		Inactive
Enquiry	☒	500
Local bit position	☒	500
Angle	☒	500
Rig sensors	☒	500
Drilling started	☒	
Drilling done	☒	
Local end pos X	0	
Local end pos Y	0	
Local end pos Z	0	
Connection timeout	4000	
Disconn. warning delay	10000	

Example of a Road model Menu

In the **Road model** menu are settings to decide which kind of information to share from the rig to a third party navigation system.

4.4.6 RHS Settings Menu



Example of a Rod Handling System (RHS) Settings Menu.

	Select to open the RHS menu.		Select to open the Joint breaking Auto RHS Parameters menu.
	Select to open the RHS Actuations menu. - Select the button to see actuated values for gripper, drill rod support, magazine rotation, and other RHS functions. - Select the button to see actuated values for break out jaws and breaking cylinders.		Select to open the RHS Parameters menu. - Threading parameters - Calibration arm parameters - Calibration gripper parameters

Settings RHS Menu

Parameter	Value
Emergency mode, RHS	☒
Emergency mode, break table	☐
Automatic switching: jaw / drill support	☐
Allow breaking of drill bit	☐
Automatic rod extraction	Extraction
Try rod add without adapter-rod-break	☒
Reverse break adapter at rod extraction	☐
Magazine direction ARHS (CCW)	☐
Initial numbers of rod-rod breaks at rod extraction	1
Rod-rod break attempts when joint is stuck	2
Number of reverse break retries	2
Rod tracker strict mode	☐

Example of a RHS menu

Emergency mode, RHS: Activate this function to deactivate all the RHS sensors. All the RHS functions are now done manually. When activated, loose grip does not work.

Emergency Mode, break table: Activate this function to enable manual control of the breakout table movement.

Automatic switching jaw/ drill support: During drilling, the button is active and controls the drill rod supports. During semi-automatic breaking, the button is deactivated and the lever then controls the breaker jaws. The button is activated when drilling starts again.

Allow breaking of drill bit: Select to allow breaking of drill bit in the correct position. To move the bit to the correct breaking position, activate button **Allow breaking of drill bit** and the extraction mode button on the right joystick, and move the feed backward.

Automatic rod extraction: Select type of rod extraction. When the rig is finished drilling the hole in automatic or the operator decides to extract the rod in automatic. There are four possible extraction actions:

- **Off** - rod is not extracted
- **Extraction** - rod is extracted with feed backward
- **Rotation** - rod is extracted with rotation
- **Rotation & air** - rod is extracted with rotation and air

Try rod add without adapter-rod-break: Select for rod-break only if the rotation does not open by rotation alone.

Reverse break adapter at rod extraction: Reverse break (M4) used when unthreading rod and adapter. It is not recommended to be active in COPROD 76, 89, 102 mm

Magazine direction ARHS (CCW): The magazine turns counterclockwise when using Auto RHS.

- Initial numbers of rod-rod breaks at rod extraction:** The initial number of times to use the breakout table to open between rod and rod in position **(M4)**. Recommended setting is 1.
- Rod-rod break attempts when joint is stuck:** The number of break attempts **(M4)** after the initial break attempt if the joint is stuck. Recommended setting is 2.
- Number of reverse break retries.:** The number of reverse break retries **(M4)** if the **Reverse break adapter at rod extraction** does not succeed. Recommended setting is 3.
- Rod tracker strict mode:** When enabled, the system shows a pop-up warning if the rod tracker loses calibration.
- Anti-jamming Extraction:** Activate **Anti-jamming Extraction** to prevent the rod getting stuck during extraction.
- Rotation pressure extraction, desired:** Use to set the rotation pressure that **Anti-jamming Extraction** aims for.
- Air flow, extraction:** Sets the air flow used at extraction.
- Flush time before start of move backward:** Set flush time before moving backward.

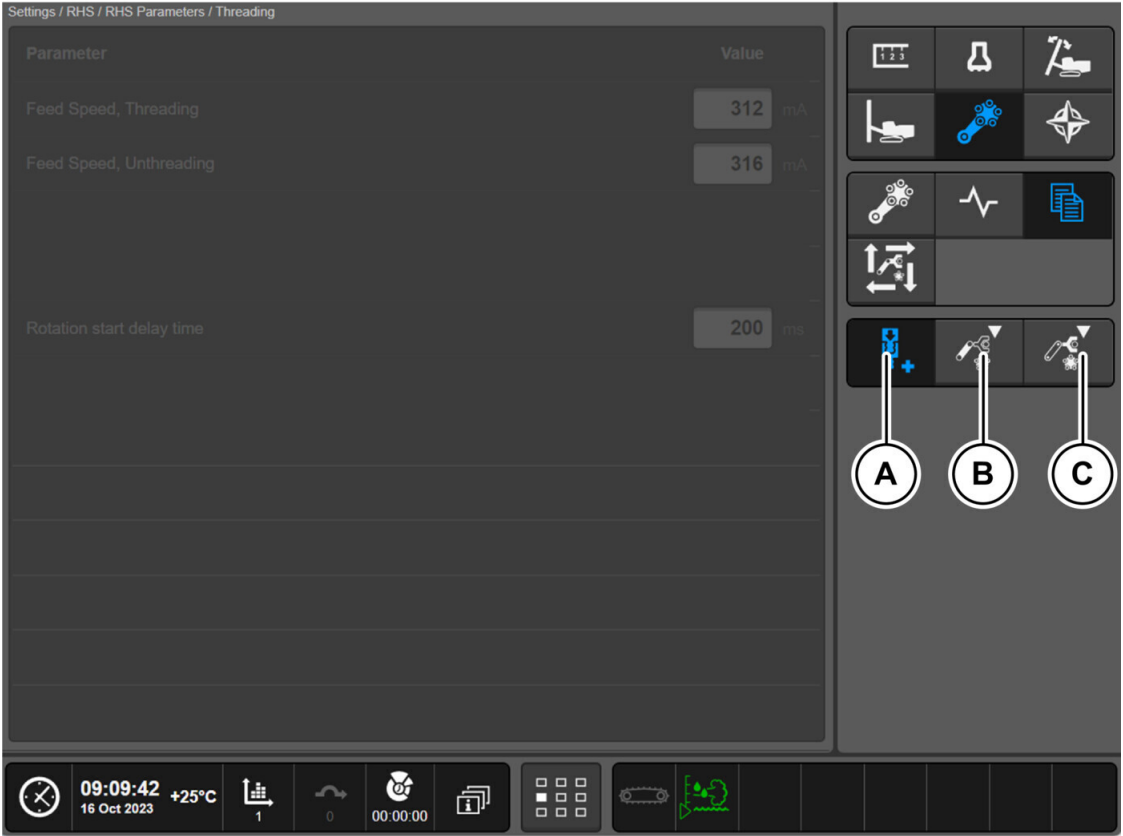
RHS Actuators Settings

Settings / RHS / Actuators / RHS

Function	Raw Value	Module	Contact	Marking
RHS on	0	D513	X11B	Y331
Grippers closed	0	D513	X12A	Y300A
Grippers open	0	D513	X12B	Y300B
Grippers loose grip	0	D513	X15A	Y306
Rotate gripper towards magazine	0	D514	X10A	Y311A
Rotate gripper towards drillcentre	0	D514	X10B	Y311B
Aim to drill center	0	D513	X17A	Y301A
Aim to magazine	0	D513	X17B	Y301B
Magazine rotation CW	0	D514	X17A	Y303A
Magazine rotation CCW	0	D514	X17B	Y303B
Magazine outlet open	0	D514	X6B	Y310
Upper Drill Support Closed	0	D514	X18A	Y361A

<

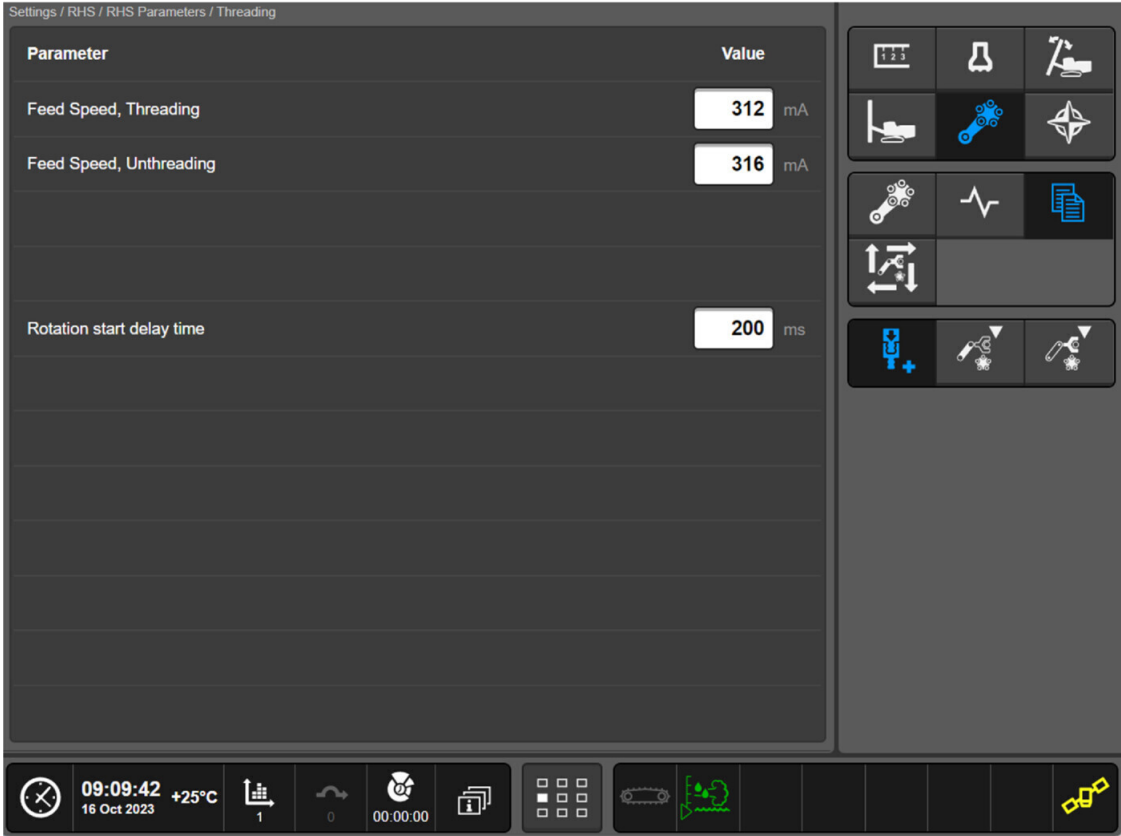
RHS Parameters



Example of an RHS Parameters Menu.

A	Select to open the Rod Handling System (RHS) Threading parameters menu.	B	Select to open the RHS Calibration arm parameters menu.
C	Select to open the RHS Calibration gripper parameters menu.		

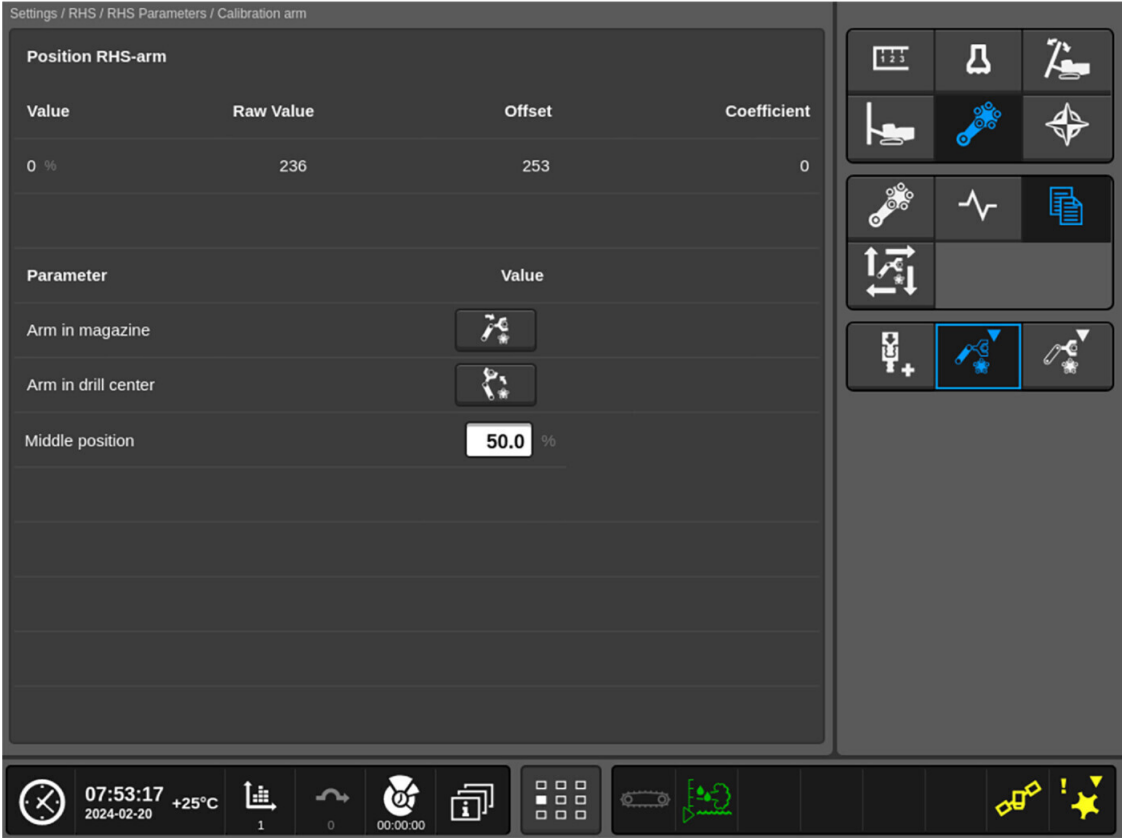
RHS Threading Parameters



Example of a Threading Menu (Access Level SE or higher)

- Feed speed, threading:** The feed speed for threading.
- Feed speed, unthreading:** The feed speed for unthreading.
- Rotation start delay time:** Set value for how long the delay time should be for rotation start.

RHS Calibration Arm Parameters



Example of a Calibration arm Menu

Arm in magazine: Tap on button  when the Rod Handling System (RHS) arms are fully in the magazine.

Arm in drill center: Tap on the button  when the Rod Handling System (RHS) arms are fully in the drill center.

Middle position: The point where the arms stop and the grippers swing around.

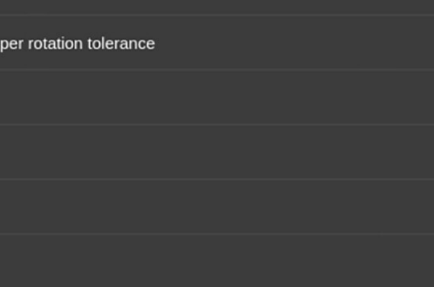
RHS Calibration Gripper Parameters

Settings / RHS / RHS Parameters / Calibration gripper

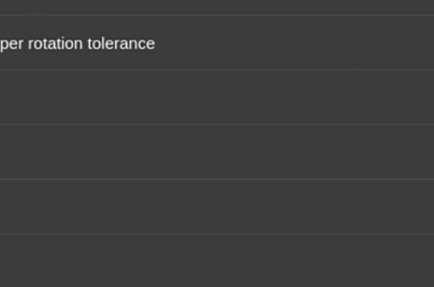
Parameter	Value	Raw Value	Offset	Coefficient
Gripper rotation	0.0	0	0	0.54932

Gripper Positions Value

Rotated towards magazine	<button style="background-color: #ccc; border: 1px solid #000; padding: 2px 10px;">Calibrate</button>
Rotated towards drill center	<button style="background-color: #ccc; border: 1px solid #000; padding: 2px 10px;">Calibrate</button>
Gripper rotation near DC	77.8 °
Gripper rotation near DC pos offset	5.0 °
Gripper rotation tolerance	5.0 °



Settings / RHS / RHS Parameters / Calibration gripper



Settings / RHS / RHS Parameters / Calibration gripper

Example of a Calibration gripper Menu

Rotated towards magazine: Tap **Calibrate** when the grippers are fully rotated towards the magazine.

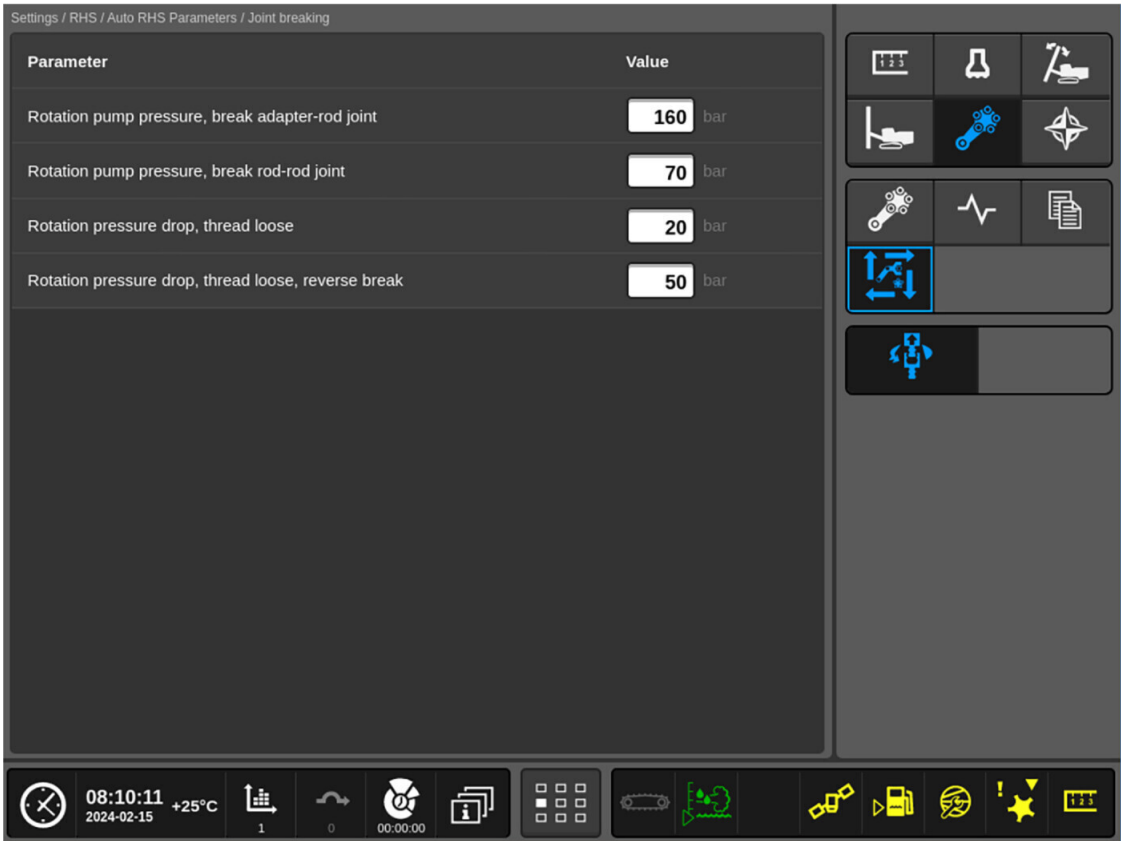
Rotated towards drill center: Tap **Calibrate** when the grippers are fully rotated towards the drill center.

Gripper rotation near DC: The maximum angle that the grippers are allowed to rotate before the rotation unit has passed **Max. cradle position, arms to DC, gripper closed**. If the sensor is correctly calibrated and no modifications have been done to the rotation unit, the recommended value is 160.

Gripper rotation near DC pos offset: Works as a brake distance for the gripper rotation.

Gripper rotation tolerance: The tolerance where the gripper rotation has to stop. If gripper rotation is outside of this tolerance, the feed cradle movement stops to avoid collision.

Joint Breaking Menu



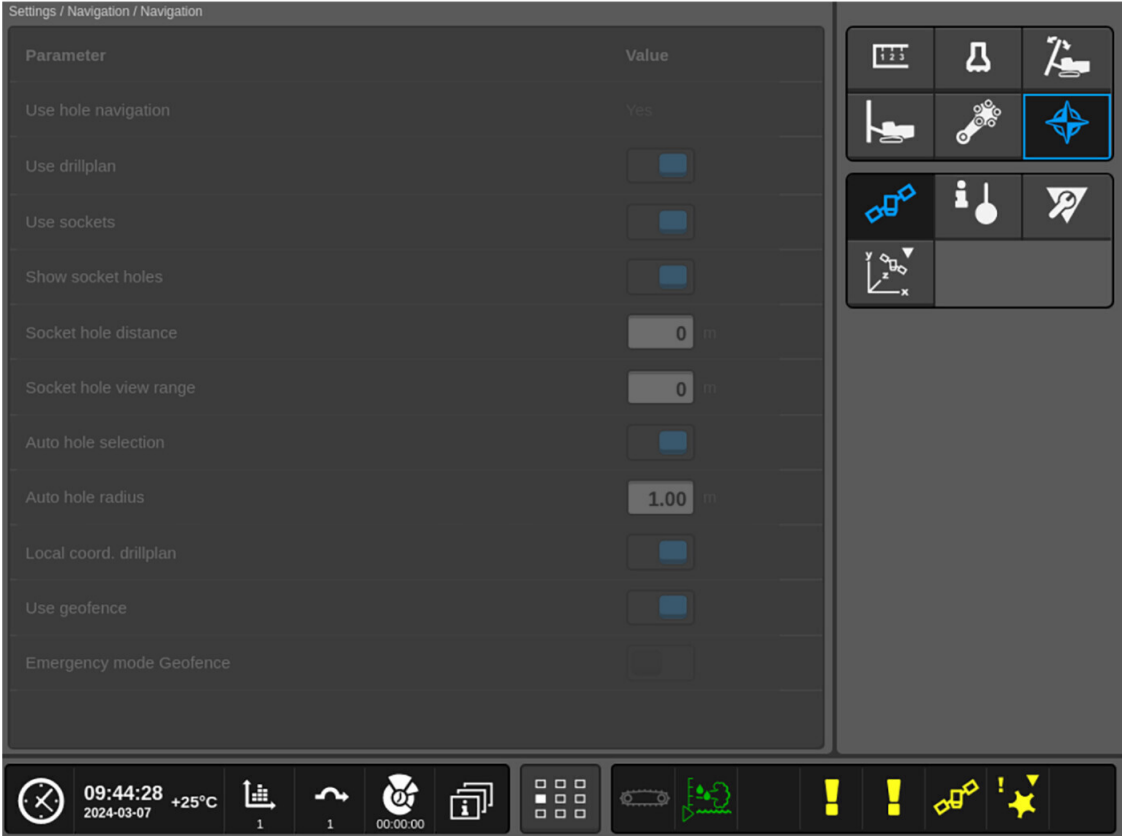
Rotation pump pressure, break adapter-rod joint: The pump pressure requested when doing an adapter-rod break.

Rotation pump pressure, break rod-rod joint: The pump pressure requested when doing a rod-rod break.

Rotation pressure drop, thread loose: The rotation pressure threshold drops when a loose thread is detected.

Rotation pressure drop, thread loose, reverse break : The rotation pressure threshold drops when the thread is detected as loose and the machine used reverse break.

4.4.7 Hole Navigation System (HNS) Settings Menu



Example of a Navigation Settings Menu

	Select the button to open the Navigation menu.		Select the button to open the Hole navigation status information menu.
	Select the button to open the Calibration coverage area menu (access level SE or higher).		Select the button to open the Link Model menu (access level SE or higher).

Navigation Menu

Settings / Navigation / Navigation

Parameter	Value
Use hole navigation	Yes
Use drillplan	☒
Use sockets	☒
Show socket holes	☒
Socket hole distance	0 m
Socket hole view range	0 m
Auto hole selection	☒
Auto hole radius	1.00 m
Local coord. drillplan	☒
Use geofence	☒
Emergency mode Geofence	☐

Navigation icons:

Bottom status bar:

Example of a Navigation Menu

Use hole navigation: Shows if hole navigation is activated or not activated.

Use Drillplan: This function must be selected for drill plan handling to work. If not selected, drill plan handling in the menu **Drill plans** is empty.

Use sockets: Select this function to use warnings for drill hole sockets. Socket holes are holes or marks that show where in the drill plan not to drill. Drill plan with socket hole must be built in the Surface Manager.

Show socket holes: Select this function to show warnings for drill hole sockets.

Socket hole distance: Set distance for warning for drill hole sockets.

Socket hole view range: Set distance from where the socket is visible in the drill plan.

Auto hole selection: If this function is selected, the system suggests a new hole for the operator to select when the drill bit is within **Auto hole radius**. To select the suggested hole, the operator has to confirm the suggestion using the Reset button on the right-hand lever. This function minimizes keystrokes in the **Drill plan** menu.

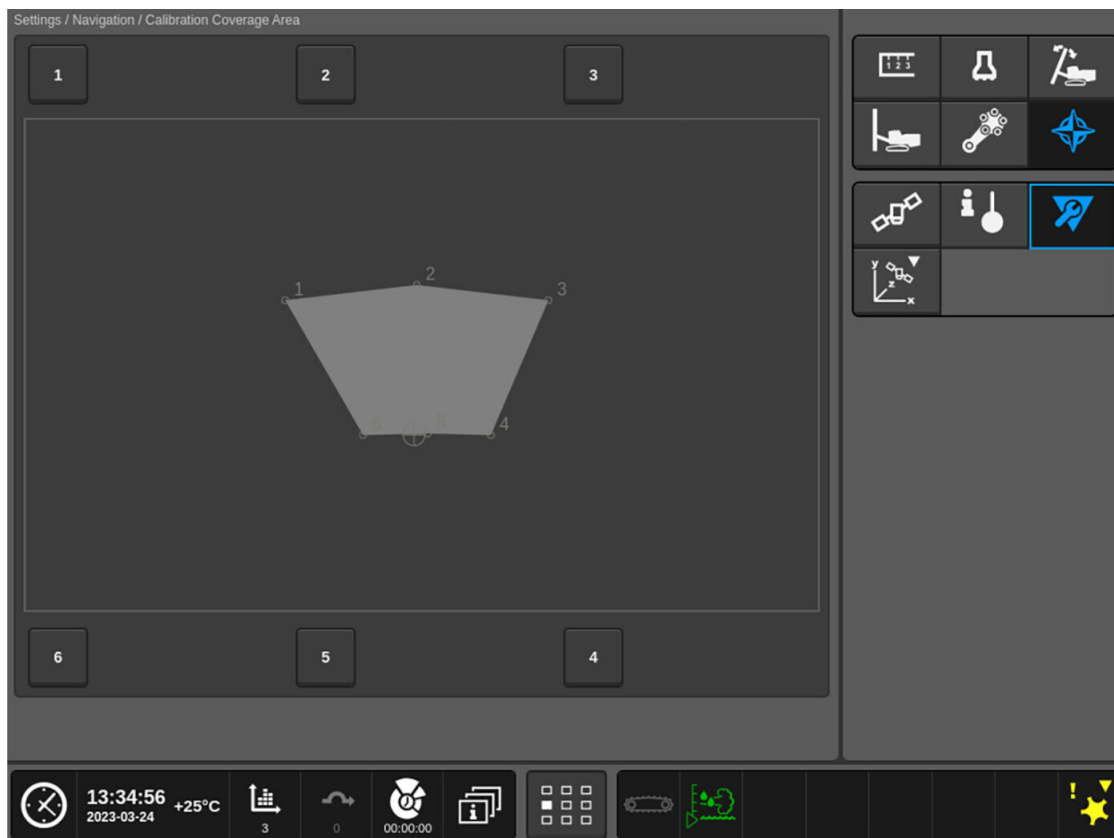
Auto hole radius: When the drill bit is within the selected distance from a hole, the system suggests the hole as next. Must be confirmed by the operator via the Reset button on the right-hand lever.

Local coord. drillplan: Must be selected to enable drilling in a local coordinate system. Must always be selected when using Hole Navigation System (HNS).

Use geofence Select to use geofence function. A geofence is a geographically limited area in which the machine must stay.

Emergency mode Geofence: If the machine goes outside the geofence a warning shows and the machine stops. Select this function to override the geofence temporarily and enable movement of the machine.

Calibration Coverage Area



Example of a **Calibration Coverage Area** menu

The **Calibration Coverage Area** menu makes it possible to calibrate the maximum coverage of the boom system. When the calibration is done the boom coverage area can be seen on the screen. The calibration coverage area is used in the navigation menu for viewing which holes can be reached from the current position.

When calibrated, the figure shows the maximum coverage area of the boom system.

Position	Position of Boom and Feed
1	• Boom: Maximum left and maximum forward position • Feed: Vertical feed and feed extension in maximum up position
2	• Boom: Straight forward and maximum forward position • Feed: Vertical feed and feed extension in maximum up position
3	• Boom: Maximum right and maximum forward position • Feed: Vertical feed and feed extension in maximum up position
4	• Boom: Maximum right position and as close to the rig as possible • Feed: Vertical feed and feed extension in maximum up position
5	• Boom: Straight forward position and as close to the rig as possible • Feed: Vertical feed and feed extension in maximum up position
6	• Boom: Maximum left position and as close to the rig as possible • Feed: Vertical feed and feed extension in maximum up position

Link Model Menu

[illegible]

Example of a Link model Menu

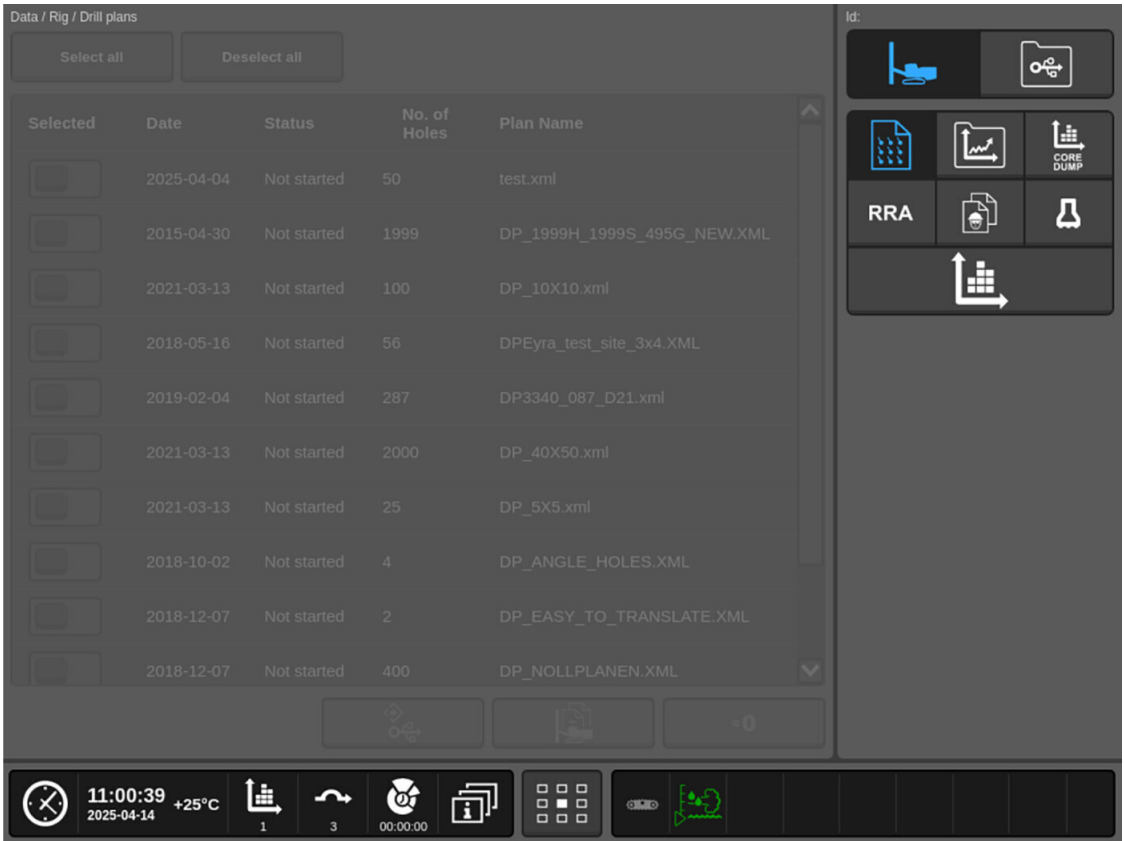
All distances in the **Link model** menu can be reset to default values except for distance **L11Z**.



NOTE: Reset can only be done at a higher access level than SE.

L11Z: Shows the distance between the front of the drill bit on the first rod and the lower part of the bulkhead on the top of the hammer.

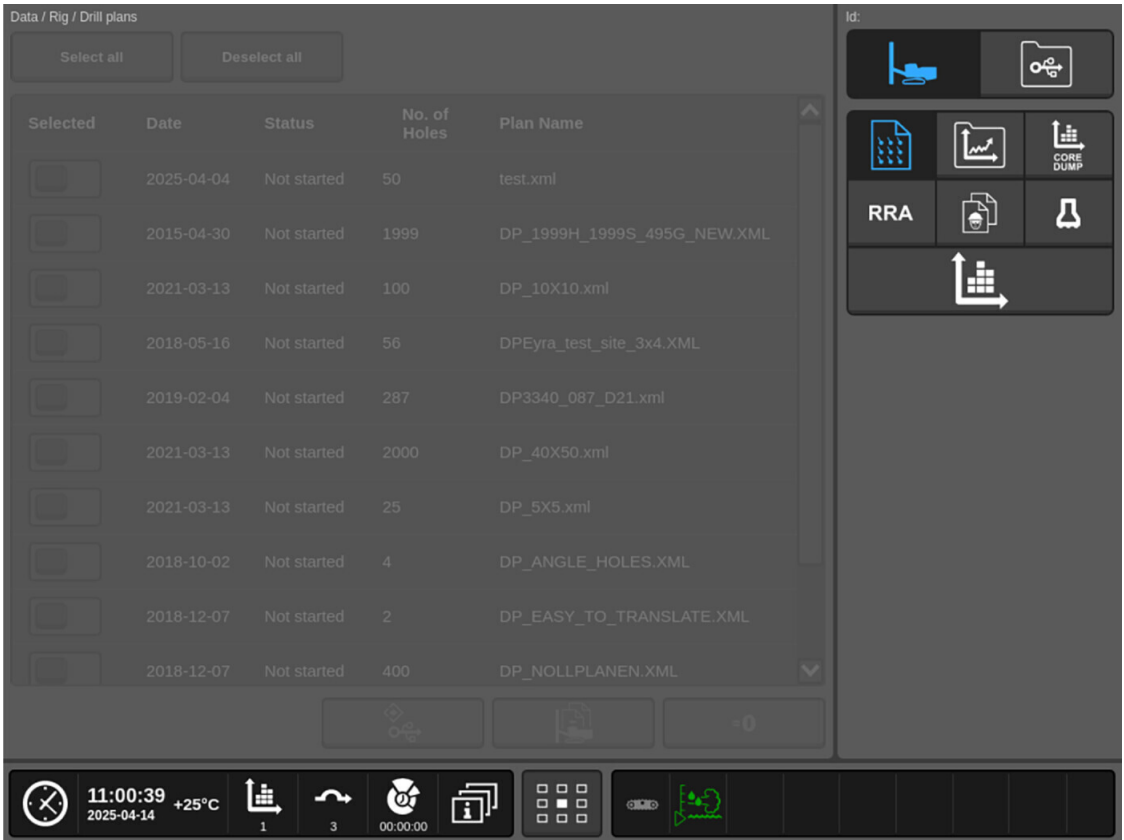
4.5 Data Menu



Example of a Data Menu

	Select this button to open the Drill plans menu. The menu is used to see the drill plans files in the rig computer memory and manage those drill plans, empty logs from the plan or transfer the drill plan to the USB.
	Select this button to open the Drill plans menu to see the Drill plans in the USB memory. This menu is used to load drill plans from a USB flash drive to the rig.

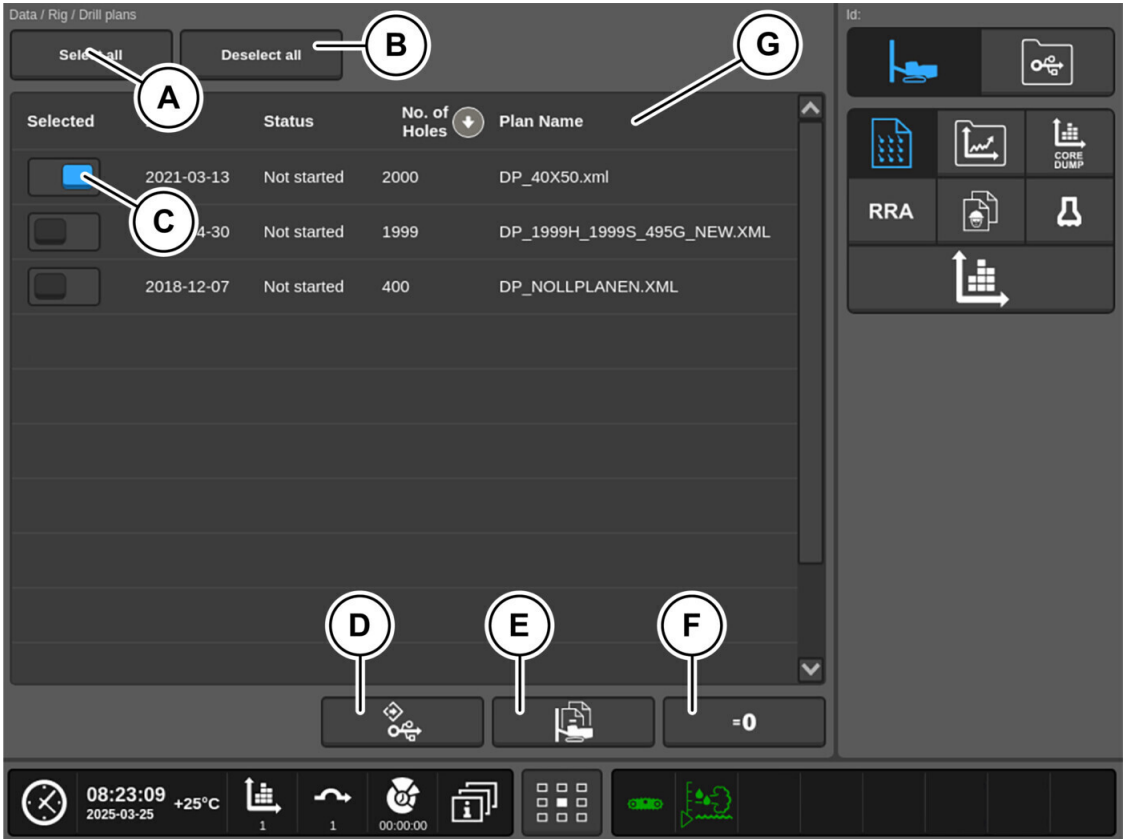
4.5.1 Drill Plans Menu for Rig



Example of a Drill plans Menu

	Select this button to open the Drill plans menu.		Select this button to open the All logs menu.
	Select this button to open the CORE DUMP menu. In this menu, core dump files can be deleted or transferred from the rig to a USB.	RRA	Select this button to open the Files for RRA menu.
	Select to see available operators.		Select this button to open the Consumables menu.
	Select this button to open the GUI logs menu.		

Drill Plans Menu



Example of a Drill plans Menu

A	Tap to Select all available drill plans.	B	Tap to deselect all drill plans file.
C	Tap to activate or deactivate a drill plan file.	D	Select to load the selected drill plan from the rig to the USB.
E	Tap to remove drill plan file from the memory. Depending on which menu used, it is either removed from the rig memory, or from the USB memory.	F	Tap to clear the drill plan history, all holes are shown as undrilled.

Tap on a column in the header (G) to sort it. The first two taps organize it alphabetically, and the third sorts it by date.

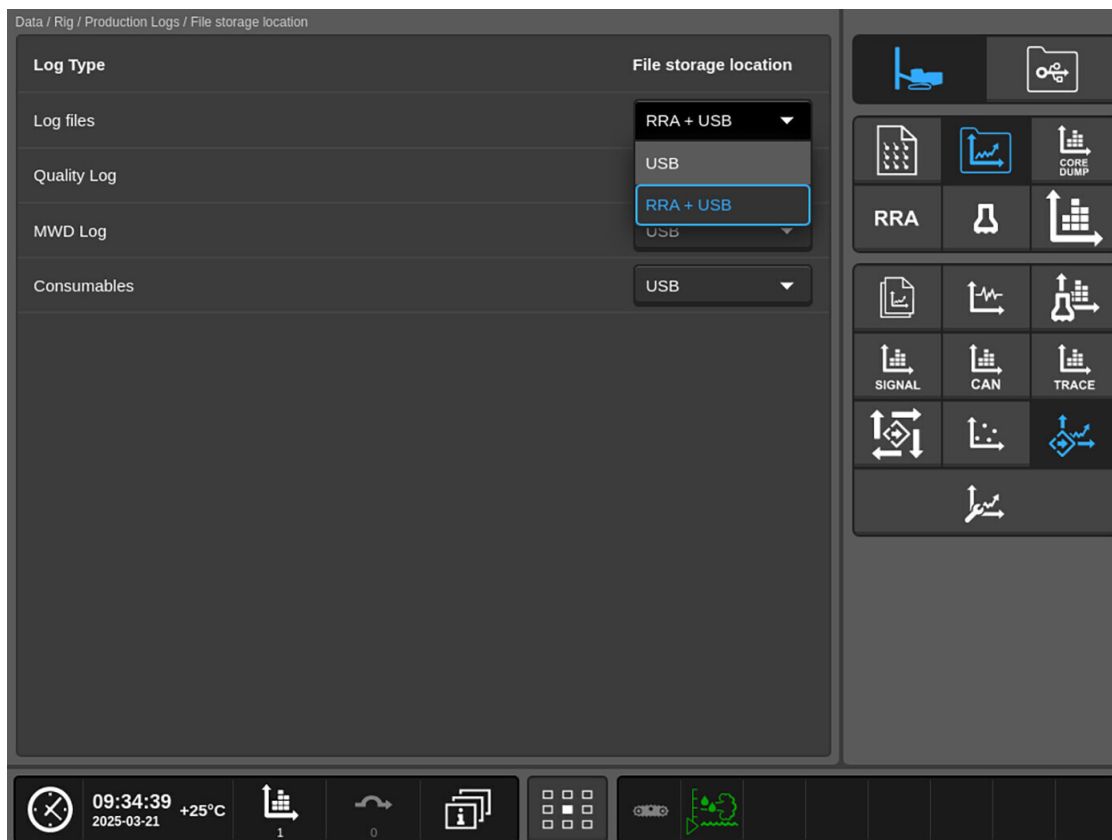
Production Logs



Example of a Production Logs Menu

A	Log Type: • Diagnostic Logs - Diagnostic data • GUI logs - Screen Graphical User Interface (GUI) data. • Statistics Log - Saved hours and meters. • Performance log - Day to day data. • Drill Plan Log - Drill plan data. • Quality Log - Data from drill holes. • MWD Log - Measure While Drilling (MWD) data.		Save data to rig and enable logs to be moved to the USB.
	Move logs by saving to the USB. The button must be selected before selecting .		

File Storage Location Menu

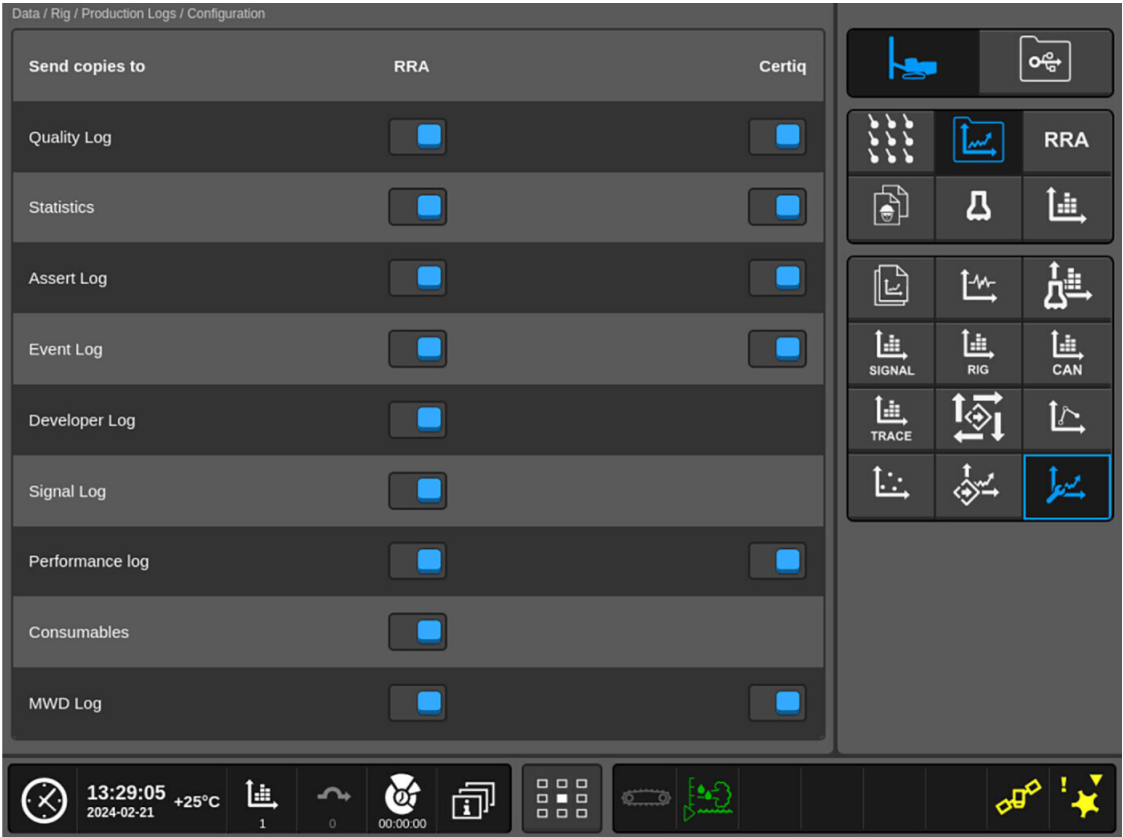


Example of a **Data>Rig>Production Logs>File Storage location Menu**

Select storage location for log files.

- **USB** - Logs to be saved later in USB
- **RRA + USB**- Logs to be sent via Wi-Fi (Rig Remote Access (RRA)) and also saved to USB.

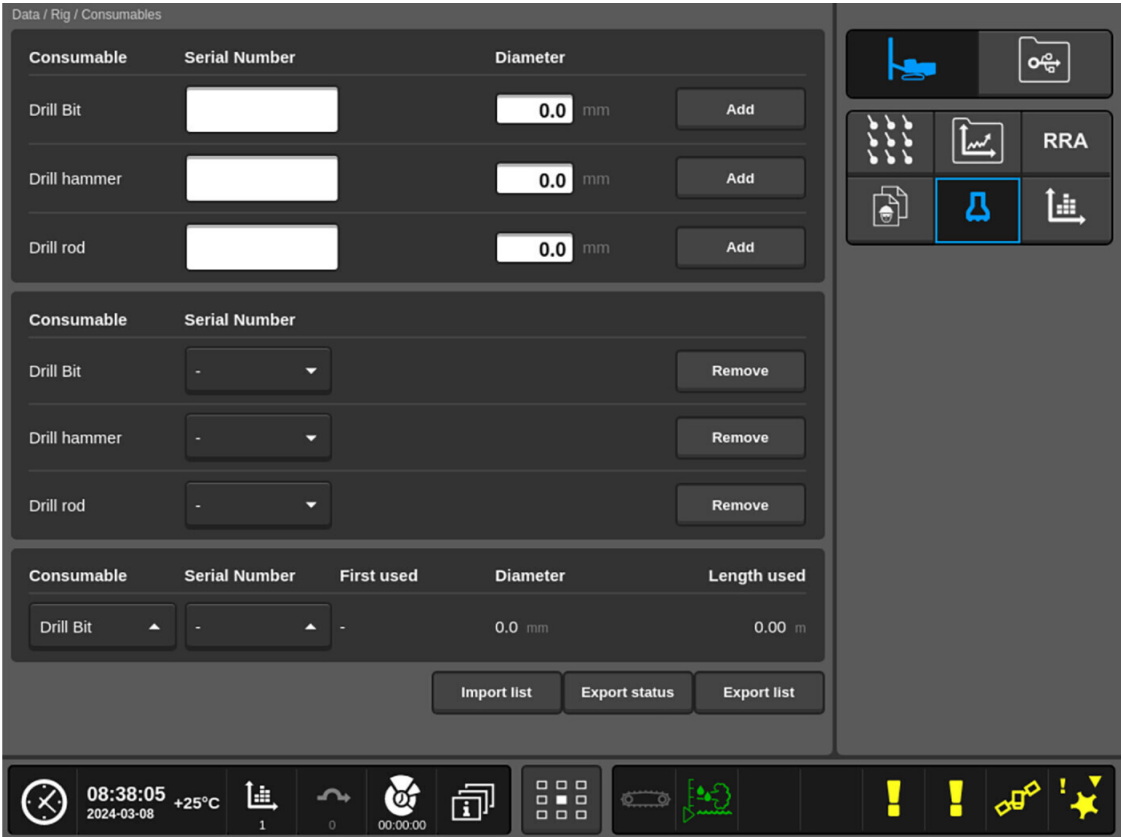
Configuration Menu



Example of a Data>Rig>Production Logs>Configuration Menu

Use the **Configuration** menu to set where the copy of the log should be sent, Rig Remote Access (RRA) or Certiq.

Consumables Menu



Example of a Consumables menu

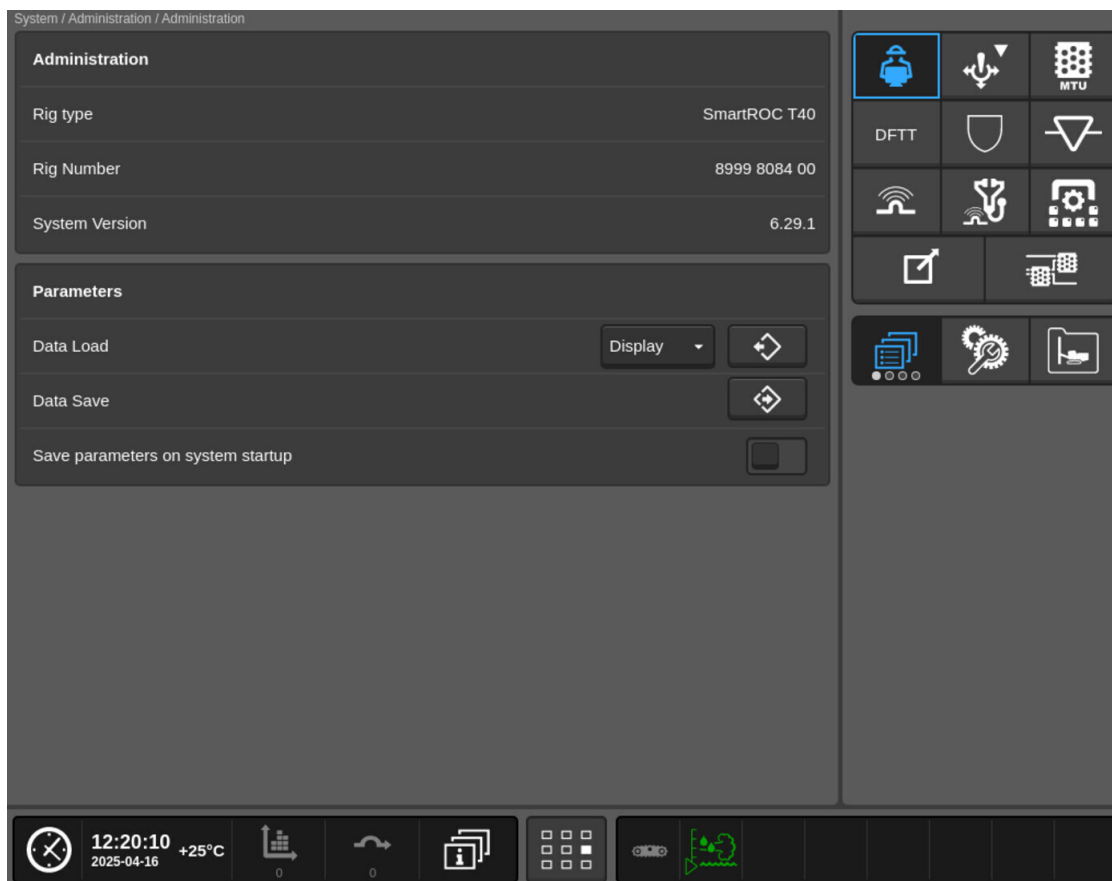
The **Consumables** menu is used to manage consumables like drill bits, rods, and hammers or shank adapters.

The consumables information can be added manually in the machine or imported by .txt files from USB and pressing **Import list**.

The list can later be exported from machine to USB by pressing **Export list**. **Export status** exports the data that is saved in the consumables list. Drill meter for each consumable.

4.6 System Menu

4.6.1 System Administration Menu



Example of an **Administration Settings** menu

The **Administration** menu contains parameters for user settings.

Rig type: Shows the model of machine.

Rig number: The machine production number.

System version: Shows the software version in use.

Data Load: Load parameter file .

- **Display:** The parameter file loads from a USB (if USB with parameter is connected to the USB port) or VCU (rig computer) storage.
- **RRA:** The parameter file loads from the Rig Remote Access (RRA)

Data save: Save parameters to USB.

Save parameters on system startup: If this function is activated, then a time stamped parameter file is sent to the Rig Remote Access (RRA) each time that the Rig Control System (RCS) starts up.

Automatic menu change: Selection of viewing mode, On/Off. When activated, there is an automatic change of menu when operator mode is changed with the mode selector. Recommend to be active during drilling operation.

Disable drillplan autopos redirect: Select this option to disable redirection to angle indication in the drill plan view during auto-positioning.

Units of measurement: Set if the system shows metric or imperial units.

Date: Shows the date. If it is necessary to change the date, tap on the button to the right of the date.

Time: Shows the time. If it is necessary to change the time, tap on the button to the right of the time.

Select Time Zone: Select the local time zone from the drop-down menu.

Enable mode switch button: Activates the mode switch button on the left joystick for readjustment of the boom and feed in drilling mode.

Lockout date and time change for operators: When selected, the possibility to change date and time settings at OP login level is removed.

Clean screen: Pressing the button temporarily disables all touch events from the screen and shows a specific cleaning screen while still having the rig up and running.

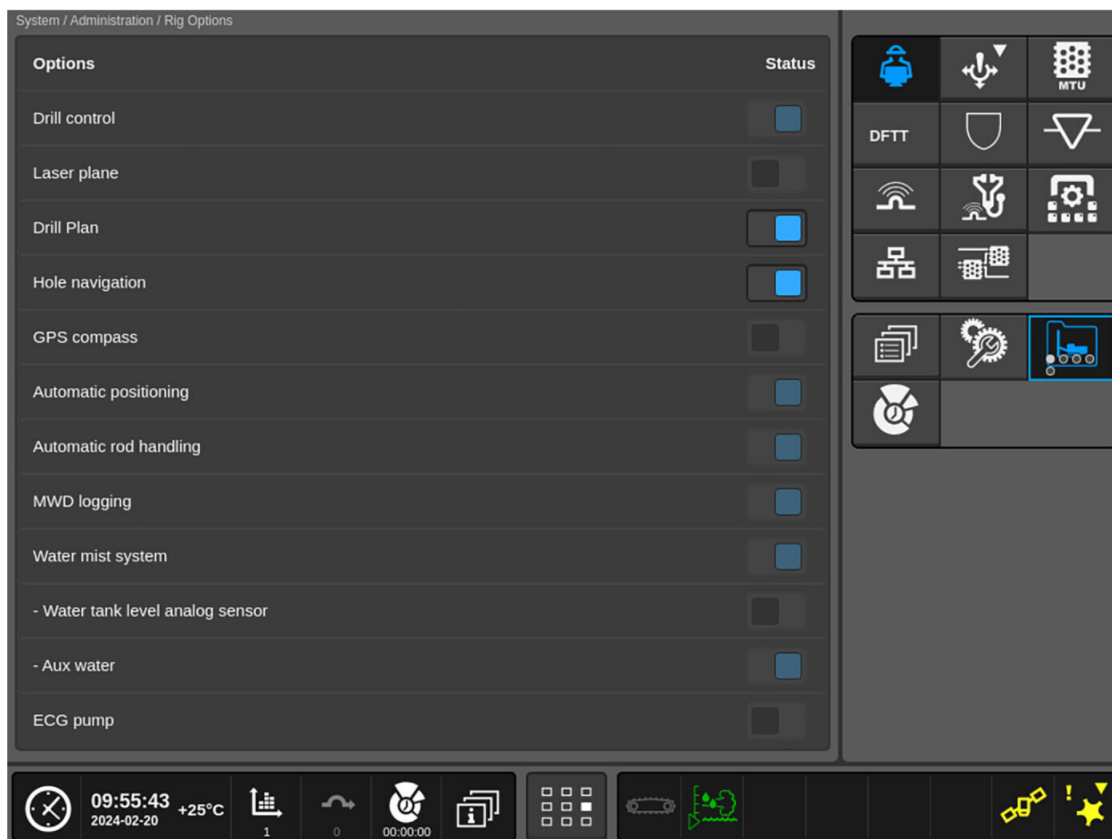
4.6.2 System Configuration Menu

System / Administration / Configuration	
Parameter	Value
Rig type	SmartROC D65
Rotation unit	DHR 6
HNS GPS receiver	Trimble SPSx50
Engine type	CAT C15 T4F/Stage V
Pump1 displacement	85 cc
Pump2 displacement	100 cc
Boom type	C20
Feeder type	BMH 8000 CFH C20 LF
Length Sensor Type	Absolute
Rod handling type	RHS-140
Magazine rotation sensor type	Can Open
RHS arms position sensor type	Analog

Example of a Configuration menu

The **Configuration** menu shows the rig configuration.

4.6.3 System Rig Options Menu

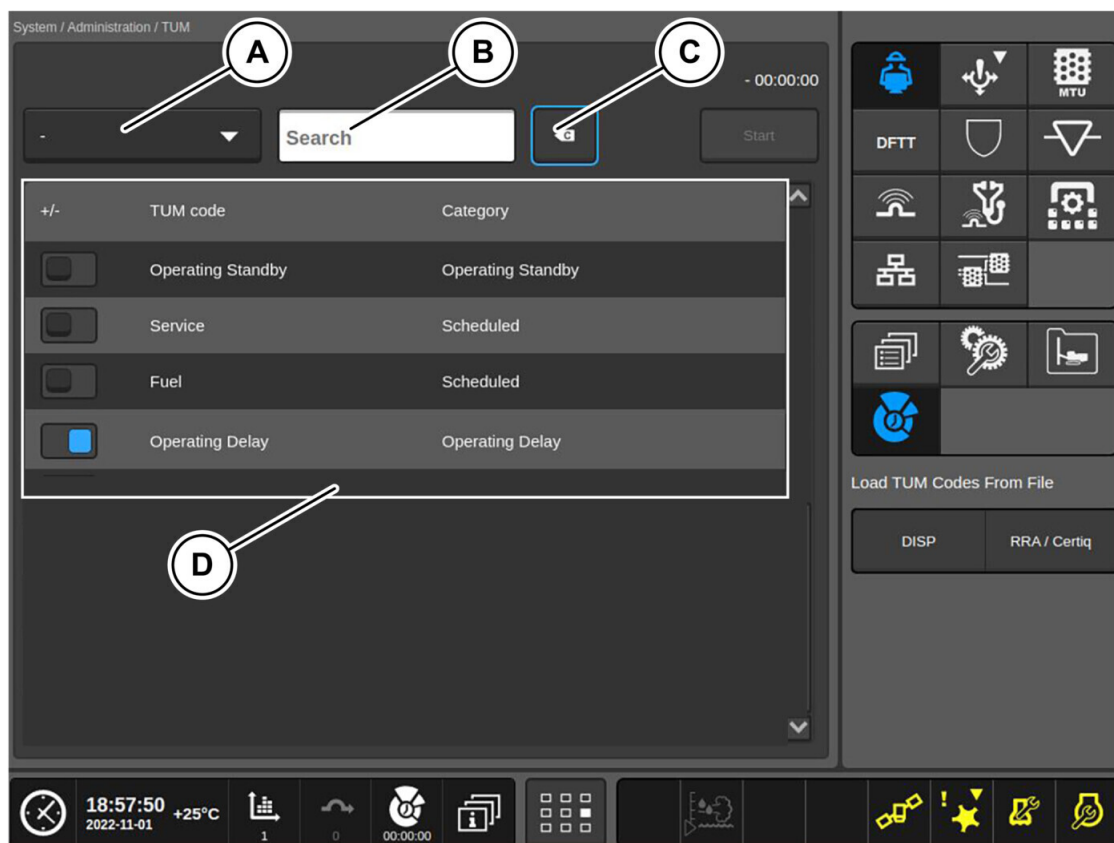


Example of a Rig Options Menu

The **Rig Options** menu shows the available rig options. Some of the options can be activated and deactivated here. If a button is gray, the function is not within the software or cannot be activated or deactivated in this menu.

4.6.4 TUM Codes Menu

A Time Usage Model (**TUM**) is a complete and consistent way to describe and classify all machine time. **TUM** codes are used to classify both operational time and machine down time. Once a **TUM** code is active, it remains active until the next **TUM** code is selected. When a new **TUM** code is selected, the previous **TUM** code ends. **TUM** codes replace delay codes.



Example of a TUM Menu

A	Select TUM category from the drop-down list. • Operating Standby • Non-Productive time • Scheduled • Operating Delay • Unscheduled Time • Productive Time • Unscheduled • Idle Time
B	Search for TUM codes
C	Tap to clear selection.
D	The available TUM codes are shown here.
DISP	Tap DISP to load TUM codes from USB on Display.
RRA/Certiq	Tap RRA/Certiq to load TUM codes through Rig Remote Access (RRA) or Certiq.

Load TUM Code

Time Usage Model (TUM) codes must be loaded onto the Rig Control System (RCS), they cannot be created on the RCS. TUM codes can be created with Surface Manager.



Example of TUM Codes Menu

Precondition ✓ The TUM codes are available on a USB flash drive that is placed in the USB port in the cab or through Rig Remote Access (RRA) or Certiq.

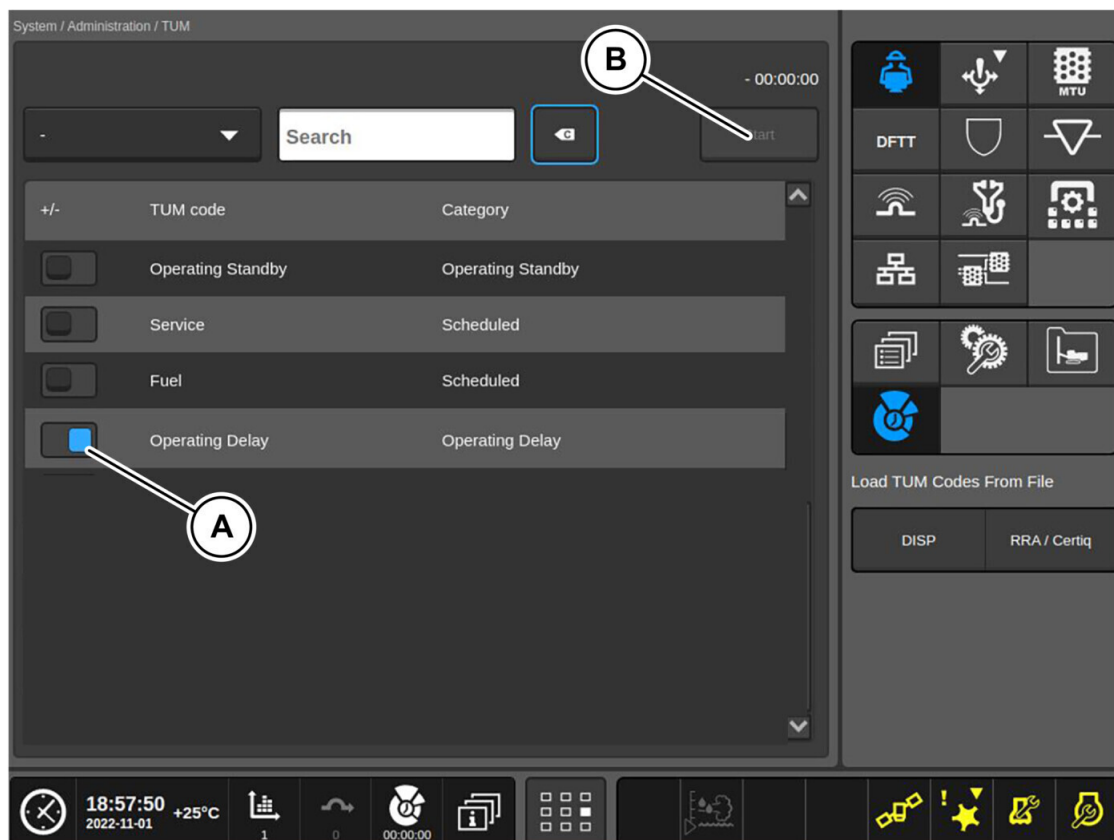
1. Open **System > Administration > TUM** menu.
2. Select button **DISP** (A) or **RRA/Certiq** (B) to load TUM codes.



NOTE: If **DISP** is selected, the TUM codes are loaded from the USB flash drive. If **RRA/Certiq** is selected, the TUM codes are loaded through RRA or Certiq.

Step result → The available TUM codes are shown in the menu (C).

Start TUM Code



1. Open the **System** > **Administration** > **TUM** menu.
2. Tap on box (A) next to the applicable Time Usage Model (TUM) code to activate the code.

Step result → The box turns blue.

3. Tap **Start** (B).

End result → The selected TUM code is started.
The timer in the  symbol starts.

4.6.5 System Levers Menu



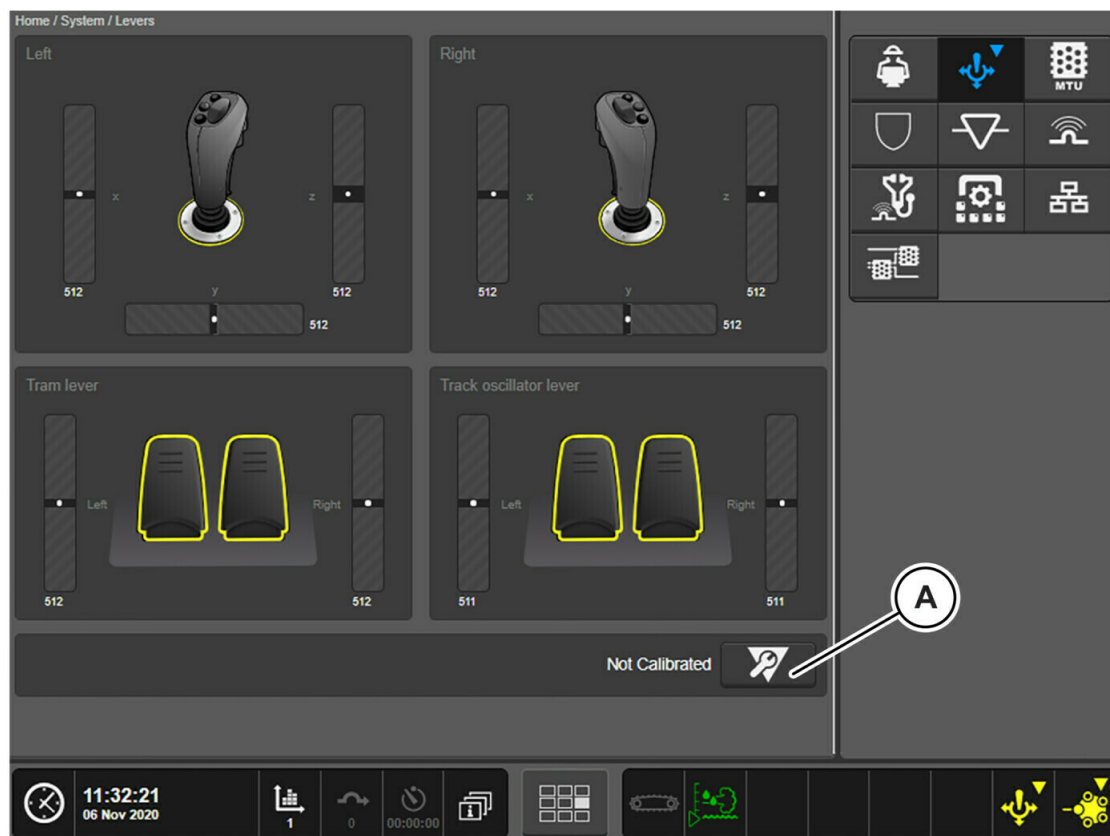
Example of a Levers Menu

In the Levers menu, it is possible to check that the operator levers are working.
Colored outlines around the lever show the status of the lever.

No outline	The lever is calibrated and fully functional.
Yellow outline	The lever is not calibrated and could indicate incorrect end position, incorrect center position or that the decoder is defective.
Red outline	The lever has no contact with the module.
Green outline	Calibration is complete. The green outline disappears on exiting calibration mode.

Calibrate the Levers

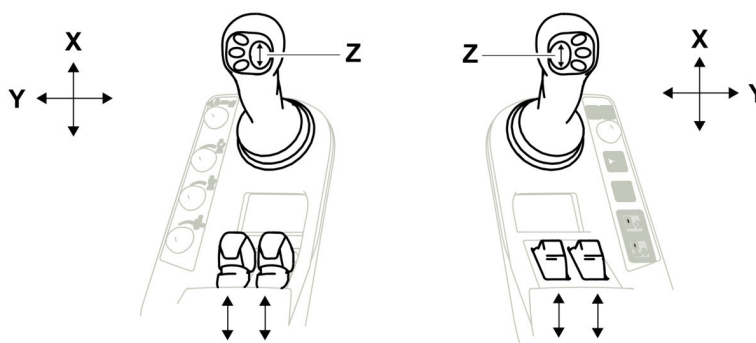
All the levers must be calibrated during the same lever calibration.



Example of a **Levers Menu**

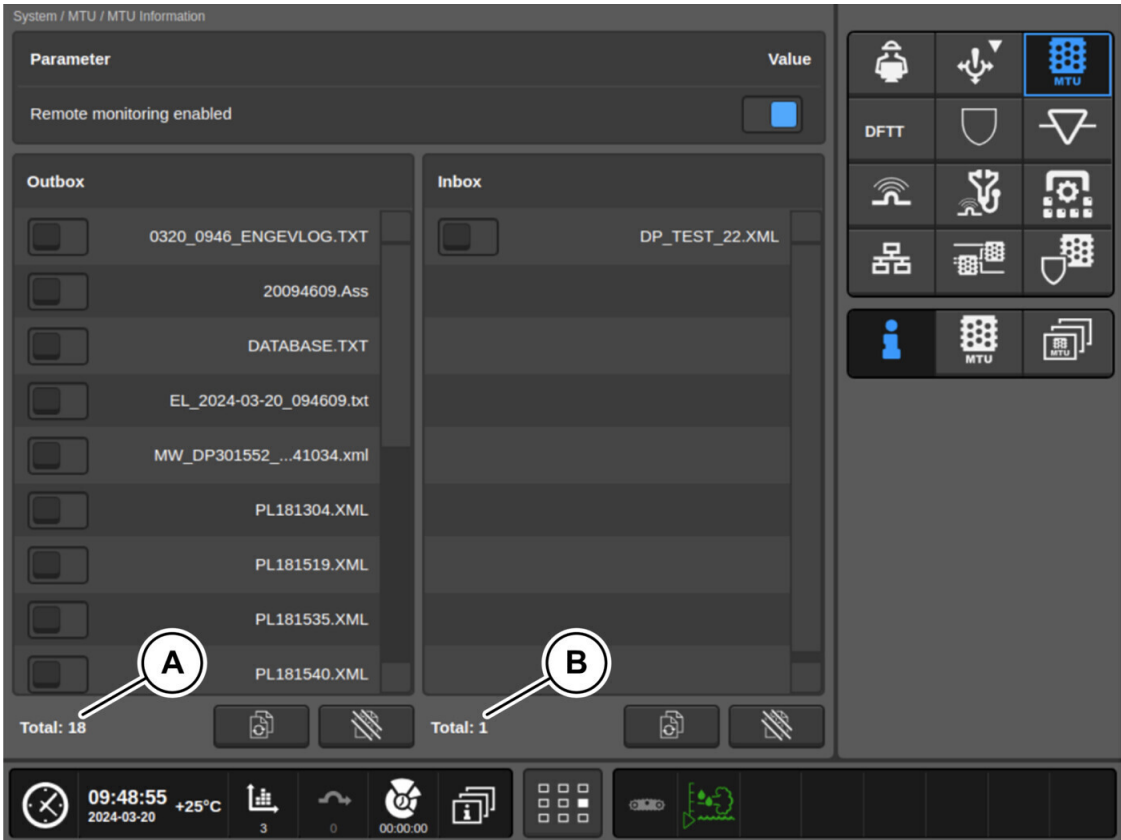
Precondition ✓ The diesel engine is off.

1. Open **System > Levers** menu and press button (A) to start calibration.
2. Move the levers in all directions to end position and then press on the button **Done**.



Step result → The status changes to **Calibrated**

4.6.6 MTU System Menu



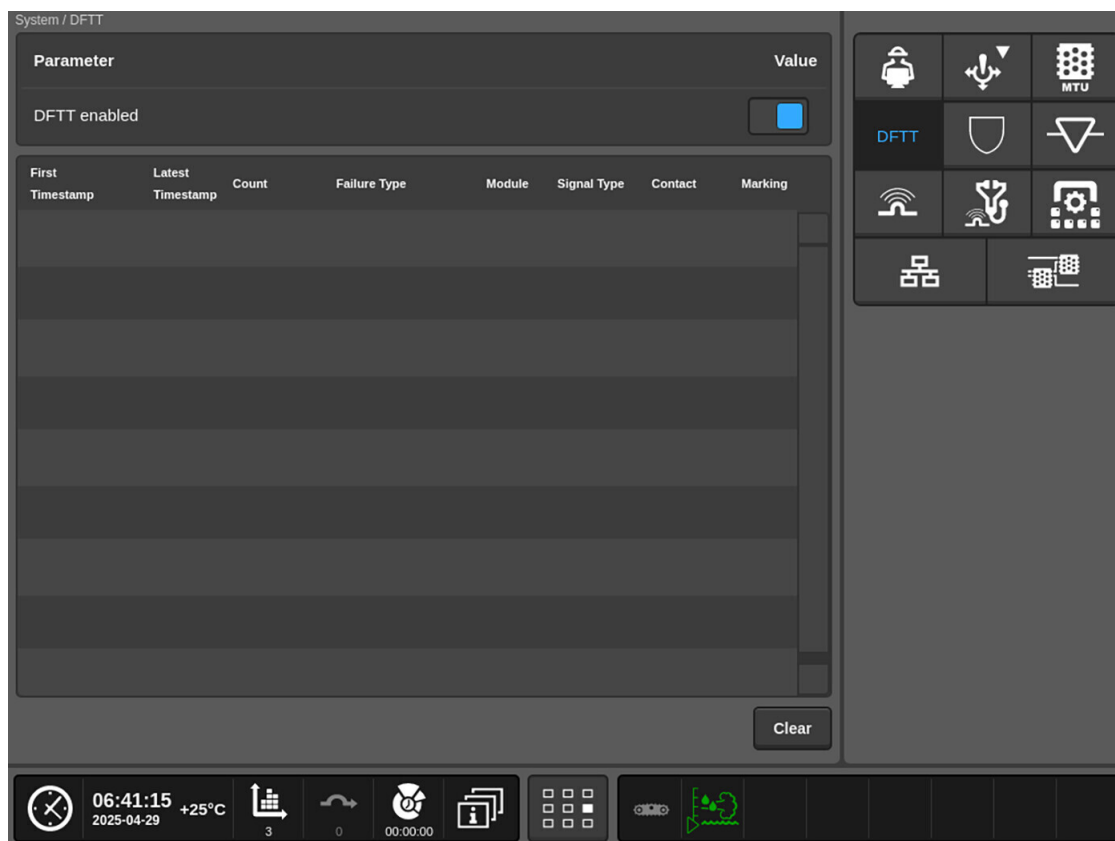
Example of a Machine Telematic Unit (MTU) System Menu

A	Shows the total number of files currently waiting in the Outbox .	B	Shows the total number of files currently waiting in the Inbox .
---	--	---	---

NOTE: The **Outbox** and **Inbox** are only shown if **Remote monitoring enabled** is enabled.

Remote monitoring enabled: Indicates if the Certiq system is activated or not. This setting is done in the factory.

4.6.7 DFTT Menu



Example of a DFTT menu

DFTT enabled: When the Diagnostic Fault Tracing Tool (DFTT) is activated, it shows the list of currently active electrical faults and faults that occurred in the past and were recorded. Use this function active to help troubleshoot the rig.

4.6.8 Guards System Menu



Example of a Guards Menu

A	The Reset Emergency stop button is used to do a manual emergency stop reset from the operator station. ! NOTE: Only shown for rigs with BenchREMOTE, Office TeleREMOTE, or Common Automation Panel (CAP).
---	--

The status of the various guards (**Active / Not active**) is shown in the **Guards** menu. The module marking and contact to each sensor is shown to make it easier to locate the source of the fault.

!	NOTE: Use the Interlocks menu to override the interlocking of a guard during a preset time.
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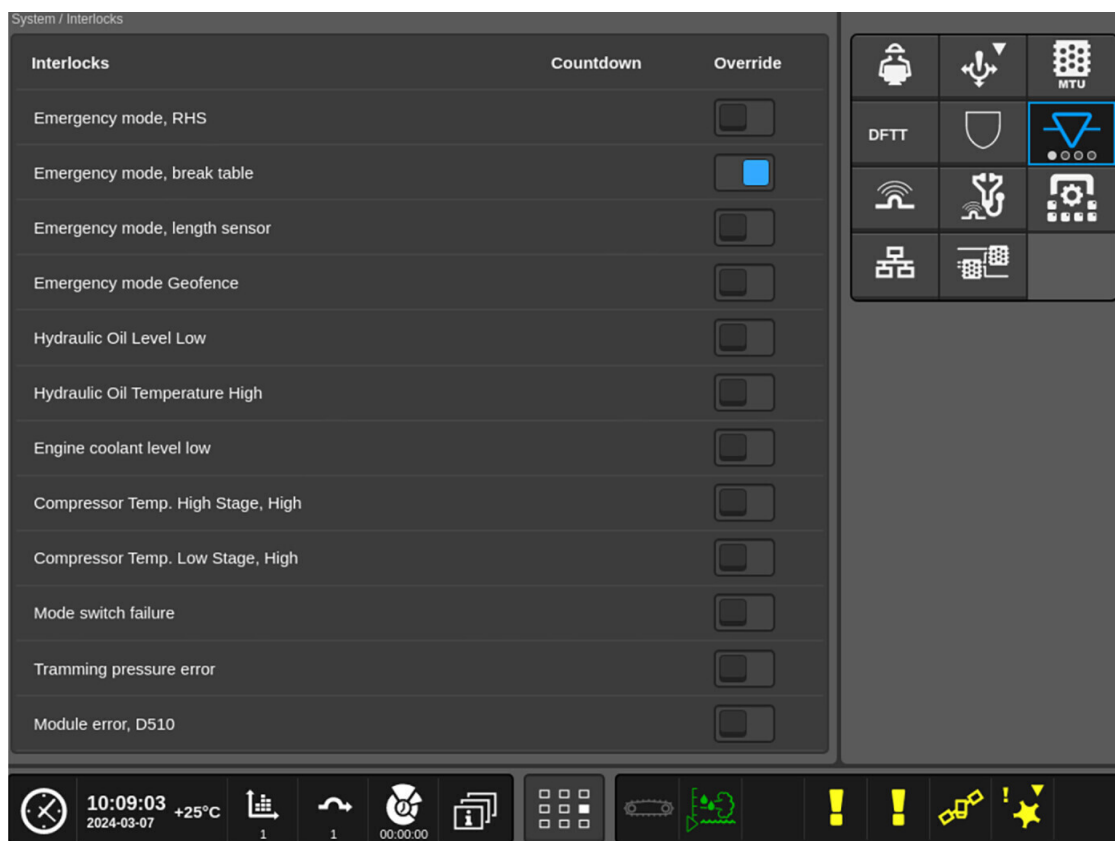
Override Interlocking Guards

NOTICE

Damage to Machine

Frequent use of the override function can damage the machine.

- Only use the override function for moving the machine to a safe place for maintenance.



1. Open **System > Interlocks** menu.
2. Press the button in the **Override** column to select the guard to override.

End result → The preset time starts to count down.

4.6.9 Sensors System Menu

System / Sensors / Pressure

Function	Value	Raw Value	Module	Contact	Marking		L-value	K-value
Pump 2 pressure	0.00 bar	102	D511	X2B	B101	=0	102	0.428
Feed Pressure	0.00 bar	100	D513	X20B	B133	=0	102	0.428
Flush air pressure	0.00 bar	409	D515	X2B	B188	=0	409	0.015
Tramming pressure	0.00 bar	102	D510	X4B	B508	=0	102	0.428
Pump 1 pressure	0.00 bar	102	D511	X3B	B512	=0	102	0.428
Pump 3 pressure	0.00 bar	102	D510	X21B	B513	=0	102	0.428
Pump 4 pressure	0.00 bar	102	D515	X3B	B512D	=0	102	0.428
Pump 5 pressure	0.00 bar	102	D515	X4B	B512E	=0	102	0.428
Compressor Vessel Pressure	0.00 bar	409	D512	X2B	B456	=0	409	0.015
Auxiliary pressure sensor	0.00 bar	0	D513	X22B	B511	=0	409	0.015

Example of a Sensors menu

In the **Sensors** menu, it is possible to view values for the sensors of the machine and, calibrate sensors if necessary. Certain sensors are preset and cannot be calibrated.

The **Sensors** menu tool panel contains several submenus (A), depending on which menu is selected different parameters are shown.

When a pressure sensor is replaced, it is important that the new sensor is the same type as the original sensor.

The new sensor must be reset and calibrated. During the reset, the sensor must be depressurized and deactivated. Reset is done by using reset, **=0**.

Calibrate Pressure Sensor

System / Sensors / Pressure

Function	Value	Raw Value	Module	Contact	Marking		L-value	K-value
Pump 2 pressure	0.43 bar	102	D511	X2B	B101	=0	101	0.428
Feed Pressure	1.28 bar	100	D513	X20B	B133	=0	97	0.428
Flush air pressure	0.00 bar	409	D515	X2B	B188	=0	416	0.015
Tramming pressure	0.86 bar	102	D510	X4B	B508	=0	100	0.428
Pump 1 pressure	1.71 bar	102	D511	X3B	B512	=0	98	0.428
Pump 3 pressure	1.71 bar	102	D510	X21B	B513	=0	98	0.428
Pump 4 pressure	0.00 bar	102	D515	X3B	B512D	=0	102	0.428
Pump 5 pressure	0.00 bar	102	D515	X4B	B512E	=0	102	0.428
Compressor Vessel Pressure	0.00 bar	409	D512	X2B	B456	=0	409	0.015
Auxiliary pressure sensor	0.00 bar	0	D513	X22B	B511	=0	0	0.015

08:42:13 2023-10-26 +25°C 3 0 00:00:00

- Precondition**
- ✓ The diesel engine is off.
 - ✓ The system must be depressurized.

1. Go to the **System > Sensor > Pressure** menu.
2. Select the box (**=0**) by the relevant pressure sensor.
3. Press Enter to reset the sensor to 0 bar.

End result → The sensor automatically receives an offset value.

Calibrate Position Sensor



NOTE: Calibrating the position sensor resets the Smart fold calibration.

System / Sensors / Feeder

Function	Value	Raw Value	Module	Contact	Marking	Offset	K-value
Feed Extend	1.00	188	D513	X21B	B305	=	-0.00320
Feed swing, cylinder position	1.105	2293	D514	X2A	B309	=	0.00015
Feed dump, cylinder position	2.514	3610	D513	X2A	B303	=	0.00035
Rod in magazine outlet	1	1	D514	X12A	B178		
Suction hood position	0	196	D513	X22A	B903	=	0.00065
Feed swing	0.00 °					=	0.000
Feed dump	110.00 °					=	110.000
Feeder inclinometer dump	0.00 °	1	D171			=	-0.001
Feeder inclinometer swing	0.00 °	1	D171			=	0.001

Right sidebar icons: DFTT, MTU, and various sensor/function icons. The aiming device icon is highlighted in blue.

Bottom status bar: 08:42:55, 2023-10-26, +25°C, 3, 0, 00:00:00, and various system icons.

1. Adjust the aiming device in the cab directly forward 90° towards the windshield.
2. Adjust the boom straight forward, parallel with the cab.
3. Reset the sensor by selecting the box for the relevant sensor, and press Enter.

Replacement of Sensors

When a sensor is replaced, the sensor must be reset and calibrated in the control system.

This reset is not a value calibration, it is just to confirm to the Rig Control System (RCS) system that the sensor is correctly connected electrically.

4.6.10 Sensor Monitoring System Menu

Home / System / Sensor monitoring / Pressure

Function	Value	Raw Value	Module	Contact	Marking	Status
Rotation Pressure	0.0 bar	102	D511	X2b	B101	●
Feed Pressure	0.0 bar	100	D513	X20b	B133	●
Flush air pressure	0.0 bar	409	D515	X2b	B188	●
Tramming pressure	1.7 bar	102	D510	X4b	B508	●
Pump 1 pressure	1.7 bar	102	D511	X3b	B512	●
Pump 3 pressure	0.0 bar	102	D510	X21b	B513	●
Pump 4 pressure	0.0 bar	102	D515	X3b	B512D	●
Pump 5 pressure	0.0 bar	102	D515	X4b	B512E	●
Compressor Vessel Pressure	0.0 bar	409	D512	X2b	B456	●
Hydraulic oil filter pressure	0.1 bar	104	D515	X20b	B139	●

=0

Tool Panel (A):

- Icons for various system functions: MTU, shield, funnel, antenna, pump, gear, and others.
- A callout 'A' points to a sub-menu icon (a blue square with a white circle and a blue dot) in the tool panel.

Status Bar:

- Time: 15:27:32, 21 Apr 2020
- Buttons for navigation and settings.
- Icons for system status: car, green signal, red signal, and warning.

Example of a Sensor Monitoring Menu

The **Sensor Monitoring** menu tool panel contains several submenus (A), depending on which menu is selected different parameters are shown.

The **Sensor Monitoring** menu shows the status of a specific sensor.

- Green light indicates that the status is OK. There is contact.
- Red light indicates that the status is not OK. Contact is broken.
- Orange light indicates that the contact has been broken but reconnection has taken place.

Reset takes place in the menu.

Replace Sensor

When a sensor is broken the symbol  shows in the status bar. The status indication for the sensor is red.

Home / System / Sensor monitoring / Pressure

Function	Value	Raw Value	Module	Contact	Marking	Status
Rotation Pressure	0.0 bar	102	D511	X2b	B101	●
Feed Pressure	0.0 bar	100	D513	X20b	B133	●
Flush air pressure	0.0 bar	409	D515	X2b	B188	●
Tramming pressure	1.7 bar	102	D510	X4b	B508	●
Pump 1 pressure	1.7 bar	102	D511	X3b	B512	●
Pump 3 pressure	0.0 bar	102	D510	X21b	B513	●
Pump 4 pressure	0.0 bar	102	D515	X3b	B512D	●
Pump 5 pressure	0.0 bar	102	D515	X4b	B512E	●
Compressor Vessel Pressure	0.0 bar	409	D512	X2b	B456	●
Hydraulic oil filter pressure	0.1 bar	104	D515	X20b	B139	●

=0

15:27:32
21 Apr 2020

1

0

00:00:00

1. Check in the **System/Sensor monitoring** menu which sensor is broken.
2. Replace the broken sensor with a new sensor.

Step result → The status indication in the menu turns orange.

3. Tap on the reset button **=0** to reset the sensor.

Step result ➔ The status indication in the menu turns green. The symbol  in the status bar goes off.

4. Go to the **System/Sensor** menu to calibrate the sensor.
5. Select the box by the relevant sensor and press Enter to reset the sensor to 0 bar.

Step result → The sensor automatically receives an offset value.

4.6.11 MMI Panel Test System Menu



Example of an MMI Panel Test Menu

In the Man Machine Interface (**MMI**) **Panel Test** menu, function tests are done for switches, encoders, and LED's.

- When a check is done on a button or knob, all its functions must be tested.
- When a panel has been tested and each function works, the panel is highlighted in green.

NOTE: The system must be restarted after a test is done.

Test the Function of the Control Buttons

NOTE: All the buttons must be tested during the same test.



Precondition ✓ The diesel engine is off.

1. Open **System > MMI panel Test** menu.
2. Press **Button Test** (A).
3. Push, turn right, and turn left on all multifunctional knobs (six pieces), one after the other.

Step result ➔ Each knob changes color to green after the test.

4. Push all other buttons one after the other.

Step result ➔ Each button changes color to green after the test.

5. Check that all buttons and knobs are green.
6. Restart the control system.

Test the Function of the LEDs



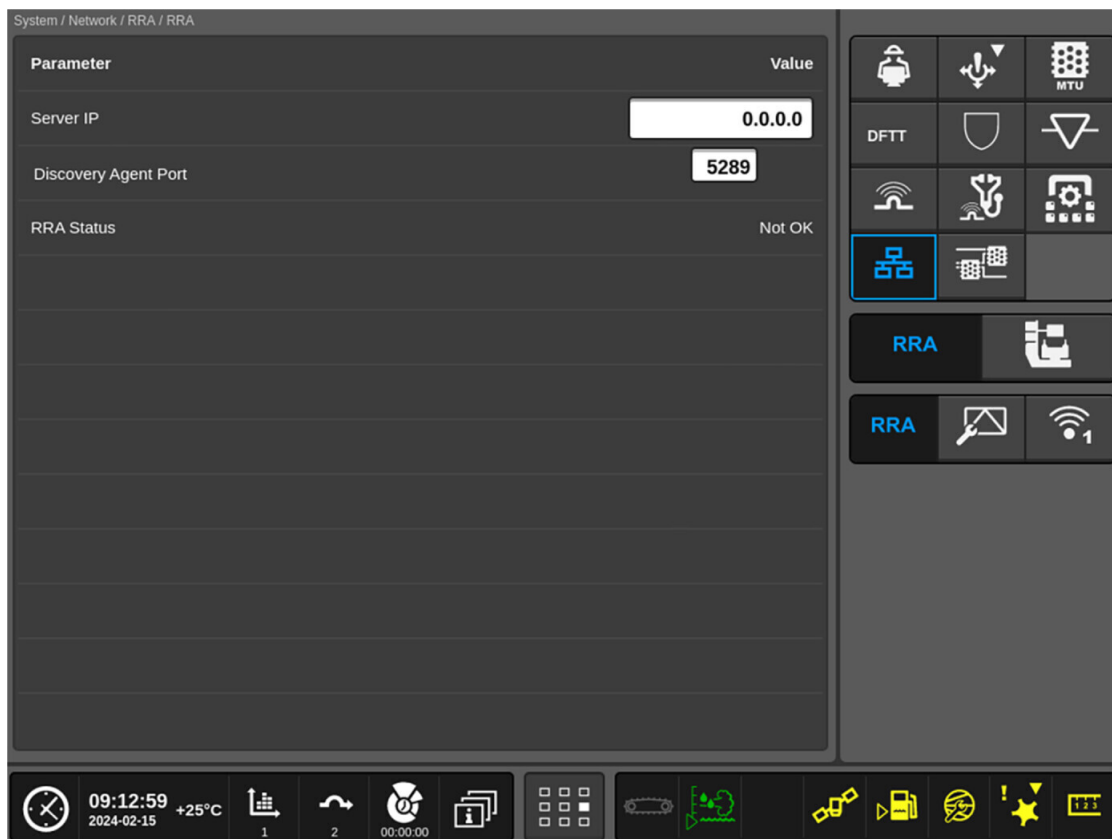
Precondition ✓ The diesel engine is off.

1. Open **System > MMI panel Test** menu.
2. Press the button **LED Test (A)**.

Step result

- ➔ The LEDs come on and go off one after the other.
After all the LEDs have been tested, all the LEDs flash for a moment.
3. Restart the control system.

4.6.12 Network System Menu



Example of a Network menu

In the Network menu, network settings can be set for Rig Remote Access (**RRA**) and **Bench remote**

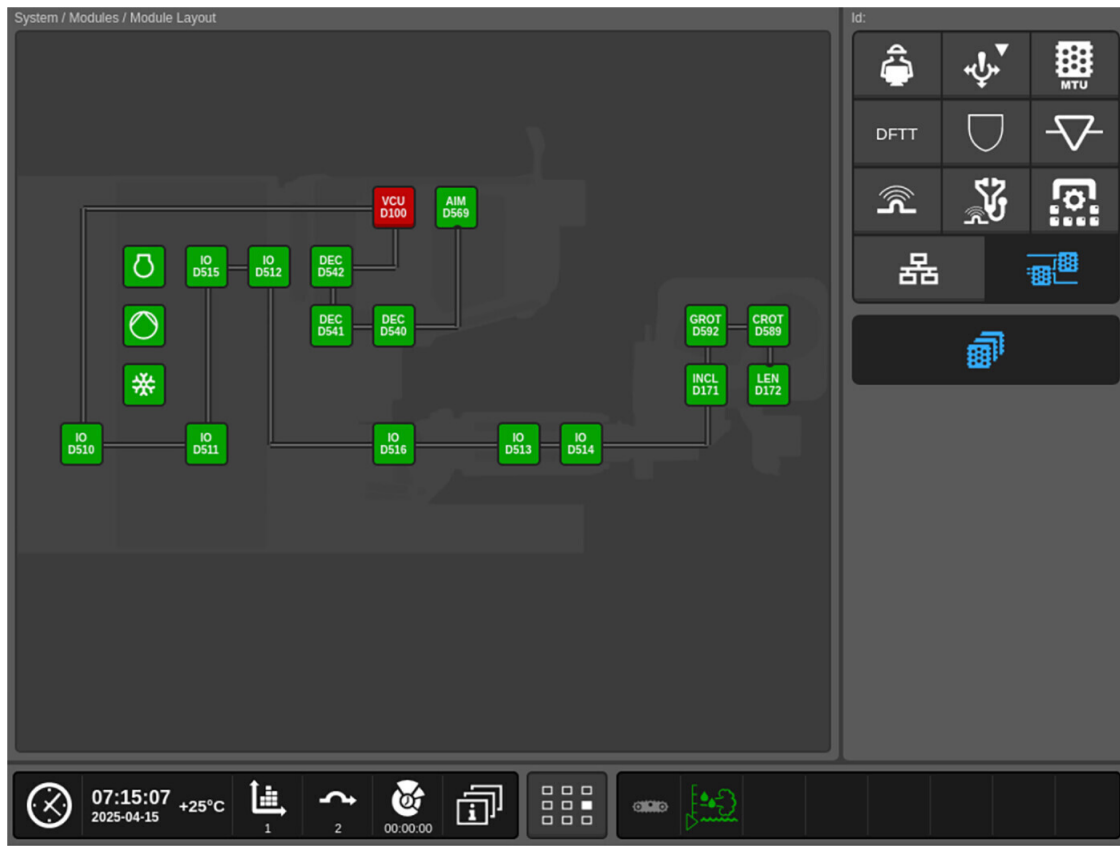
RRA parameters:

- **Server IP:** Shows the IP address of the RRA server. The end user must provide this address.
- **Discovery Agent Port:** Shows the port that is used to communicate the production files. Default Port: 5289.

Bench remote parameters:

- **Ethernet Card:** Identifies the network adapter that is installed in the Rig Control System (RCS) Graphical User Interface (GUI).
- **Link Status:** Shows the condition of the connection between the ethernet card and network hardware.
 - **Link Established:** Connection to network switch detected.
 - **Link Not Established:** Connection to network switch not detected.
- **IP Address:** IP address for RCS GUI. Default IP: 192.168.100.030
- **Subnet Mask:** Subnet mask of Epiroc network. Default: 255.255.255.000
- **Gateway:** Gateway of Epiroc network. Default: 192.168.100.001

4.6.13 Modules System Menu



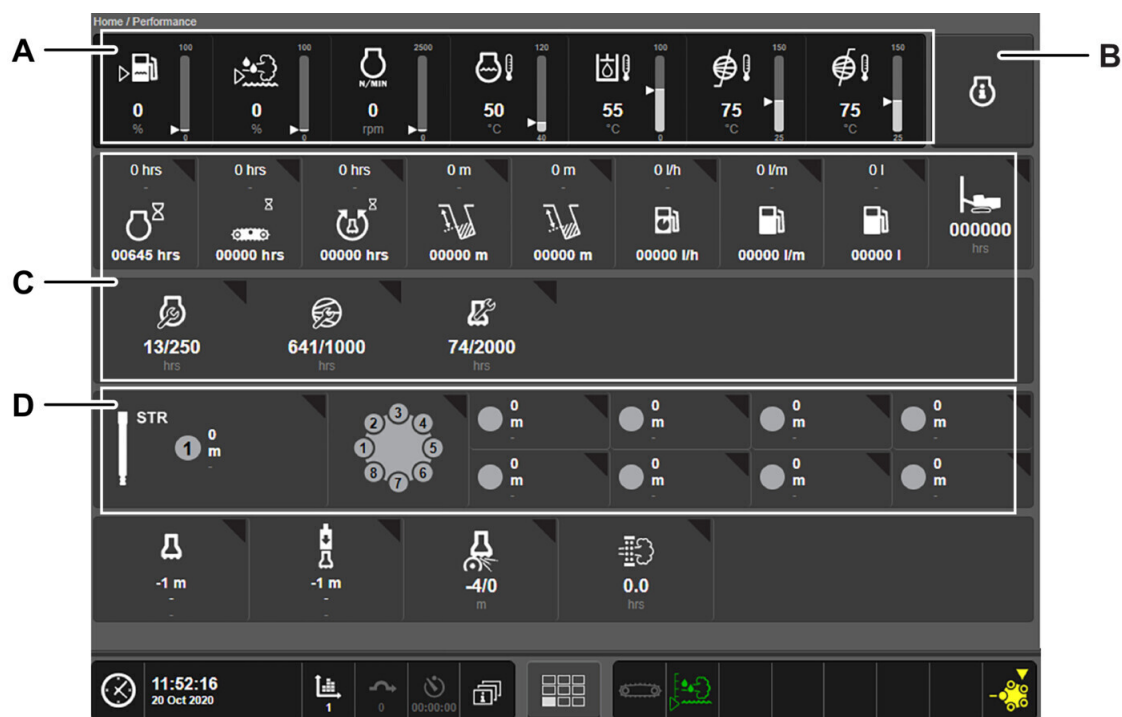
Example of a Modules Menu

The **Modules** menu displays **Module Layout**, showing the electronic modules that are part of the Rig Control System (RCS), and their status.

- Modules highlighted in green are functioning properly.
- Modules highlighted in red are indicating a fault, such as a breakdown in the CAN communication or power supply.

By selecting the respective module, more information is available.

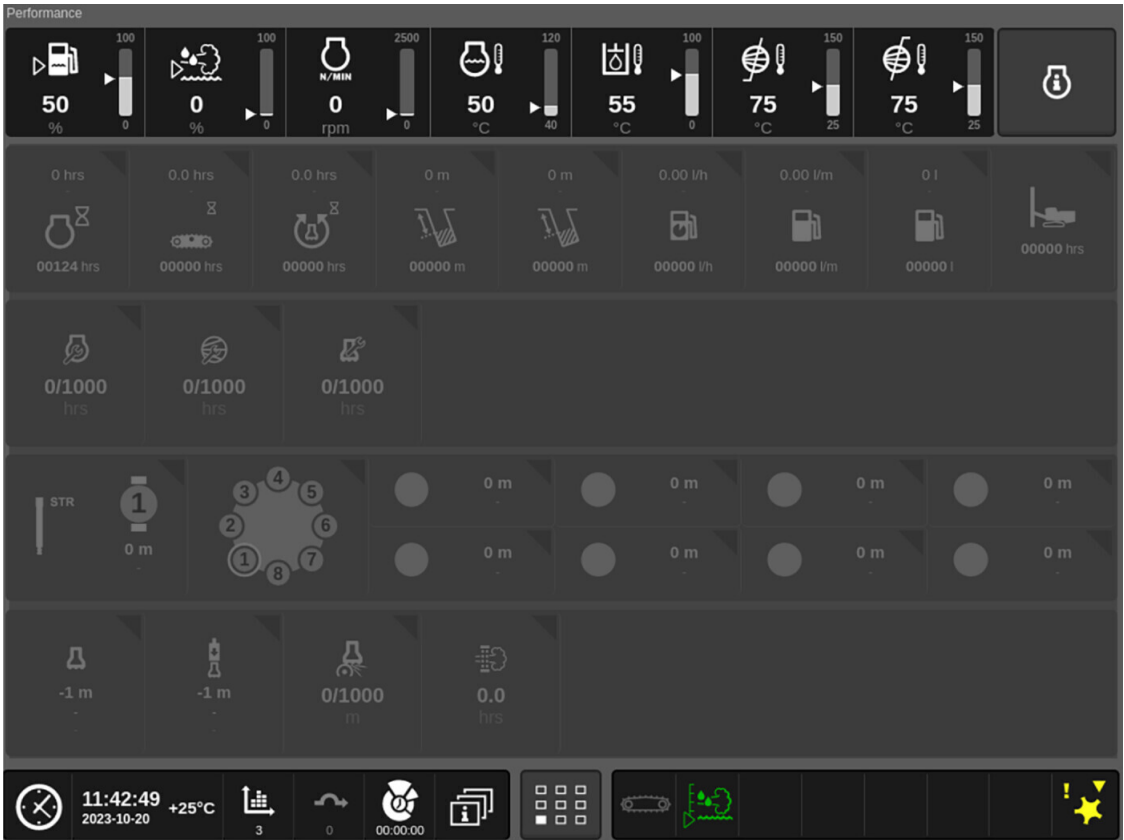
4.7 Performance Menu



Example of a **Performance Menu**

A	Performance gauges	B	Engine performance information
C	Machine performance statistics	D	Tap on these buttons to open pop-up menu to see rod tracker settings.
	Diameter and drilled meters for selected drill bit		Diameter and drilled meters for selected hammer
	Grind interval for drill bit		Time until the next exhaust gas purge.

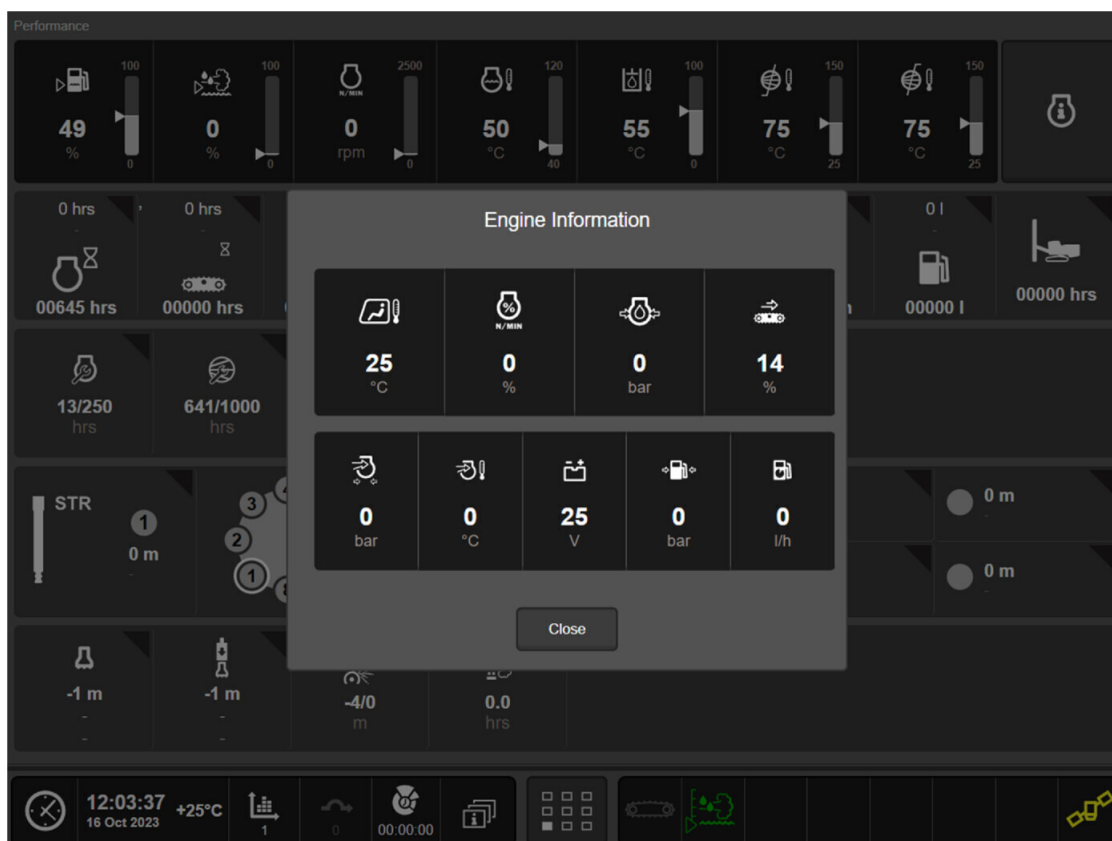
4.7.1 Performance Menu Gauges



Performance Gauges

	The fuel gauge displays the fuel level.		The Diesel Exhaust Fluid (DEF) gauge displays the diesel exhaust fluid.
	Engine revolutions		The coolant temperature gauge displays the temperature of the coolant
	The hydraulic oil temperature gauge displays the temperature of the hydraulic oil.		Compressor temperature low step
	Compressor temperature high step		

4.7.2 Engine Performance Information



Example of an Engine Information pop-up Menu

	Outdoor temperature Shows the temperature outdoors.		Engine load Displays the instantaneous load the diesel engine is exerted to in percent.
	Oil pressure Displays the engine oil pressure when running. Low oil pressure automatically turns off the diesel engine and an information box is shown on the display.		Tramming mode Slow tramming
	Tramming mode Fast tramming		Boost pressure Displays the instantaneous boost pressure when running.
	Charge air temperature Displays the instantaneous charge air temperature when running.		Battery status Displays the charge status of the battery. If there is no charge, an information box is shown in the status bar.
	Fuel pressure Shows the fuel pressure.		Fuel Consumption Shows information about fuel consumption.

4.7.3 Machine Performance Statistics



A pop-up menu is opened when a machine performance statistic icon is selected.

- The **Current** value is resettable and shows the value since last reset.

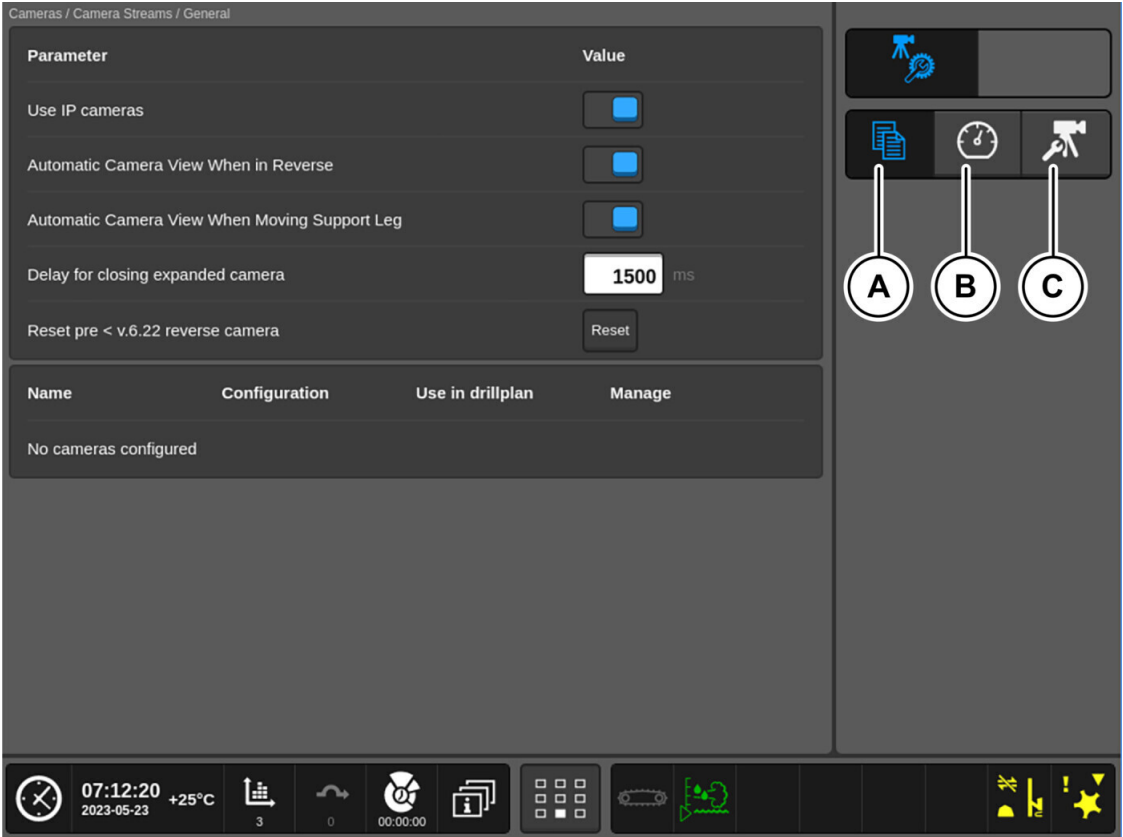
Tap on **=0** to reset the current value.

- The **Accumulated** value shows the total value during the life of the rig. The operator cannot reset the accumulated value.

	Tap to open pop-up menu for Total engine hours .	Shows the current and accumulated engine hours that have been run.
	Tap to open pop-up menu for Total tramming hours .	Shows the current and accumulated number of tramming hours. ! NOTE: Leaving the machine in tramming mode does not increase the total number of tramming hours.
	Tap to open pop-up menu for Total Drilling hours .	Shows the current and accumulated drilling hours.

		 NOTE: Leaving the machine in drilling mode does not increase the total number of drilling hours.
	Tap to open pop-up menu for Hole length trip 1 .	Shows the current and accumulated hole length.
	Tap to open pop-up menu for Hole length trip 2 .	Shows the current and accumulated hole length.
	Tap to open pop-up menu for Fuel rate per hour .	Shows the current and accumulated average fuel consumption per hour.
	Tap to open pop-up menu for Fuel rate per drill distance .	Shows the current and accumulated total fuel consumption for a drilled distance.
	Tap to open pop-up menu for Total fuel consumption .	Shows the current and accumulated total fuel consumption.
	Tap to open pop-up menu for Total machine hours .	Shows the accumulated machine hours.
	Tap to open pop-up menu for Engine Service interval .	Shows the time elapsed and when the next engine service is due.
	Tap to open pop-up menu for Service interval 641/1000 .	Shows the Compressor service hours.
	Tap to open pop-up menu for rock drill Service interval .	Shows the time elapsed and when the next rotation unit service is due.

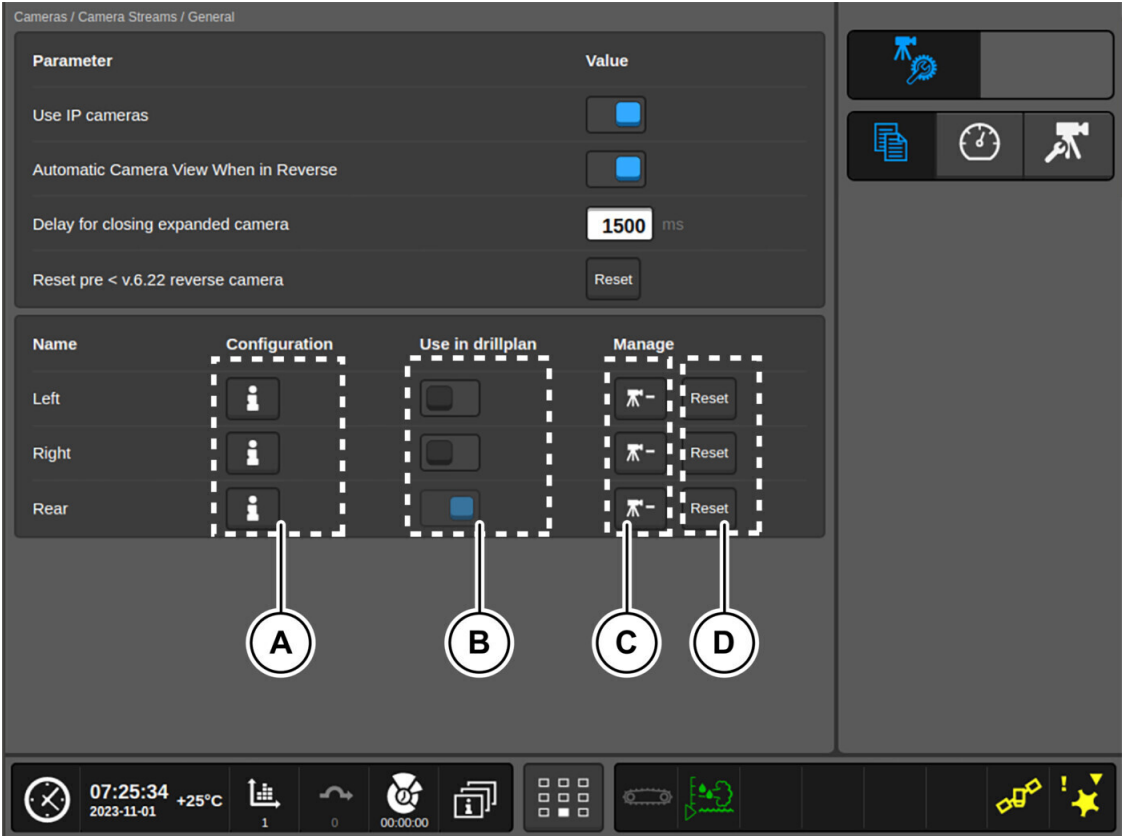
4.8 Camera Menu



Example of a Cameras Menu

A	Select to open General Camera Streams menu.	B	Select to open Camera layout menu.
C	Select to open Configure Camera Streams menu.		

4.8.1 Camera Streams Menu



Example of a Camera Streams Menu

A	Select to see the configuration of the camera and change the name if necessary.	B	Use to set if a camera is shown or not shown in the Drill Plan menu.
C	Select to remove the camera completely from the system. ! NOTE: Only use if the camera is no longer working.	D	Select for Factory reset of camera. ! NOTE: A factory reset removes the camera from the system and the camera has to be reconfigured.

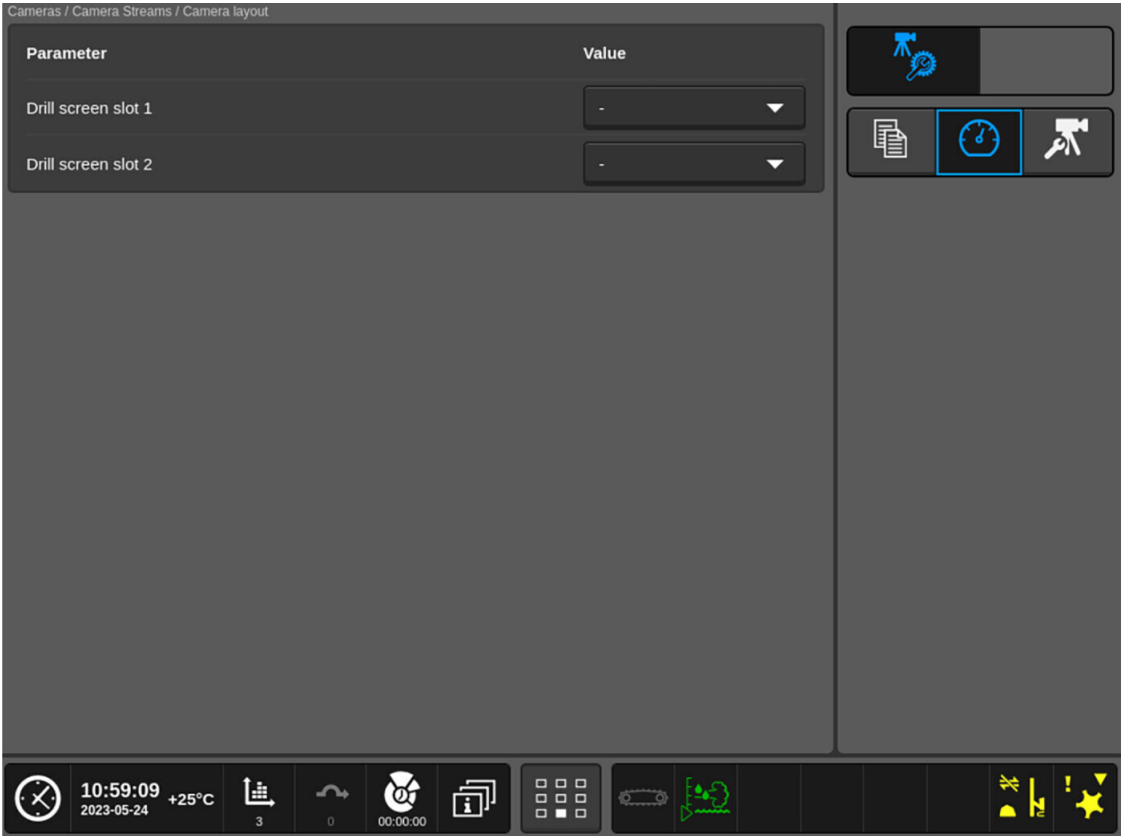
Use IP cameras: The traffic from the cameras is streamed through the router. The video starts streaming when the camera is powered on. If the option is not on, all camera features in the application such as the camera view in the **Drill plan** menu are disabled and not displayed.

Automatic Camera View When in Reverse: Select to show the rear camera view when the rig is tramping backward.

Delay for closing expanded camera: The parameter is used to remove the flickering that occurs when operators operate the support leg and reverse tram causing the cameras to expand. This parameter is used in a timer that is triggered when the expanding action stops. The operator now has x number of seconds defined by the parameter to reinitiate the action without the camera expanding.

Reset pre < v.6.6.22 reverse camera: This button is used to factory reset a reverse camera that is configured from 6.21 and earlier to factory settings. All cameras with a software version older than 6.22 must be reset to factory settings before they are reconfigured.

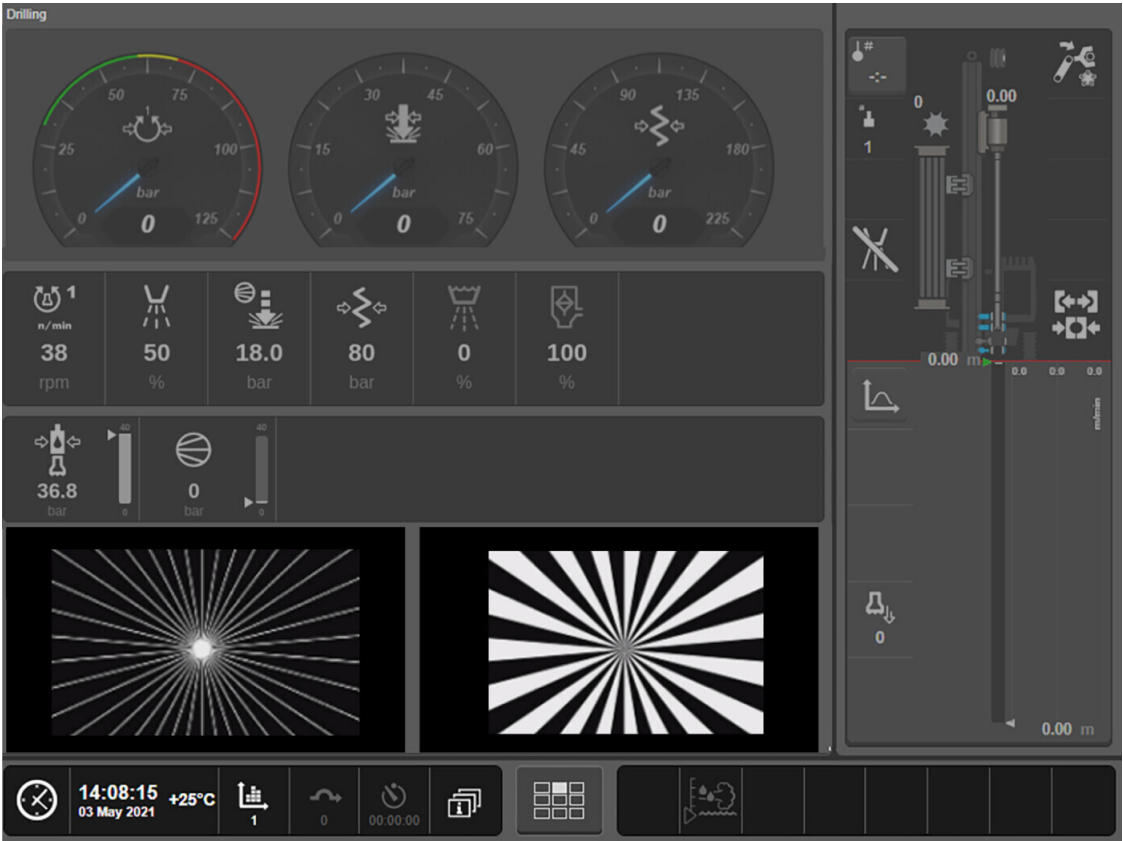
4.8.2 Camera Layout Menu



Example of a Camera Layout Menu

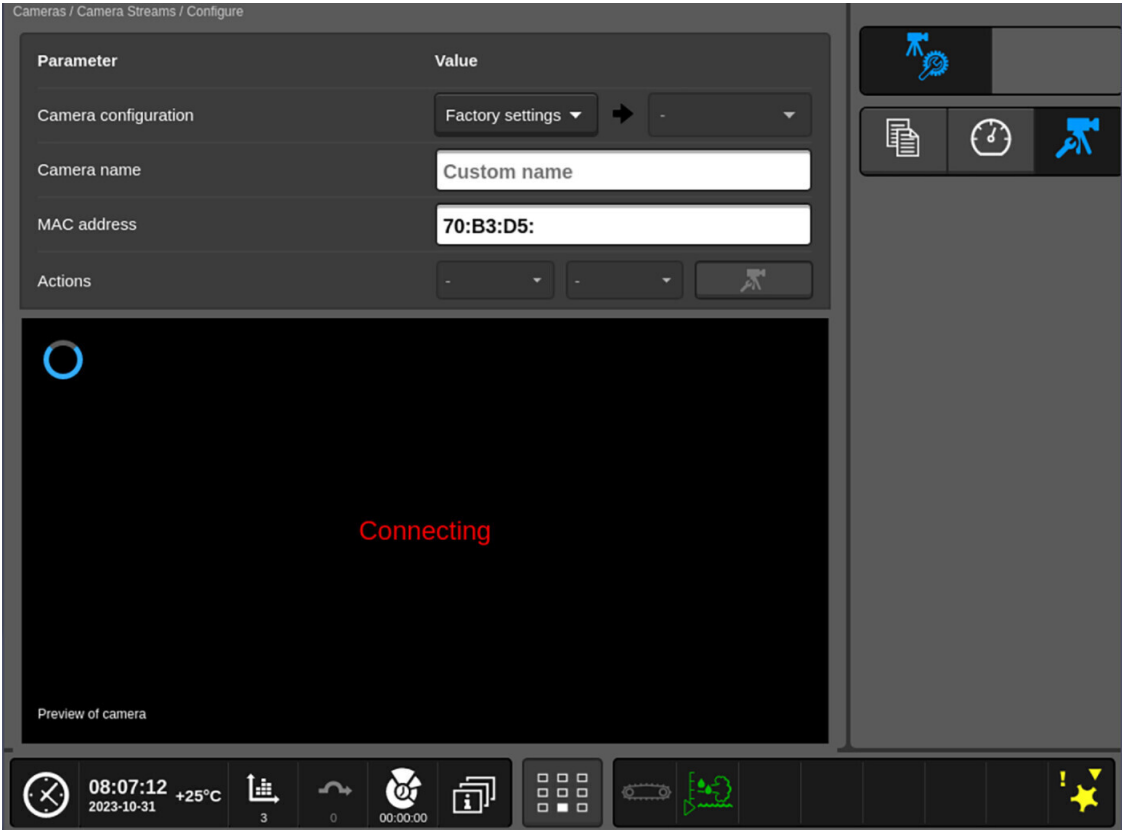
All the cameras that have been configured are shown in these drop-down boxes.

Drill screen slot 1	Select the video stream to be shown in the bottom left corner of the Drilling menu.
Drill screen slot 2	Select the video stream to be shown in the bottom right corner of the Drilling menu.



Drilling Menu with Camera Screens

4.8.3 Configure Camera Streams Menu



Example of a Camera Streams Configure Menu

Camera configuration	In the first drop-down box, select the camera position to configure from. In the second drop-down box, select how to configure the camera (IP address, functions connected to the video stream, and so on).
Camera name	Enter a name that is used for the camera in the system. This name is also visible in all camera views.
MAC address	Enter the address that is marked on the rear side of the camera in this text field. This is done to be able to check directly how the camera is configured.
Actions	In the first drop-down box, choose to flip or not flip the camera view. In the second drop-down box, choose to mirror or not mirror the video stream from the camera. Both of these selections are mandatory. Press the button to send the configuration to the camera that is selected from the first drop-down box in this form. A pop up is shown that states if the configuration was ok or an issue has occurred.
Preview window	Shows a preview of the video stream from the camera that is selected for configuration.

5 Software Management

The Rig Control System (RCS) and related software are supplied and installed from the factory when the equipment is new. A USB flash drive is included with the delivery. This is used for system software backup, parameter backup and storage of the log file. A new USB flash drive is supplied whenever the equipment is upgraded with new options and functions. An upgrade or update can be sent electronically and then loaded using the existing USB flash drive for installation.

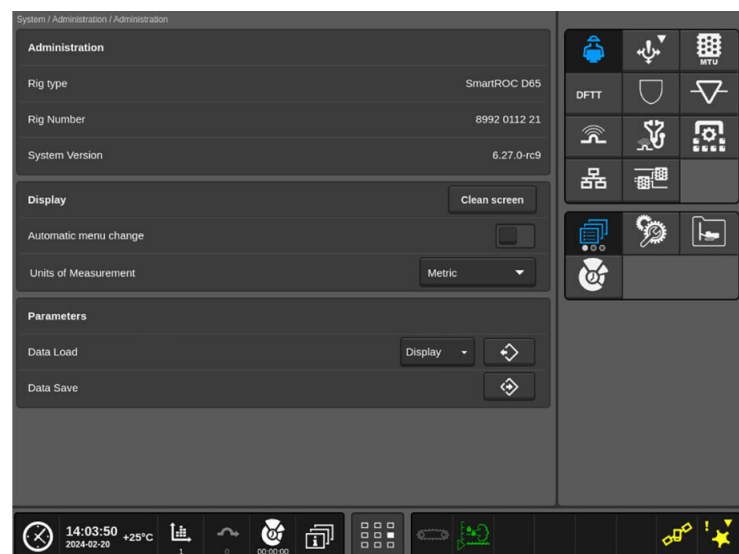
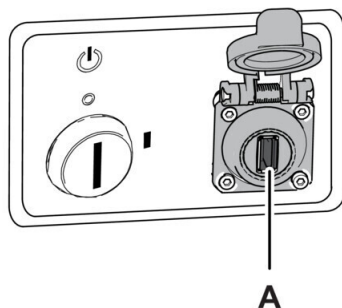
5.1 Software Installation

5.1.1 Necessary Equipment to Install New Software

The following equipment is required for installation:

- USB flash drive for the Rig Control System (RCS).

5.1.2 Save Existing Parameter Data



1. Start the control system.
2. Insert an empty USB flash drive into the USB-port (A).
3. Go to the **System** menu.
4. Tap on **Data Save**  to save existing data.
5. Remove the USB flash drive.
6. Switch off the control system.

5.1.3 Install New Software

- Precondition**
- ✓ The existing rig parameter data must be saved.
 - ✓ The control system is off.

1. Connect the USB flash drive with a software file to the USB port of the control panel. The file name must have the rig number and is a .pkg file



NOTE: Do not load software files from other rigs. Each rig has a unique software file.

2. Turn on the ignition key to start the control system.

Step result

→ The system starts up and presents different options.

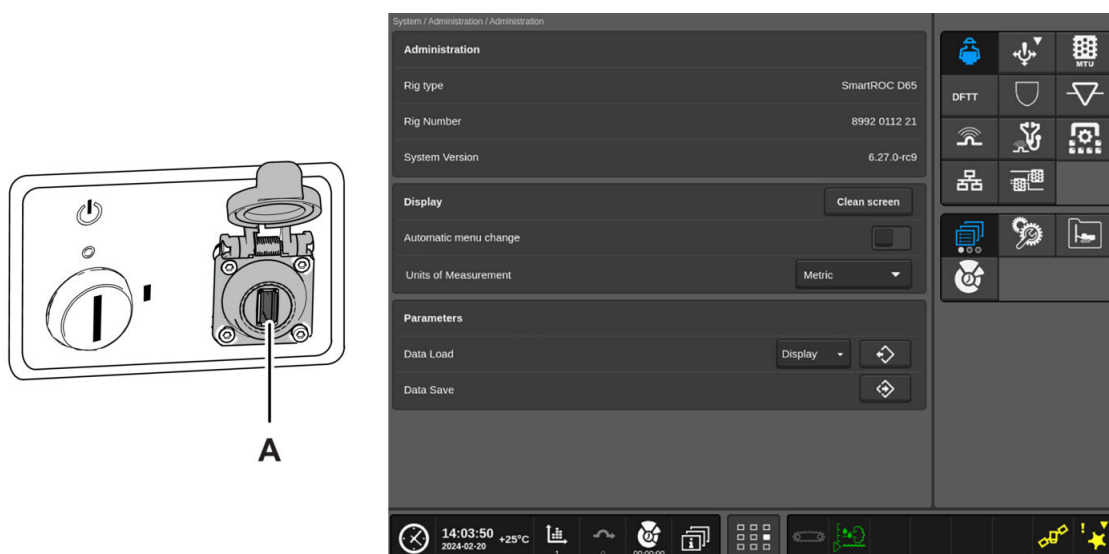
3. Select **UPDATE**.
4. Respond to the question on whether the rig parameters have been saved.



NOTE: It is important that the rig parameters have been saved. If they have not been saved, then the control system must be switched off and the installation restarted from the **System** menu.

5. Select **UPDATE** to load the new software. The installation takes about 5 minutes.
6. Wait until the message **CODE LOAD COMPLETE** is shown and then remove the USB flash drive.
7. Switch off the control system.

5.1.4 Load Parameter Data to USB Flash Drive



1. Start the control system.
2. Insert a USB flash drive with previously saved parameters into the USB-port (A).

3. Go to the **System** menu.
4. Tap on **Data Load**  to load parameter data.
5. Remove the USB flash drive.
6. Switch off the control system.

5.2 After Installing the Software

1. Open the upper and lower breakout jaws for the first time to teach the correct position.
2. Calibrate the rod tracker in the **Performance** menu, click **Rod Setting**, and calibrate.

